

RKNN3 SDK Development Guide

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Preface

Overview

This document is intended for users of the RKNN3 SDK. It systematically introduces the SDK's model conversion, quantization, evaluation, and on-device deployment methods for both coprocessor mode (RK1820 / RK1828) and non-coprocessor mode (RK3572). The document is organized into five parts: fundamentals, conversion and evaluation, on-device deployment, advanced features, and diagnostics and optimization.

Intended Audience

This document is primarily intended for the following engineers:

- Technical support engineers
- Software development engineers

Revision History

Version	Modified By	Date	Description of Changes	Approved By
V1.0.0	HPC / NN	2026-01-01	1. Initial release (RK182X)	熊伟
V1.0.4	HPC / NN	2026-05-12	1. Added RK3572 support 2. Added LLM inference pause and resume 3. Added KV Cache import and export 4. Supported LoRA 5. Supported encrypted model deployment	熊伟
V1.1.0	HPC / NN	2026-08-08	1. Restructured the document 2. Updated rkllm3-server usage 3. Corrected quantization descriptions and API consistency	熊伟

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1. Overview and Development Preparation

1.1 RKNN3 SDK Overview

1.1.1 SDK Positioning and Typical Application Scenarios

The RKNN3 SDK is an AI model deployment development kit from Rockchip for the RK1820 / RK1828 coprocessors and the RK3572 SoC. It provides a complete toolchain covering model conversion, quantization and evaluation, and on-device deployment, and supports efficient NPU inference for various model types such as CNN / LLM / MLLM.

Typical application scenarios include:

- **Visual recognition and detection:** Deploy CNN models such as MobileNet / YOLO on RK182X or RK3572 for tasks such as image classification and object detection.
- **Large language model inference:** Deploy large language models (LLMs) such as Qwen, with support for multi-turn dialogue, streaming generation, KVCache acceleration, LoRA fine-tuning adaptation, and more.
- **Multimodal understanding:** Deploy vision-language models such as Qwen2.5-VL, with support for image-text understanding, video understanding, and more.
- **OpenAI-compatible services:** Provide OpenAI-compatible APIs through rkllm3-server to integrate with the existing application ecosystem.

The SDK supports two deployment modes:

- **Coprocessor mode (RK182X):** The host SoC (such as RK3588 / RK3576) connects to the RK1820 / RK1828 coprocessors via PCIe / USB / Ethernet. Applications run on the host SoC and control the coprocessors to perform AI inference through the RKNN3 API.
- **Non-coprocessor mode (RK3572):** The RK3572 SoC integrates a high-performance NPU. Applications run directly on the RK3572 and control the local NPU to perform AI inference through the RKNN3 API.

1.1.2 Supported Hardware Platforms and Deployment Modes

This document applies to the following hardware platforms:

- **RK1820 / RK1828 (coprocessor mode):** Serve as AI computing acceleration units. They are scheduled by the host SoC and share the computation workload via PCIe / USB / Ethernet interfaces.
- **RK3572 (non-coprocessor mode):** An SoC platform with an integrated high-performance NPU, where the NPU shares memory space with the CPU / GPU.

The main differences between the two modes:

Comparison Item	Coprocessor Mode (RK182X)	Non-coprocessor Mode (RK3572)
Host SoC	RK3588 / RK3576, etc.	None (RK3572 itself is the host)
NPU	RK1820 / RK1828 coprocessors	RK3572 integrated NPU

Comparison Item	Coprocessor Mode (RK182X)	Non-coprocessor Mode (RK3572)
Communication method	PCIe / USB / Ethernet	Unified memory architecture
Communication service	rknn3_transfer_proxy	rknn3-server
Supported systems	Android / Linux / Windows	Android / Linux (aarch64 only)

1.1.3 Software Stack and Component Responsibilities

The overall RKNN3 software stack architecture is divided into three layers: the PC side, the device-side API and application services, and the Runtime and NPU driver.

Coprocessor mode architecture

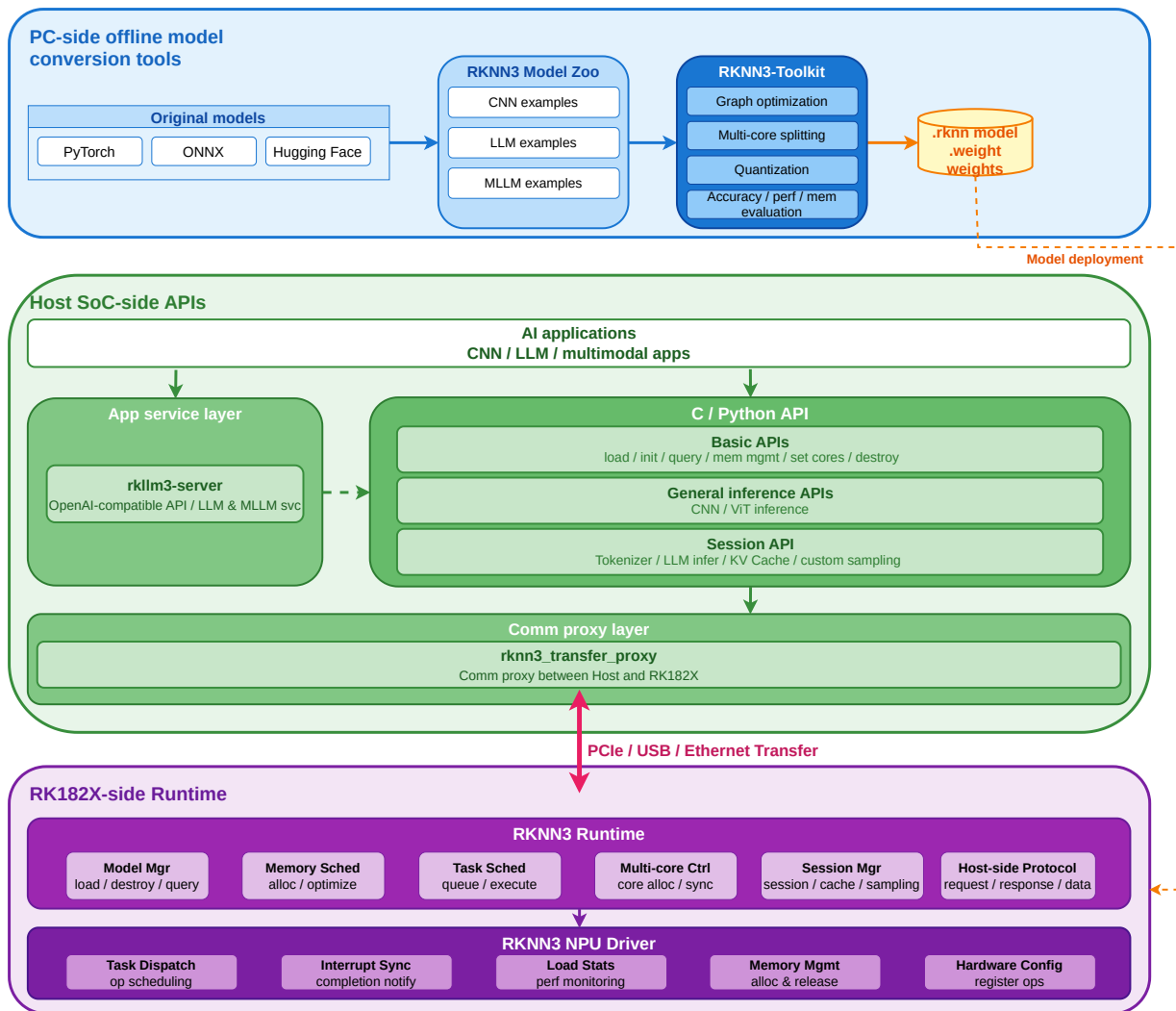


Figure 1-1 Coprocessor mode architecture

Non-coprocessor mode architecture

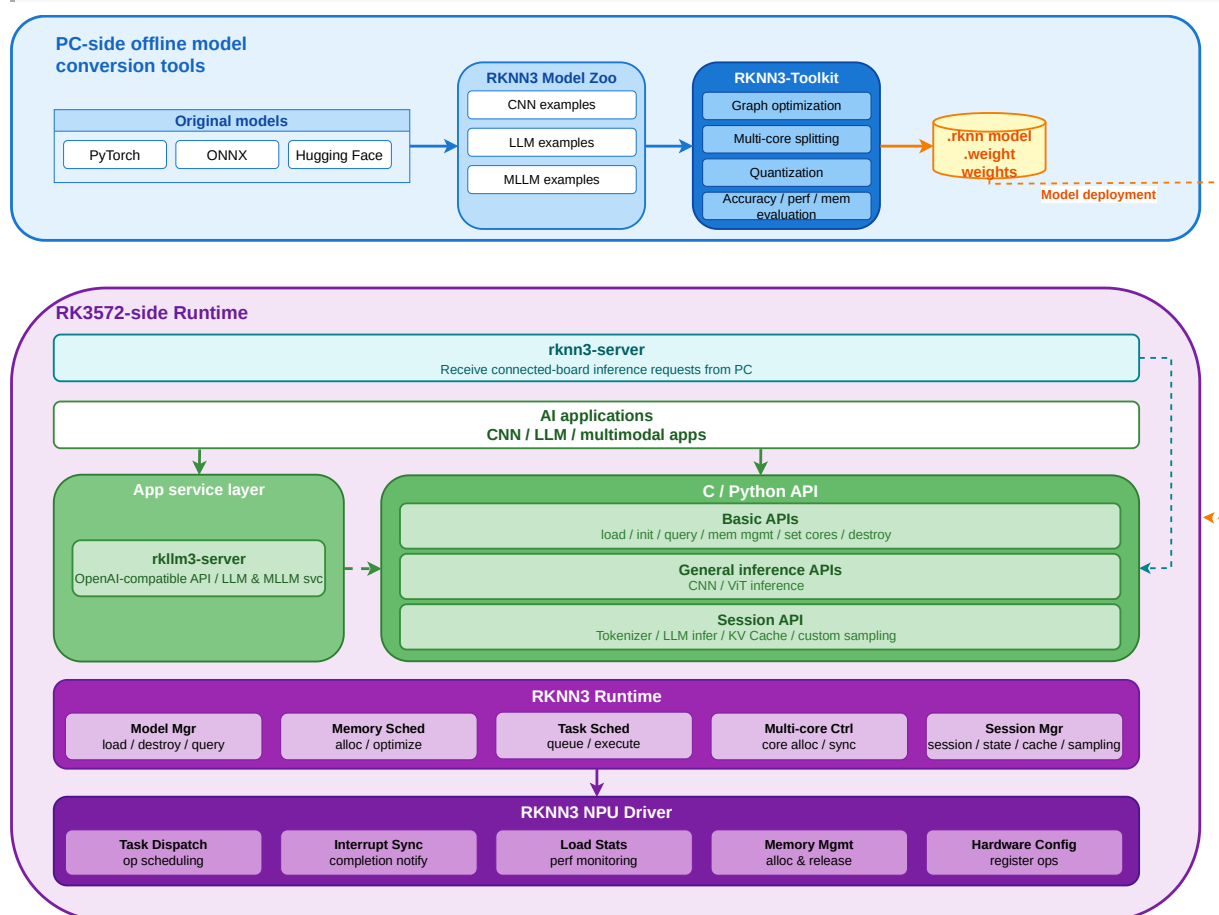


Figure 1-2 Non-coprocessor mode architecture

The responsibilities of each component are as follows:

1. PC side

- **RKNN3 Model Zoo:** Provides model conversion and deployment examples covering multiple model types, including CNNs (MobileNetV2 / ResNet50 / YOLOv5/v6/v8, etc.), LLMs (Qwen2.5 / Qwen3, etc.), and MLLMs (FastVLM / Qwen2.5-VL / Qwen3-VL, etc.).
- **RKNN3-Toolkit:** The core model conversion tool. It converts ONNX models into `.rknn` and `.weight` formats and performs graph optimization, multi-core partitioning, quantization, accuracy analysis, and visualization analysis. For LLM / MLLM models, first use the export scripts from the Model Zoo to convert the model to ONNX, and then use the Toolkit to convert it to RKNN.

2. Device-side API and application services

- **RKNN3 API:** The core interface layer, available in both C and Python, categorized by function into:
- **Basic API:** Fundamental operations such as model loading, initialization, query, memory management, setting the running cores, and destruction.
- **General inference API:** Inference interfaces for general vision models such as CNN / ViT.

- **Session API:** Inference interfaces for LLMs, supporting KVCache, custom sampling strategies, and other features.
- **rkllm3-server:** Provides an OpenAI-compatible API service with support for text, image, and audio input, simplifying the integration and invocation of large models.

3. Communication services

- **Coprocessor mode:** Uses **rknn3_transfer_proxy**, which provides a communication proxy between the host SoC and the RK1820 / RK1828 coprocessors and supports three connection types: PCIe / USB / Ethernet.
- **Non-coprocessor mode:** Uses **rknn3-server**, which runs on the RK3572 side, handles connected-board inference requests from the PC, and manages NPU resources.

4. Runtime and NPU driver

- **RKNN3 Runtime:** Responsible for model management, memory scheduling, inference task scheduling, multi-core parallel control, and Session management.
- **RKNN3 NPU driver:** Responsible for low-level operations such as task dispatch, interrupt synchronization, and load statistics.

In coprocessor mode, applications run on the host SoC and control model loading and inference execution on the RK1820 / RK1828 coprocessors through the RKNN3 API and **rknn3_transfer_proxy**. In non-coprocessor mode, applications run directly on the RK3572 and control the local NPU through the RKNN3 API. Users can perform model inference directly through the RKNN3 API, or invoke large model services through the OpenAI-compatible interface provided by **rkllm3-server**.

1.1.4 Relationship Among Toolkit / Runtime / Toolkit Lite / Server

The RKNN3 SDK provides four core components, each corresponding to a different stage of the model lifecycle. Users choose among them based on the use case:

Component	Running Location	Main Responsibilities	Typical Scenarios
RKNN3-Toolkit	PC	Model conversion, quantization, evaluation, connected-board debugging	Model conversion and evaluation stage
RKNN3 Runtime	On-device (RK182X / RK3572)	Model execution, memory management, multi-core scheduling	On-device inference (C API)
RKNN3-Toolkit Lite	On-device	Python inference interface	Quick on-device verification in Python
rkllm3-server	On-device	OpenAI-compatible API service	Service-oriented deployment of large models

The relationships and data flow among the components:

1. **Model conversion stage:** On the PC, the user uses RKNN3-Toolkit to convert an ONNX model into `.rknn` and `.weight` files. During this stage, quantization, performance

evaluation, and accuracy analysis can be performed.

2. **On-device deployment stage:** Deploy the conversion outputs to the device, and load and run the model through RKNN3 Runtime (C API) or RKNN3-Toolkit Lite (Python). For LLM / MLLM, rkllm3-server can also provide an OpenAI-compatible service.
3. **Connected-board debugging stage:** RKNN3-Toolkit supports connected-board inference, where inference is initiated on the PC side and the model runs on the device-side NPU. This is used for model verification and accuracy analysis.

Note: The versions of RKNN3-Toolkit and Runtime / Toolkit Lite must match. Otherwise, a model may convert successfully but fail to run on the device. For details, see 1.2.1 Version Compatibility Matrix.

1.1.5 Terms, Abbreviations, and Document Conventions

This section introduces the terms and abbreviations commonly used in this document.

Hardware and Deployment

- **Host SoC:** The main processor that runs user applications (such as RK3588 / RK3576), involved only in coprocessor mode.
- **Coprocessor:** A dedicated acceleration unit (such as RK1820 / RK1828) scheduled by the host SoC and sharing the computation workload via PCIe / USB / Ethernet interfaces, involved only in coprocessor mode.
- **RK182X:** The collective term for the RK1820 and RK1828 coprocessors. The two share the same architecture, and RK182X is used throughout this document to refer to both.
- **Connected-board inference:** The PC connects to the development board through the host SoC or via USB / network (adb connect). Inference is initiated by RKNN3-Toolkit on the PC side, and the model runs on the NPU. This is used for model verification and accuracy analysis.

Models and Tensors

- **RKNN model:** The model files that run on the NPU, consisting of two files: `.rknn` (model structure) and `.weight` (weights data).
- **LLM (large language model):** Large Language Model, a large-scale language model based on the Transformer architecture, such as Qwen2.5 / Qwen3, used for tasks such as text generation and dialogue.
- **MLLM (multimodal large language model):** Multimodal Large Language Model, a large model that extends the LLM to support multiple input modalities such as image, audio, and video, such as Qwen2.5-VL / Qwen3-VL.
- **Tensor:** A data structure representing a multi-dimensional array, consisting of data memory and attribute information (shape / dtype / layout, etc.).
- **Data type (dtype):** The data type of tensor elements, such as FP32 / FP16 / INT8 / UINT8.
- **Data layout (layout):** The arrangement of a tensor in memory, such as NCHW / NHWC / NC1HWC2.

- **NC1HWC2**: An NPU-optimized memory layout format that stores the channel dimension in blocks (16 sub-channels for FP16, 32 sub-channels for INT8) to improve NPU access efficiency.

Quantization

- **Quantized model**: A model that uses low-precision data types such as INT8 and requires scale and zero_point parameters for quantization/dequantization.
- **Non-quantized model**: A model that uses the FP16 data type and requires no quantization parameters.

API and Features

- **Basic API**: Inference interfaces for general vision models such as CNN / ViT.
- **Session API**: Dedicated interfaces for LLMs, providing session management, KVCache / LoRA, sampling control, and other features.
- **Core mask (core_mask)**: Specifies which NPU cores the model runs on. For example, 0x3 indicates that core 0 and core 1 are used.
- **Dynamic shape**: A single model supports multiple input size combinations, which can be switched at runtime via `rknn3_set_shape()`.
- **KVCache**: The cached key-value attention states in a large language model, used to accelerate autoregressive generation.
- **LoRA**: Low-Rank Adaptation, an efficient model fine-tuning method.

Document Conventions

- Lines starting with `#` in code blocks are comments.
- Content enclosed in `<>` is a parameter placeholder and must be replaced according to the actual situation.
- Interface function names, parameter names, and file names are enclosed in backticks (```).
- For detailed descriptions of the API parameters involved in this document, refer to the corresponding API reference manuals.

1.2 SDK Components and Environment Constraints

Note: This section describes the version compatibility, functional responsibilities, and environment constraints of the SDK components. For specific procedures such as Toolkit installation, hardware connection, and on-device environment deployment, refer to *RKNN3_SDK_Quick_Start*.

1.2.1 Version Compatibility Matrix

The versions of the RKNN3 SDK components (RKNN3-Toolkit, Runtime, communication services, and rkllm3-server) must match. Otherwise, problems may occur, such as a model converting successfully but failing to run on the device, or abnormal inference results.

Component Overview and Version Consistency

Component	Running Location	Function Description	Version Consistency Requirement
RKNN3-Toolkit	PC	Model conversion, quantization, evaluation	Same version as Runtime and the communication services
RKNN3 Runtime (librknn3_api.so)	On-device	Model execution, memory management, multi-core scheduling	Same version as Toolkit
rknn3_transfer_proxy	On-device	Communication proxy between the host SoC and the RK1820 / RK1828 coprocessors (coprocessor mode)	Same version as Toolkit / Runtime
rknn3-server	On-device	Handles connected-board inference requests from the PC and manages RK3572 NPU resources (non-coprocessor mode)	Same version as Toolkit / Runtime
rklm3-server	On-device	OpenAI-compatible API service	Same version as Toolkit / Runtime

Scenarios Requiring a Version Consistency Check

- When switching to a new development machine or Docker environment
- After updating or reinstalling RKNN3-Toolkit (the communication services and Runtime must be updated in sync)
- When pulling code or models from a project / SDK shared by others
- When runtime errors or model loading failures occur

Note: The RKNN model file records the Toolkit version used for conversion. If it does not match the on-device Runtime version, there is a risk of runtime errors. It is recommended to reconvert the model with a Toolkit of the same version as the Runtime. For version query methods, see 1.2.4 Environment Verification.

1.2.2 Host System Requirements

RKNN3-Toolkit runs on the PC for model conversion, quantization, and evaluation. The host environment must meet the following requirements:

Item	Requirement
Operating system	Ubuntu 20.04 / Ubuntu 22.04 / macOS (Beta)
Python	3.10 / 3.12 (only cp310 / cp312 supported as of V1.1.0)
Architecture	x86_64 (Ubuntu) / ARM64 (macOS Apple Silicon)
Core dependencies	onnx>=1.16 / onnxruntime>=1.22.0 / torch>=2.2 / numpy>=2.0

Note: The requirements file shipped with the Toolkit is the authoritative source for the complete dependency list; only the core dependencies that deserve special attention are listed above. As of V1.1.0, Python support has converged to 3.10 / 3.12, and the cp36/cp37/cp38/cp39/cp311 requirements files from older versions are no longer provided. If dependency version conflicts cause installation failures, try switching the pip index (for example, <https://pypi.tuna.tsinghua.edu.cn/simple/>) or run `pip install <package> --no-deps` to skip dependency checks, but make sure the runtime dependencies are complete on your own. It is recommended to run GRQ quantization (RKQuantizer) in a CUDA or MPS environment; running it on the CPU takes a very long time.

1.2.3 On-Device Environment Requirements

The on-device environment differs depending on the deployment mode:

Coprocessor mode (RK182X)

Item	Requirement
Host SoC	RK3588 / RK3576, etc.
Coprocessor	RK1820 / RK1828
Host system	Android / Linux / Windows
Communication service	rknn3_transfer_proxy
Runtime	librknn3_api.so + librknn3_api_rkcp.so
Communication interface	PCIe / USB / Ethernet
Number of NPU cores	8

Non-coprocessor mode (RK3572)

Item	Requirement
SoC	RK3572
System	Android / Linux
Communication service	rknn3-server
Runtime	librknn3_api.so + librknn3_api_native.so
Number of NPU cores	1 (single core)

Note: RK3572 supports only Linux on the aarch64 architecture and does not support x86_64. In RK182X coprocessor mode, the Runtime library depends on the host SoC architecture; select the library that matches the host.

1.2.4 Environment Verification

After the environment is deployed, verify that each component is ready using the following methods:

1. Verify the Toolkit installation

```
# Enter Python interactive mode and import the RKNN class
python
>>> from rknn.api import RKNN
# No errors indicate a successful installation
```

2. Verify the on-device communication service

- Coprocessor mode: Run `rknn3_transfer_proxy devices` in the host SoC terminal. The service works normally if the coprocessor devices are listed.
- Non-coprocessor mode: Run `rknn3-server -v` in the RK3572 terminal. The service works normally if the version number is printed.

3. Verify the device connection

```
# Query adb-connected devices on the PC
adb devices
# Device IDs listed indicate a normal connection
```

4. Verify version consistency

Confirm that the versions of Toolkit, Runtime, and the communication services match. Query the versions as follows:

```
# Query the rknn3-toolkit version
pip list | grep rknn3-toolkit

# Query the rknn3-toolkit version used for model conversion
strings xxxx.rknn | grep rknn3-toolkit

# Query the Runtime library version
strings /usr/lib/librknn3_api.so | grep -i "librknn3_api version"

# Coprocessor mode (RK182X): query the rknn3_transfer_proxy version
strings /usr/bin/rknn3_transfer_proxy | grep "Transfer version"

# Non-coprocessor mode (RK3572): query the rknn3-server version
/usr/bin/rknn3-server -v
```

Android platform paths: On the Android platform, the library and executable paths differ from those on Linux. The query methods are the same, but the paths must be replaced: - Runtime library: `/system/lib64/librknn3_api.so` or `/vendor/lib64/librknn3_api.so` - rknn3-server: `/system/bin/rknn3-server` or `/vendor/bin/rknn3-server` - rknn3_transfer_proxy: `/system/bin/rknn3_transfer_proxy` or `/vendor/bin/rknn3_transfer_proxy`

1.3 Development Workflow and Interface Selection

1.3.1 General Model Development Workflow

The typical development workflow for vision models such as CNN / ViT consists of three stages: model conversion, model evaluation, and on-device deployment and execution. The overall workflow is shown in the following figure:



Figure 1-3 General model development workflow

1. Model conversion

Complete the following on the PC side with RKNN3-Toolkit:

- Prepare the original model (in ONNX format).
- Configure parameters such as normalization, quantization, and target platform in RKNN3-Toolkit.
- Build the model via `rknn.build()`. Quantization can be enabled for higher performance and lower memory usage.
- Export the `.rknn` and `.weight` model files via `rknn.export_rknn()`.

1. Model evaluation

Connect the PC to the development board via USB or network (adb connect), and use the RKNN3-Toolkit Python API for connected-board evaluation:

- Initialize the runtime environment for connected-board use via `rknn.init_runtime()`, and perform connected-board inference with `rknn.inference()` to verify functional correctness and refine pre-processing and post-processing logic.
- Accuracy analysis: Use `rknn.inference(accuracy_analysis=True)` on the simulator or a connected board to analyze the quantization error and the sources of on-device inference error.
- Performance analysis: Use `rknn.eval_perf()` to evaluate operator latency and optimize running-core scheduling.

1. On-device deployment and execution

Integrate inference into on-device applications using the RKNN3 Runtime C API:

- Initialize the context via `rknn3_init()`, and load the `.rknn` and `.weight` files via `rknn3_load_model_from_path()`.
- Complete the model initialization configuration via `rknn3_model_init()` (including the multi-core mask `run_core_mask`), and query the input and output attributes via `rknn3_query()`.
- Allocate input and output memory via `rknn3_create_mem()`, synchronize the data via `rknn3_mem_sync()`, and invoke `rknn3_run()` to execute inference.
- Implement input pre-processing (data type conversion, layout

conversion) and output post-processing (dequantization, business logic), and integrate them into the actual business workflow.

Note: For specific operations on model conversion, quantization, and evaluation, see Chapter 2 (Sections 2.1-2.4). For on-device deployment, see Chapter 3 (Sections 3.1-3.5).

1.3.2 LLM Development Workflow

The LLM development workflow differs from the CNN workflow mainly in the special handling of HuggingFace models, and in that on-device deployment uses the Session API, which differs from that of CNN models.

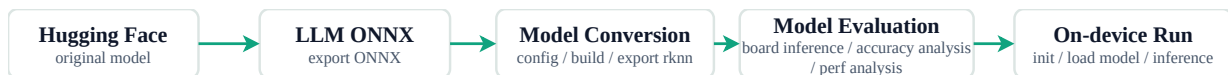


Figure 1-4 LLM model development workflow

The LLM model development workflow consists of three stages:

1. Model conversion

Complete the following on the PC side with RKNN3-Toolkit:

- Export the HuggingFace model to ONNX. GRQ quantization with RKQuantizer is optional.
- Load the LLM model via `rknn.load_llm()`.
- Build the model via `rknn.build()`. Quantization can be enabled for higher performance and lower memory usage.
- Export deployment outputs such as `.rknn` / `.weight` / `.embed.bin` / `.tokenizer.gguf` via `rknn.export_rknn()`.

1. Model evaluation

Connect the PC to the development board via USB or network (adb connect), and use the RKNN3-Toolkit Python API for connected-board evaluation:

- Initialize the runtime environment for connected-board use via `rknn.init_runtime()`, and perform connected-board inference with `rknn.inference()` to verify LLM model functionality and generation quality.
- Verify that features such as multi-turn dialogue and streaming generation work correctly.

1. On-device deployment and execution

Integrate inference on the device using the RKNN3 Runtime C API (Session interfaces) or rkllm3-server:

- Load the model via `rknn3_init()` / `rknn3_load_model_from_path()`, and complete initialization via `rknn3_model_init()`.
- Create a session via `rknn3_session_init()`, and configure the Chat Template, callbacks, and sampling parameters.
- Perform session inference via `rknn3_session_run()`, with support for KVCache / LoRA, multimodal input, and more.

Note: For detailed requirements of LLM model conversion (ONNX input conventions, KVCache, multiple shapes, etc.), see 2.2 LLM and MLLM Model Conversion. For Session deployment, see 3.3 LLM and MLLM Session Deployment.

1.3.3 MLLM Development Workflow

An MLLM (Multimodal Large Language Model) extends the LLM with non-text input capabilities such as vision and audio. The VLM (vision-language model) is the most common form.

An MLLM consists of a multimodal encoding part (such as Vision / Audio) and an LLM part. The overall development workflow consists of three stages: model conversion, model evaluation, and on-device deployment and execution. Each stage requires separate handling of the multimodal encoding part and the LLM part:

Multimodal encoding part development workflow

The multimodal encoding part is essentially a general model (such as a Vision model). Its development workflow follows 1.3.1 General Model Development Workflow: load it via `rknn.load_onnx()`, build it via `rknn.build()`, and export it via `rknn.export_rknn()`. On the device, run it through the general C API to obtain the multimodal embedding.

LLM part development workflow

The development workflow of the LLM part is identical to the 1.3.2 LLM Development Workflow. The difference is that during inference, the embedding output by the multimodal encoder must be fed as input together with the text prompt.

Note: For conversion of the MLLM multimodal encoding part, see 2.2.2 Multimodal Encoding Part Export and 2.1 General Model Conversion. For multimodal input deployment, see 3.3.6 Multimodal Input.

1.3.4 Interface Selection

The RKNN3 SDK provides three on-device invocation methods. Users choose among them based on the scenario:

Method	Language	Applicable Models	Applicable Scenarios
RKNN3 Runtime (C API)	C / C++	General models, LLM / MLLM	Production deployment, performance-sensitive scenarios
RKNN3-Toolkit Lite	Python	General models, LLM / MLLM	Quick on-device verification, prototyping
rkllm3-server	HTTP API	LLM / MLLM	Service-oriented deployment, OpenAI ecosystem integration

Differences among the three:

- **Performance:** The C API delivers the best performance, followed by rkllm3-server (which incurs network and serialization overhead). Toolkit Lite is slightly lower due to the overhead of the Python wrapper.
- **Development convenience:** rkllm3-server is the most convenient (no programming required; start the service and invoke it via HTTP), followed by Toolkit Lite (Python programming, no cross-compilation required). The C API requires cross-compilation and integration.
- **Feature completeness:** The C API supports the full set of advanced features (custom sampling, KVCache management, dynamic LoRA switching, etc.); rkllm3-server supports common configurations; Toolkit Lite targets verification and covers fewer advanced features.

Selection recommendations:

- **General models:** Use the C API for production deployment and Toolkit Lite for on-device verification.
- **LLM / MLLM:** Use rkllm3-server to integrate with the OpenAI ecosystem or to quickly set up serving; use the Session interfaces of the C API when deep control over the inference flow is required.

1.3.5 Model Conversion and Deployment Outputs

The deployment outputs generated after model conversion are as follows:

General models (CNN / ViT)

File	Description
<code>.rknn</code>	RKNN model structure file
<code>.weight</code>	RKNN model weights file

LLM / MLLM models

File	Description
<code>xxx.rknn</code>	RKNN model structure file
<code>xxx.weight</code>	RKNN model weights file
<code>xxx.embed.bin</code>	Embedding layer weights; not required in Tied word embeddings mode
<code>xxx.tokenizer.gguf</code>	Tokenizer file, used to initialize the tokenizer during on-device deployment
<code>xxx.safetensors</code>	External ROPE cache file (required by some models, such as Gemma4)
<code>xxx_per_layer_inputs. embed.bin</code>	PLE (per_layer_inputs embed) file, the extra input for each layer (required by some models, such as Gemma4)

Note: Transfer the above outputs to the device for deployment. C API deployment requires `.rknn` + `.weight` (for LLMs, `.embed.bin` + `.tokenizer.gguf` are also required). For Server deployment, specify the corresponding files using the parameters in 3.5 RKLLM3 Server Integration.

2. Model Conversion, Quantization, and Evaluation

2.1 General Model Conversion

This section describes the conversion flow for general models such as CNN / ViT, using RKNN3-Toolkit to convert ONNX models into RKNN models. The overall flow is: prepare the ONNX model → configure conversion parameters → load and build the model → export the RKNN model → verify the conversion result.



Figure 2-1 General model conversion flow

2.1.1 Model Preparation

General model conversion takes ONNX format as input. Users must first export the model to ONNX from the original training framework (PyTorch / TensorFlow, etc.):

- **ONNX export:** Complete the export on the original framework side. An opset version ≥ 11 is recommended. If the original model is in PyTorch, refer to the ONNX export scripts in the RKNN3 Model Zoo.
- **Input constraints:** When RKNN3-Toolkit loads an ONNX model, the model input name / shape / dtype must be specified. This information is used for subsequent `config` configuration and `load_onnx` loading.
- **Model inspection:** After export, use ONNX tools (such as `onnxruntime` or Netron) to check the model structure and operator support to ensure a correct export.

Note: RKNN3-Toolkit currently supports only ONNX model loading. Models from other frameworks (TensorFlow / PyTorch, etc.) must be exported to ONNX first.

2.1.2 Conversion Configuration

During model conversion, the parameters of `rknn.config()` and `rknn.build()` affect the conversion result. `rknn.config()` configures parameters such as normalization, quantization, and target platform, while `rknn.build()` controls whether quantization is applied and specifies the quantization dataset.

Basic flow for determining conversion parameters

Refer to the following steps to determine the conversion parameters:

1. Determine the NPU platform (RK1820 / RK1828 / RK3572), and set `target_platform`.
2. Confirm whether the input is RGB or BGR, and set `quant_img_RGB2BGR`.
3. Confirm the normalization parameters, and set `mean_values` and `std_values`.
4. Confirm the model input size, which is used for `load_onnx()` loading.

5. Confirm the quantization bit width, and set `quantized_dtype`.
6. Confirm the quantization algorithm, and set `quantized_algorithm`.
7. Confirm whether to quantize, and set `do_quantization` of `rknn.build()`. When quantizing, specify the calibration dataset through `dataset`.

Parameter details

- **target_platform**: Target NPU platform. Supports `rk1820`, `rk1828`, and `rk3572`.
- **quant_img_RGB2BGR**: Whether quantization images undergo RGB2BGR conversion. Default is `False`. It takes effect only during the quantization stage. If the model is trained with RGB, set it to `False` or leave it unset, and input RGB images at inference; if the model is trained with BGR, set it to `True`, and input BGR images at inference. If the calibration data uses the npy format, do not use this parameter to avoid confusion. **Caution**: This parameter affects only the channel conversion when calibration set images are read during quantization; it does not affect input processing at inference. At inference, the input data is only mean-subtracted and divided by the standard deviation, and channel conversion must be completed by the user.
- **mean_values / std_values**: Mean and normalization values of the input, both in the same format. For a single input with N channels, the format is `[[channel_1, ..., channel_n]]`; for multiple inputs, `[[channels of input 1], [channels of input 2], ...]`. The normalization formula is $\text{output} = (\text{input} - \text{mean}) / \text{std}$, and the Toolkit normalizes the input according to this formula at inference. After configuration, no mean normalization is needed during the C API inference stage, reducing deployment time.
- **quantized_dtype**: Quantization type. Supports `w4a16` / `w6a16` / `w8a16` / `w16a16` / `w8a8`. Default is `w16a16`.
- **quantized_algorithm**: Quantization algorithm. Default is `normal`. `normal` is fast; a calibration set of 20-100 images is recommended. `mmse` achieves higher accuracy but is slower and memory-intensive; a calibration set of 20-50 images is recommended. `kl_divergence` takes time in between and suits uneven feature distributions; a calibration set of 20-100 images is recommended. Start with `normal`; try other algorithms only if the results are unsatisfactory.
- **quantized_method**: Quantization method. Supports `layer` / `channel` / `group{SIZE}`. Default is `None` (must be configured when `quantized_dtype` is not `w16a16`). `group{SIZE}` is valid only under `w4a16` / `w6a16` / `w8a16` and supports only MatMul / Linear type operators.
- **optimization_level**: Model optimization level. Default is `3` (all optimizations enabled). Values 2/1 disable some optimizations; 0 disables all optimizations.
- **input_attrs**: Input attributes, such as `dtype` / `layout` / `shape` / `stride`.
- **dynamic_input**: Dynamic input configuration. Specifies multiple groups of input shapes to simulate dynamic input. Default is `None`.

For the format and how to fill in the quantization calibration dataset, see section 2.3 Model Quantization. For more detailed parameter descriptions, refer to the *RKNN3_Toolkit_Python_API_Reference*.

Example:

```
rknn.config(
    mean_values=[[103.94, 116.78, 123.68]],
    std_values=[[58.82, 58.82, 58.82]],
    quant_img_RGB2BGR=False,
    input_attrs={'pixel': {'dtype': 'uint8', 'layout': 'NHWC'}},
    target_platform='rk1820')
```

2.1.3 Model Loading and Building

1. Initialize the RKNN object

The first step of the conversion flow is to initialize the RKNN object. Pass in `verbose` to control log output:

```
from rknn.api import RKNN
rknn = RKNN(verbose=True, verbose_file='./mobilenet_build.log')
```

2. Load the ONNX model

Invoke `rknn.load_onnx()` to load the ONNX model. Provide the model file path:

```
rknn.load_onnx(model='./arcface.onnx')
```

3. Build the RKNN model

Invoke `rknn.build()` to build the model. Choose whether to quantize:

- `do_quantization`: Whether to quantize. Default is `True`.
- `dataset`: Quantization calibration dataset in text file format, with one calibration sample path per line.

dataset.txt example:

```
./imgs/ILSVRC2012_val_00000665.JPEG
./imgs/ILSVRC2012_val_00001123.JPEG
./imgs/ILSVRC2012_val_00001129.JPEG
```

Build example:

```
rknn.build(do_quantization=True, dataset='./dataset.txt')
```

Note: If `do_quantization=False`, quantization is skipped and the model is built in floating-point (w16a16) form. No dataset is required.

2.1.4 RKNN Model Export

After the build completes, invoke `rknn.export_rknn()` to export the RKNN model files for on-device deployment. The `export_path` parameter specifies the export file path. Based on this path, the Toolkit generates two files: `.rknn` (model structure) and `.weight` (weights data):

```
rknn.export_rknn(export_path='./mobilenet_v1.rknn')
```

The example above generates `mobilenet_v1.rknn` and `mobilenet_v1.weight` in the current directory. After the conversion flow completes, invoke `rknn.release()` to release resources:

```
rknn.release()
```

2.1.5 Conversion Result Verification

After export, invoke `rknn.inference()` to run inference verification on the RKNN model. Compare the outputs with the original ONNX model to confirm that the conversion result is correct.

Note: For more detailed evaluation methods such as accuracy analysis and performance evaluation, see section 2.4 Model Evaluation.

2.2 LLM and MLLM Model Conversion

This section describes the conversion flow for LLM and MLLM models.

LLM / MLLM models are usually stored in SafeTensors format (a safe tensor format developed by Hugging Face) and are loaded and managed through the HuggingFace ecosystem. Therefore, conversion takes HuggingFace-format models as input. Because RKNN3-Toolkit cannot convert SafeTensors / HuggingFace models directly, first export them to ONNX and accompanying files, and then convert them into RKNN models.

Differences from general models

- **Input format:** General models take ONNX as input; LLM / MLLM models take HuggingFace-format models (usually SafeTensors) as input and require an additional export step.
- **Export step:** An LLM must first export 4 types of files from the HuggingFace model (ONNX model, config, tokenizer, and embedding weights). The export depends on `transformers` (to load the model) and `torch` (`torch.onnx.export`); `export_llm.py` in `rknn3-model-zoo` wraps this export logic for direct use, and users can also refer to its implementation and export on their own.
- **Conversion tool:** Conversion still uses RKNN3-Toolkit, but it loads models with `rknn.load_llm()` and configures quantization, core, and attention parameters with `llm_config`, unlike the `rknn.load_onnx()` / `rknn.config()` used for general models.

MLLM specifics

An MLLM consists of a multimodal encoding part (such as Vision / Audio) and an LLM part. The two parts are converted separately and independently:

- **LLM part:** The export and conversion methods are the same as for LLM models (see 2.2.1~2.2.3).
- **Multimodal encoding part:** It is essentially a general model. Export it as an independent ONNX and convert it following the general model conversion method in 2.1 (see 2.2.2 Multimodal Encoding Part Export for details).
- **Joint usage:** The two parts are used together at on-device inference, where the LLM receives the embeddings output by the multimodal encoder.

The overall flow of LLM model conversion is shown below:



Figure 2-2 LLM model conversion flow

2.2.1 Hugging Face Model Preparation

LLM / MLLM model conversion takes HuggingFace-format models as input. Users must first prepare HuggingFace-format model files (including model weights, config.json, tokenizer.json, etc.), which can be loaded with `AutoModelForCausalLM` / `AutoConfig` from the `transformers` library:

```
from transformers import AutoModelForCausalLM, AutoConfig
config = AutoConfig.from_pretrained(args.model_path, **kwargs)
model = AutoModelForCausalLM.from_pretrained(args.model_path, **kwargs)
```

Note: An MLLM model consists of a multimodal encoding part (such as Vision / Audio) and an LLM part. Prepare and convert them separately. For multimodal encoding part export, see 2.2.2; for its conversion, see 2.1.

2.2.2 ONNX, Config, Tokenizer, and Embedding Export

Before LLM model conversion, export 4 types of files from the HuggingFace model: ONNX model, config, tokenizer, and embedding weights. `rknn3-model-zoo` wraps these processes in `export_llm.py`, with the corresponding interfaces `causal_llm_to_onnx`, `export_llm_config`, `export_tokenizer`, and `export_embed_weight`.

1. Export the ONNX model

Convert the LLM Torch model into an ONNX model. Note that inputs must match the original torch definition; some models may not follow the `input_ids` / `attention_mask` / `position_ids` order:

```

dummy_input = torch.zeros(1, in_len), dtype=torch.long)
attention_mask = torch.ones(1, in_len), dtype=torch.float)
position_ids = torch.arange(0, in_len, dtype=torch.long).unsqueeze(0)
inputs = (dummy_input, attention_mask, position_ids)
input_names = ["input_ids", "attention_mask", "position_ids"]

torch.onnx.export(
    model, inputs, args.export_llm_path,
    export_params=True, opset_version=19,
    do_constant_folding=True,
    input_names=input_names, output_names=output_names,
    dynamic_axes=dynamic_axes)

```

Note: Among the ONNX model inputs, `attention_mask` is required; the other inputs (`input_ids` / `position_ids` / `num_logits_to_keep` / `per_layer_inputs`, etc.) depend on the actual model. RKNN3-Toolkit classifies attention ops into N types based on the connection relationships of `attention_mask` and configures each type separately. The ONNX model must not have KVCache inputs/outputs (usually exported with `use_cache` set to False in the config).

For MLLM models, in addition to the LLM part ONNX export above, the multimodal encoding part (such as Vision / Audio) must also be exported separately as an independent ONNX model. `rknn3-model-zoo` wraps this process in scripts such as `export_vision.py`. The export method is similar to general model ONNX export and is completed with `torch.onnx.export`. The multimodal encoding part is converted the same way as a general model; see 2.1 General Model Conversion for details.

2. Export the config file

The config file is required by RKNN3-Toolkit for RKNN model conversion. It contains model specification parameters such as chat template, vocab size, and hidden size, as well as quantization-related configuration:

```

export_llm_config(args.model_path, os.path.splitext(args.export_llm_path)[0] +
    '.config.pkl', chat_context, prompt)

```

3. Export the tokenizer file

The tokenizer converts between text and token ids. On-device inference requires the `tokenizer.gguf` file and the `libtokenizer.a` library:

- `tokenizer.gguf`: Exported with `export_tokenizer`. It contains tokenization and vocabulary information and corresponds to the `tokenizer.json` of the LLM model.
- `libtokenizer.a`: Parses the vocabulary information in `tokenizer.gguf` and provides interfaces such as text-to-token-id and token-id-to-text conversion. `rknn3-model-zoo` uses the llama.cpp tokenizer; the source code is at `rknn3_model_zoo/tokenizer`, and the build artifact `libtokenizer.a` is in `rknn3_model_zoo/3rdparty/tokenizer`. To adjust it, modify the code under `tokenizer/src` and rebuild.

```
export_tokenizer(args.model_path, os.path.splitext(args.export_llm_path)[0] +
'.tokenizer.gguf')
```

4. Export the embedding file

The embedding file contains the embedding data of the LLM model (Float16 type), i.e., the vocabulary vectors, stored in the original layout of the embedding weights in the Torch model, with shape (vocab_size, embedding_dim):

```
export_embed_weight(model.model.embed_tokens.weight,
os.path.splitext(args.export_llm_path)[0] + '.embed.bin')
```

Note: Embedding layer operations run on the CPU side, which looks up the embedding data by token id and transfers it to the NPU. If the model's embedding layer operations are complex, embed.bin can be omitted and the user can manage the embedding in application logic. embed.bin is also not required in Tie Word mode.

2.2.3 LLM and MLLM Model Conversion

2.2.3.1 LLM MODEL CONVERSION

The LLM model conversion flow is: configure `llm_config` -> load with `rknn.load_llm()` -> build with `rknn.build()` -> export with `rknn.export_rknn()`.

1. Configure llm_config

For LLM models, `rknn.config()` adds the `llm_config` parameter (Python dictionary type). Users can first load the default configuration `DEFAULT_RKNN_LLM_CONFIG` and then modify it as needed:

```
from rknn.api import RKNN, DEFAULT_RKNN_LLM_CONFIG
rknn = RKNN(verbose=True)
my_config = DEFAULT_RKNN_LLM_CONFIG.copy()
rknn.config(target_platform='rk1820', llm_config=my_config)
```

`llm_config` mainly has 4 dictionary sub-items:

- **attention_config**: Configures the inference mechanism and KVCache of attention ops. The model must have a mask input with the attention structure. The Toolkit classifies attention ops into N types based on the mask connection relationships and configures each separately. Common fields:
 - `mask_name`: Mask input name (required, e.g., `attention_mask`).
 - `kvcache_buffer_len`: KVCache buffer length. Once exceeded, entries are overwritten cyclically in FIFO order. Must be 32-aligned.
 - `kvcache_dtype`: KVCache data type. Supports `Float16` / `Int8_to_F16` / `Int4_to_F16`, with memory usage decreasing in that order.
 - `attention_type`: Attention type. Supports `FullAttention` / `SlidingAttention`.

- `position_name` : Position encoding input name, used to extract RoPE. Set to None when the model has no position encoding.
- `max_position_embeddings` : Maximum position encoding length. Must be 32-aligned.
- `position_embeddings_host_storage` : Storage location of position encodings (RoPE): `False` stores them in device memory, `True` in Host memory.
- `mrope_type` : mrope type. Supports `Qwen2.5-VL / Qwen3-VL / Qwen3.5` . When enabled, `position_name` must be configured.
- `mrope_section` : mrope section information. For Qwen2.5-VL / Qwen3-VL / Qwen3.5 models, the corresponding configuration values can be found in config.json.
- `mrope_new_id_name` : Name of the newly generated mrope position input. After mrope is enabled, a new input with [seq_len, 3] dimensions is generated.
- **llm_head**: Configures the optimization behavior of the LLM head layer (usually a large Linear layer). Common fields:
 - `device` : Compute device. Supports `rk1820 / rk3588 / rk3576` .
 - `quantized_dtype` : Quantization type. Supports `w16a16 / w4a16 / w6a16 / w8a8` . Forced to `w8a8` on rk3576/rk3588.
 - `quantized_method` : Quantization method. Supports `layer / channel / group{SIZE}` . Forced to `channel` on rk3576/rk3588.
- **vocab**: Configures vocabulary layer optimization. When `tie_embeddings` is True, vocab shares weights with llm_head (Tie Word mode) and embed.bin is not required.
- **keep_one_logit**: Configures taking a single value along the specified output axis (a gather op is inserted) to reduce lm_head computation in the Prefill stage. `idx_name` specifies the gather indices input name.

Note: For the complete fields and default values of each `llm_config` sub-item, see `DEFAULT_RKNN_LLM_CONFIG` . For more detailed parameter descriptions, refer to the *RKNN3_Toolkit_Python_API_Reference*.

2. Load the LLM model

Invoke `rknn.load_llm()` to load the ONNX model. Parameters include `model` (ONNX path), `config` (config.pkl path), and `seq_lens` (list of input token counts supported per inference):

```
rknn.load_llm(model='./Qwen2.5-3B.onnx',
              config='./Qwen2.5-3B.config.pkl',
              seq_lens=[1, 128])
```

Note: `seq_lens` configures multiple shape inputs (e.g., `[1, 64]` generates a model with prefill length 64 and decoder length 1). It applies only to models with standard inputs (input_ids / attention_mask / position_ids / num_logits_to_keep); for models with non-standard inputs, use the `dynamic_input` parameter of `rknn.config()` .

3. Build and export

Invoke `rknn.build()` to build the model (quantization can be enabled), then invoke `rknn.export_rknn()` to export the deployment artifacts:

```
rknn.build(do_quantization=True, dataset=args.dataset_path)
rknn.export_rknn(args.rknn_path)
```

Exported artifacts include `.rknn`, `.weight`, `.embed.bin`, `.tokenizer.gguf`, etc. (see 1.3.5 Model Conversion and Deployment Artifacts). For loading LoRA weights, see 4.2.7 LoRA.

2.2.3.2 MLLM MODEL CONVERSION

An MLLM model consists of a multimodal encoding part (such as Vision / Audio) and an LLM part, and the two parts are converted separately and independently. The multimodal encoding part is essentially a general model: it is loaded with `rknn.load_onnx()`, built, and exported (see 2.1 for the conversion method). The LLM part is converted following the LLM model conversion flow above (configure `llm_config` -> `rknn.load_llm()` -> `rknn.build()` -> `rknn.export_rknn()`). The two parts are used together at on-device inference.

Taking the Qwen2.5-VL-3B model as an example, rknn3-model-zoo exports two files: `Qwen2.5-VL-3B-vision.onnx` (the vision encoder for image feature extraction) and `Qwen2.5-VL-3B-llm.onnx` (the LLM part for text generation). They are converted separately as follows:

```
# Build the vision model
rknn.config(target_platform='rk1820',
            quantized_dtype='w4a16', quantized_algorithm='normal',
            quantized_method='group32', core_num=8)
rknn.load_onnx(model='../model/vision/Qwen2.5-VL-3B-vision.onnx')
rknn.build(do_quantization=True, dataset=args.dataset_path)
rknn.export_rknn('../model/vision/Qwen2.5-VL-3B-vision.rknn')

# Build the LLM model
rknn.config(target_platform='rk1820',
            quantized_dtype='w4a16', quantized_algorithm='normal',
            quantized_method='group32')
rknn.load_llm(model='../model/llm/Qwen2.5-VL-3B-llm.onnx',
             config='../model/llm/Qwen2.5-VL-3B-llm.config.pkl')
rknn.build(do_quantization=True,
          dataset='../data/llm/dataset.txt')
rknn.export_rknn('../model/llm/Qwen2.5-VL-3B-llm.rknn')
```

Note: At on-device inference, the vision model uses the general RKNN3 API (`rknn3_run()`, etc.) and obtains image embeddings (`img_embeds`) through inference. At LLM inference, the prompt and `img_embeds` are passed into the LLM model together for multimodal inference. For multimodal input deployment, see 3.3.6 Multimodal Input.

2.2.4 Model and Tokenizer Adaptation

This section describes how to use the tokenizer to convert between text and token ids when deploying LLM / MLLM models on device. The tokenizer converts text into token ids for model inference, and converts the output token ids back to text.

Tokenizer files and library

On-device inference requires the following two files:

- `tokenizer.gguf`: Tokenizer file containing tokenization rules and vocabulary information. It corresponds to the `tokenizer.json` of the LLM model and is exported with `export_tokenizer` during model conversion (see 2.2.2).
- `libtokenizer.a`: Tokenizer library that parses `tokenizer.gguf` and provides interfaces such as text-to-token-id and token-id-to-text conversion. rknn3-model-zoo uses the

tokenizer from [llama.cpp](#), located at `rknn3_model_zoo/3rdparty/tokenizer`. To adjust it, modify the source code and rebuild.

Tokenizer class

rknn3-model-zoo wraps the `Tokenizer` class (header file `Tokenizer.h`). Its main interfaces are as follows:

- `Tokenizer(type, tokenizer_path)`: Constructor. Loads `tokenizer.gguf` and initializes the tokenizer. `type` takes `TOKENIZER_BACKEND_LLAMA`.
- `GetVocabInfo(&info)`: Gets vocabulary information (`vocab_size`, `special_bos_id`, `special_eos_id`, `linefeed_id`, etc.) for LLM session initialization.
- `Tokenize(text, text_len, tokens, n_tokens_max)`: Tokenizes text into a token id sequence and returns the number of tokens.
- `TokenToPiece(token)`: Converts a single token into a string piece for streaming output.
- `Decode(tokens, n_tokens)`: Restores a token sequence into a complete string for non-streaming output.

Using the tokenizer with the model

In on-device deployment, the tokenizer is registered to the LLM session via `tokenizer_callback` of `RKLLMCallback`, and the Runtime invokes it at inference to convert text into token ids. The usage flow is as follows:

1. Create a `Tokenizer` object and get the vocabulary information:

```
#include "Tokenizer.h"

// tokenizer_path is the path to the tokenizer.gguf file
Tokenizer* tokenizer = new Tokenizer(TOKENIZER_BACKEND_LLAMA, tokenizer_path);
VocabInfo vocab_info;
tokenizer->GetVocabInfo(&vocab_info);
```

1. Implement `tokenizer_callback`, and invoke `Tokenize` in the callback to convert text into token ids:

```
int tokenizer_callback(void *userdata, const char *text, int32_t text_len,
                      int32_t *tokens, int32_t n_tokens_max)
{
    Tokenizer *tokenizer = (Tokenizer *)userdata;
    int n_tokens = tokenizer->Tokenize(text, text_len, tokens, n_tokens_max);
    return n_tokens;
}
```

1. Register the tokenizer to the session callback. At inference, the Runtime automatically invokes `tokenizer_callback` for tokenization:

```

RKLLMCallback callback;
memset(&callback, 0, sizeof(RKLLMCallback));
callback.tokenizer_callback = tokenizer_callback;
callback.tokenizer_userdata = tokenizer;
rknn3_session_set_callback(session, &callback);

```

1. In the result callback, use `TokenToPiece` or `Decode` to convert output tokens back to text:

```

// Streaming output: convert to string token by token
std::string piece = tokenizer->TokenToPiece(result->token_ids[0]);
// Non-streaming output: batch convert to string
std::string text = tokenizer->Decode(result->token_ids, result->num_tokens);

```

Note: `tokenizer_callback` is an optional callback, used only when a custom tokenizer is needed. If not registered, the Runtime uses the default tokenization logic. For detailed descriptions of tokenizer-related interfaces, see `LLMTokenizerCallback` in the *RKNN3_Runtime_C_API_Reference*.

2.2.5 Conversion Result Verification

After LLM / MLLM model conversion, verify the results with `rknn3-toolkit-lite` (the on-device Python inference library). LLM models generate autoregressively. Set `llm_mode=True` at initialization, load the model, run inference verification, and check whether the model generates text (or image + text results) correctly:

```

from rknn3lite.api import RKNN3Lite

rknn_lite = RKNN3Lite(llm_mode=True)
rknn_lite.load_rknn(model_path='./Qwen2.5-3B-Instruct.rknn',
                    weight_path='./Qwen2.5-3B-Instruct.weight')
# Run inference verification
...
rknn_lite.release()

```

Note: For detailed usage of `rknn3-toolkit-lite`, see 3.4 Toolkit Lite Python Deployment. For more detailed evaluation methods such as accuracy analysis and performance evaluation, see section 2.4 Model Evaluation.

2.3 Model Quantization

2.3.1 Quantization Basics

2.3.1.1 QUANTIZATION DEFINITION

Model quantization converts the floating-point parameters or activation values of a deep learning model into fixed-point representations, such as converting FP32 to INT8. Quantization reduces

memory usage and enables model compression and inference acceleration, but usually introduces some accuracy loss.

Based on the form of the mapping function between floating-point parameters and fixed-point numbers, quantization falls into linear quantization and non-linear quantization. Common non-linear quantization includes exponential quantization, codebook quantization, and piecewise linear quantization. RKNN3-Toolkit supports only linear quantization. In linear quantization, the intervals between fixed-point numbers are uniform. For example, INT8 linear quantization divides the quantization range evenly into 256 values, so it is also called uniform quantization. Depending on whether the quantization zero point is 0, linear quantization falls into symmetric quantization and asymmetric quantization. Symmetric quantization is a simplified form of asymmetric quantization. Asymmetric quantization usually handles uneven data distributions better, so asymmetric quantization schemes are more commonly used in practice.

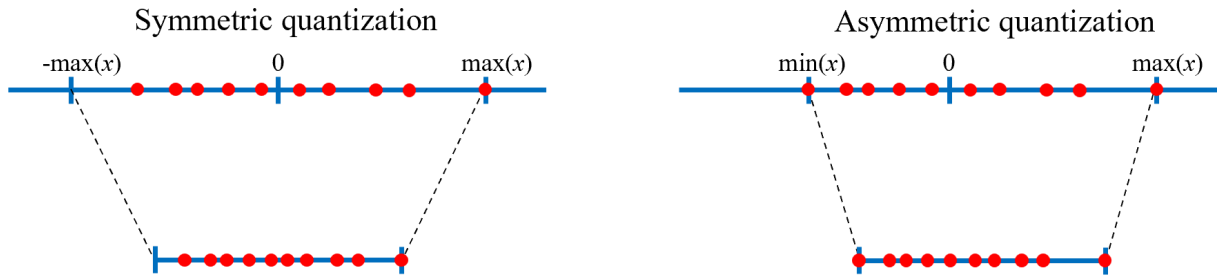


Figure 2-3 Linear symmetric quantization and linear asymmetric quantization

2.3.1.2 QUANTIZATION COMPUTATION

Taking linear asymmetric quantization as an example, the computation for quantizing a floating-point number into a signed fixed-point number is as follows:

$$x_{int} = \text{clamp}(\lfloor \frac{x}{s} \rceil + z; -2^{b-1}, 2^{b-1} - 1) \quad \text{tag}\{5-1\}$$

where x is the floating-point number, x_{int} is the quantized fixed-point number, $\lfloor \cdot \rceil$ is the round-to-nearest operation, s is the quantization scale factor, z is the quantization zero point, and b is the quantization bit width (for the INT8 data type, b is 8); $\text{clamp}(x; a, c)$ is the clamping operation, defined as follows:

$$\text{clamp}(x; a, c) = \begin{cases} a, & x < a \\ x, & a \leq x \leq c \\ c, & x > c \end{cases} \quad \text{tag}\{5-2\}$$

Converting a fixed-point number back into a floating-point number is the dequantization process, defined as follows:

$$x \approx \widehat{x} = s(x_{int} - z) \quad \text{tag}\{5-3\}$$

where $\widehat{x}$ is also called the fake-quantized floating-point number. Clearly, due to the rounding and clamping operations in quantization computation, $\widehat{x}$ does not equal x , and the difference between them is called the quantization error.

Given the quantization range $(q_{\min}, q_{\max})$ and the clamping range $(c_{\min}, c_{\max})$, the quantization parameters s and z are computed as follows:

$$s = \frac{q_{\max} - q_{\min}}{c_{\max} - c_{\min}} = \frac{q_{\max} - q_{\min}}{2^b - 1} \quad \text{tag}\{5-4\}$$

$$z = c_{\max} - \lfloor \frac{q_{\max}}{s} \rceil \quad \text{or} \quad z = c_{\min} - \lfloor \frac{q_{\min}}{s} \rceil \tag{5-5}$$

The clamping range is determined by the quantized data type; for example, the clamping range of INT8 is (-128, 127). The quantization range is determined by the specific quantization algorithm. Setting $z=0$ in the definition formulas above yields the computation definition of linear symmetric quantization.

2.3.1.3 QUANTIZATION GRANULARITY

Quantization granularity specifies which parameters in a model's floating-point tensors are quantized as a whole. RKNN3-Toolkit supports three quantization granularities: Per-Layer / Per-Channel / Group. Per-Layer quantization quantizes all channels of a network layer's weights or activations as a whole, with all channels sharing the same quantization parameters. Per-Channel quantization quantizes each output channel of a network layer's weights independently, with each channel having its own quantization parameters. Group quantization applies only to linear layer weights. Based on Per-Channel quantization, it groups the elements within each output channel, and each group is quantized independently.

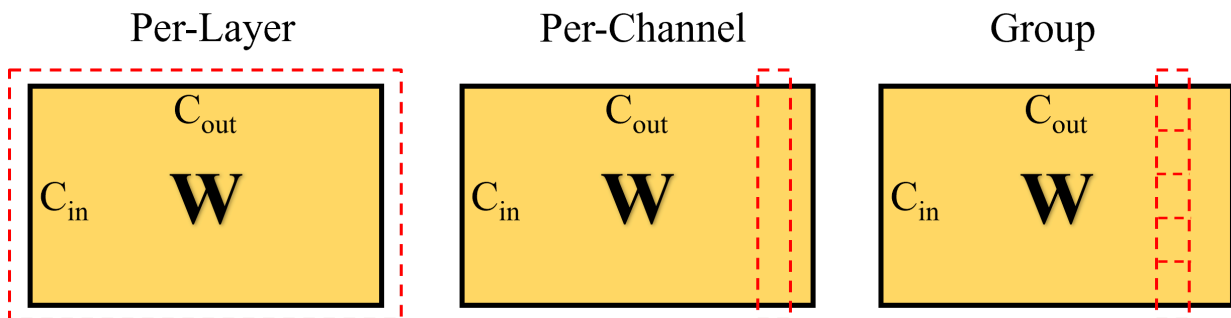


Figure 2-4 Per-Layer quantization, Per-Channel quantization, and Group quantization

Note: In RKNN3-Toolkit, Group quantization and Per-Channel quantization apply only to weights; activations and intermediate values still use Per-Layer quantization; Group quantization is valid only for linear layers under a16 conditions.

2.3.1.4 QUANTIZATION ALGORITHMS

A quantization algorithm is a scheme that, under a given quantization form (such as linear quantization), solves for the low-bit fixed-point representation of a floating-point tensor and its quantization parameters. By implementation, quantization falls into two categories: quantization-aware training (QAT) and post-training quantization. QAT fine-tunes model parameters and quantization parameters during training to improve accuracy. It usually requires in-depth understanding of the training pipeline and large amounts of training data, so it is rarely used directly in model deployment toolchains. Post-training quantization usually needs only a small amount of calibration data, or even none, and obtains quantization parameters through low-cost computation. RKNN3-Toolkit supports four post-training quantization algorithms: Normal / KL-Divergence / MMSE, and GRQ.

2.3.2 Toolkit Quantization Flow Overview

RKNN3-Toolkit provides two flows: internal quantization and RKQuantizer quantization. The internal quantization flow first exports the floating-point model to a floating-point ONNX model, then exports the RKNN model through `rknn.config()` → `rknn.load_llm()/rknn.load_onnx()` → `rknn.build()` → `rknn.export_rknn()`. This flow offers strong compatibility, supports models such as CNN / ViT / LLM / MLLM, and can use the three quantization algorithms Normal / KL-Divergence / MMSE, but it does not support the GRQ quantization algorithm.

The RKQuantizer quantization flow mainly targets standard-architecture models such as ViT / LLM / MLLM. This flow uses RKQuantizer, the quantization component in the Toolkit, to run quantization algorithms such as GRQ, replacing the floating-point weights in the model with fake-quantized weights and exporting a fake-quantized ONNX model; the RKNN model is then exported through the same `rknn.config()` → `rknn.load_llm()/rknn.load_onnx()` → `rknn.build()` → `rknn.export_rknn()` chain. Importantly, when converting ONNX to RKNN, `rknn.config()` must set `quantized_algorithm='normal'` — MMSE or KL-Divergence must not be used — and must set the correct `quantized_dtype` (`w4a16` , `w6a16` , or `w8a16`) and `quantized_method='group32'` . This chapter gives a brief introduction to RKQuantizer usage. For more detailed interface and usage instructions (including mixed-precision quantization, calibration data preparation, and GPU memory usage), refer to the *RKNN3_Toolkit_Python_API_Reference*.

Internal quantization flow:

```
Floating-point PyTorch model → export floating-point ONNX → rknn.config() →  
rknn.load_llm()/rknn.load_onnx() → rknn.build() → rknn.export_rknn()
```

RKQuantizer quantization flow:

```
Floating-point PyTorch model → RKQuantizer → fake-quant PyTorch model →  
export fake-quant ONNX → rknn.config() → rknn.load_llm()/rknn.load_onnx() →  
rknn.build() → rknn.export_rknn()
```

2.3.3 Internal Quantization Flow

2.3.3.1 DATA TYPES

Internal quantization supports the quantization data types `w4a16` , `w6a16` , `w8a16` , `w16a16` , and `w8a8` , with the default `w16a16` . Here, w4/w6/w8/w16 mean the weights are quantized to the corresponding bit width (int4/int6/int8/int16 or float16), and a8/a16 mean the activations are quantized to int8/float16.

2.3.3.2 INTERNAL QUANTIZATION ALGORITHMS

Internal quantization supports three quantization algorithms: Normal / KL-Divergence / MMSE:

- Normal: Determines the quantization range directly from the maximum and minimum floating-point values. This algorithm produces no clamping error but is sensitive to outliers. It runs fast and suits general scenarios.

- **KL-Divergence:** Determines the quantization range by comparing the distributions of floating-point and fixed-point values, adjusting thresholds, and minimizing the KL divergence. This algorithm adapts better to non-uniform data distributions, mitigates the impact of a few outliers, and runs slightly slower than Normal.
- **MMSE:** Determines the quantization range by minimizing the mean squared error between floating-point values and dequantized floating-point values. This algorithm usually achieves higher accuracy but requires more computing resources, runs slower, and has higher memory overhead. Compared with KL-Divergence, it often delivers better accuracy in outlier scenarios.

Users are advised to use the Normal algorithm first. When quantization accuracy issues arise, try KL-Divergence or MMSE.

2.3.3.3 CALIBRATION DATA PREPARATION

In the internal quantization flow, the calibration dataset is used to compute the quantization ranges of activations. Choose data that covers the different data distributions of the model's actual application scenarios; for example, the calibration data of a classification model should include images of different categories from real scenes. A calibration dataset of 20~200 images is generally recommended; adjust this as appropriate based on the running time of the quantization algorithm. Note that under a16 conditions, Normal quantization requires no calibration dataset. For the other quantization algorithms, increasing the number of calibration samples increases running time but does not necessarily improve accuracy.

Internal quantization calibration data currently supports the text file format. Users can write the paths of the calibration images (jpg or png format) or npy files into a `.txt` file. Each line in the text file is one path entry, for example:

```
a.jpg  
b.jpg
```

For multiple inputs, separate the files of each input with spaces, for example:

```
a0.jpg a1.jpg  
b0.jpg b1.jpg
```

2.3.3.4 SPECIFIC EXAMPLES

Example 1: ResNet50 internal quantization flow example:

```

from rknn.api import RKNN

model_path = "resnet50-v2-7.onnx" ## Floating-point ONNX model exported in advance
output_path = 'resnet50-v2-7.rknn'
DATASET_PATH =
'../../../../datasets/imagenet/ILSVRC2012_img_val_samples/dataset_20.txt'

# Create RKNN object
rknn = RKNN(verbose=False)

# Pre-process config
print('--> Config model')
rknn.config(mean_values=[[255*0.485, 255*0.456, 255*0.406]],
            std_values=[[255*0.229, 255*0.224, 255*0.225]],
            target_platform="rk1828",
            input_attrs={'input': {'dtype': 'uint8', 'layout': 'NHWC'}},
            quantized_dtype='w8a8',          ## optional: "w8a16", "w4a16", etc.
            quantized_algorithm='normal',     ## optional: "MMSE", "KL-Divergence"
            quantized_method='channel',)
print('done')

# Load model
print('--> Loading model')
ret = rknn.load_onnx(model=model_path, inputs=['data'], input_size_list=[[1, 3,
224, 224]])

if ret != 0:
    print('Load model failed!')
    exit(ret)
print('done')

# Build model
print('--> Building model')
ret = rknn.build(do_quantization=True, dataset=DATASET_PATH)
if ret != 0:
    print('Build model failed!')
    exit(ret)
print('done')

# Export rknn model
print('--> Export rknn model')
ret = rknn.export_rknn(output_path)
if ret != 0:
    print('Export rknn model failed!')
    exit(ret)
print('done')

```

2.3.4 RKQuantizer Quantization Flow

2.3.4.1 DATA TYPES

RKQuantizer supports the quantization data types `w4a16` , `w6a16` , and `w8a16` . The weight quantization method (`quantized_method`) supports only group32.

2.3.4.2 RKQUANTIZER QUANTIZATION ALGORITHMS

RKQuantizer quantization supports two algorithms: GRQ and Normal. GRQ is based on backpropagation. It minimizes the mean squared error between the outputs of each block in the network (such as a DecodeLayer in an LLM) before and after quantization, and solves the quantization parameters of each linear layer within the block. Unlike algorithms such as KL-Divergence and MMSE that minimize tensor quantization errors, GRQ aims to make the outputs of the quantized model closer to those of the floating-point model, so it usually achieves higher quantization accuracy and is the preferred algorithm. Because GRQ uses backpropagation, run it in a GPU / CUDA environment.

2.3.4.3 CALIBRATION DATA PREPARATION

The Normal quantization algorithm requires no calibration dataset. The GRQ algorithm requires a calibration dataset to collect the activation distributions of each layer and fine-tune the weights. The calibration dataset is provided as a **JSON file** whose content is a list, with each element being one sample. The quantization component RKQuantizer reads the samples one by one and constructs the model inputs. Two sample formats are currently supported; users can choose based on the model type.

Method 1: Hook-captured inputs (general, recommended)

Capture the real inputs of the submodule to be quantized during forward passes of the floating-point model with PyTorch `register_forward_pre_hook`, save them with pickle serialization, and build the JSON index file. This applies to all models, including MLLM, speech, and other models with non-pure-text inputs.

Each sample in the JSON file is a dictionary containing only one `sample` field, whose value is the relative path of a pickle file (relative to the directory of the JSON file). The pickle file contains a dictionary with two keys, `args` (positional arguments) and `kwargs` (keyword arguments), corresponding to the positional and keyword inputs for the quantized model's forward pass. The format is as follows:

```
[
  {"sample": "model_inputs/sample_0"},
  {"sample": "model_inputs/sample_1"}
]
```

The standard way to obtain this data is: on the floating-point model to be quantized, register `register_forward_pre_hook` on the module to be quantized (enable `with_kwargs=True` to also capture keyword arguments); in the hook, store the inputs into a list and raise an exception to terminate the forward pass early (avoiding a full inference), then dump the captured inputs to disk with pickle. `rknn3-model-zoo` wraps this mechanism as the public utility `py_utils/cali_data.py`, providing the `capture_module_input(hook_module, run, *args, **kwargs)` function: it packs "register forward pre hook → invoke the model → capture (`args`, `kwargs`) → raise `StopForward` to short-circuit the forward → remove the hook" into one line, so callers need no awareness of hook details. The internal hook raises `StopForward` to terminate the forward of the captured submodule. Therefore, even when `model.generate` is passed in, only one forward pass runs before the interruption, and no tokens are actually generated. For example:

```

import os
import json
import pickle
import torch
from transformers import AutoProcessor
# Import the common hook capture utility (adjust the path to your project
structure)
from py_utils.cali_data import capture_module_input

model_path = 'Qwen/Qwen2.5-VL-3B-Instruct'
model = ... # Load the floating-point model; keep torch_dtype
consistent with quantization
processor = AutoProcessor.from_pretrained(model_path)

# Forward per sample; capture the inputs of the submodule to be quantized in one
call
info = []
datasets = json.load(open('./dataset.json', 'r')) # Raw annotated data (images,
questions, etc.)
for idx, data in enumerate(datasets):
    inputs = processor(...) # Construct the complete input as
required by the model
    inputs = inputs.to(model.device)
    with torch.inference_mode():
        # Pass the quantized model itself to hook_module (i.e., the submodule
passed to load_model; pass model.visual to quantize the vision part),
        # The captured inputs are those at the outermost entry of this submodule,
not the inputs of its first internal transformer block.
        # Pass the model call to trigger via run (wrapped in a lambda; may include
preceding steps such as prepare_inputs_embeds)
        temp = capture_module_input(
            model.visual,
            lambda: model(**inputs)
        )
        # temp is {"args": (...), "kwargs": {...}}; it is None if this input does
not flow through the submodule

    # Serialize the captured inputs with pickle and record them in the index JSON
    sample_name = 'sample_{}'.format(idx)
    os.makedirs('./model_inputs', exist_ok=True)
    with open(os.path.join('./model_inputs', sample_name), 'wb') as f:
        pickle.dump(temp, f)
    info.append({"sample": "model_inputs/{}".format(sample_name)})

with open('./cali_data.json', 'w', encoding='utf-8') as f:
    json.dump(info, f, indent=2, ensure_ascii=False)

```

Method 2: Pure-text JSON (for pure-text LLMs only)

For LLM models whose inputs are text only, a simpler method is available: each sample in the JSON file is a dictionary containing two keys, `input` and `target`, and `RKQuantizer` automatically encodes it into model inputs with `tokenizer(input + target)`. This method requires neither loading the floating-point model nor hooks, and has the lowest preparation cost. The format is as follows:

```
[
  {
    "input": "Cysteine and tyrosine can be derived from what in the human body, respectively.\nA. Cystine and methionine\nB. Methionine and phenylalanine",
    "target": "\nAnswer:\n B"
  },
  {
    "input": "Natalia sold clips to 48 of her friends in April, and then she sold half as many in May. How many altogether?",
    "target": ""
  }
]
```

Here, `input` is the prompt text and `target` is the expected output text (it can be an empty string). The two are concatenated and encoded as a whole by the tokenizer. `target` can include supplementary context such as answers or chains of thought, making the calibration distribution closer to real generation scenarios. For example:

```
import json

# Write samples directly or organize them from an existing dataset, then save as JSON
samples = [
    {"input": "Briefly describe Newton's first law.", "target": "\nAnswer:\n Newton's first law states..."},
    {"input": "What is 1+1?", "target": "\nAnswer:\n 2"},
]
with open('./cali_data.json', 'w', encoding='utf-8') as f:
    json.dump(samples, f, ensure_ascii=False, indent=2)
```

Note: This method applies only to LLMs with pure-text inputs. For multimodal (MLLM), speech, and other models with non-text inputs, Method 1 must be used.

The quality of the calibration data greatly affects the accuracy of the GRQ-quantized model. Follow the core principle when selecting data: **use diverse data from real application scenarios**, so that the calibration data distribution stays as close as possible to the actual input distribution after the model is deployed. Specific recommendations:

- **Cover real scenarios:** Calibration samples should come from real inputs in the model's actual deployment scenarios, such as actual system prompts and function-tools. For agent tasks, such inputs are especially important and can be loaded through the `load_model` interface.
- **Maintain diversity:** Samples should be diverse and avoid highly homogeneous content, so that the quantization parameters adapt to different input distributions. Preparing 20~200 samples is generally recommended.

2.3.4.4 MIXED-PRECISION QUANTIZATION SETTINGS

RKQuantizer provides two mixed-precision quantization methods: manual per-layer specification (`layer_quant_config`) and automatic proportional selection (`auto_hybrid_rate`).

Method 1: Manual per-layer specification (layer_quant_config)

The user explicitly specifies which layers use w6a16 or w8a16. `layer_quant_config` accepts a dictionary or a JSON file path in the format `{layer_pattern: "w8a16"}`, where `layer_pattern` is a shell-style wildcard pattern for matching Linear layer names and supports the `*` wildcard (for matching same-named sublayers across all transformer blocks). Matched layers use the quantization type specified in the configuration; unmatched layers are quantized with `quantized_dtype`. Currently `layer_quant_config` supports only "w8a16" or "w6a16", and it must not be used together with `auto_hybrid_rate`.

Taking the Qwen3.5 model as an example, its `layer_quant_config.json` is as follows, specifying the linear-attention-related qkv/b/a/out_proj layers and the v_proj and down_proj layers as w8a16 (or w6a16):

```
{
  "model.layers.*.linear_attn.in_proj_qkv": "w8a16",
  "model.layers.*.linear_attn.in_proj_b": "w8a16",
  "model.layers.*.linear_attn.in_proj_a": "w8a16",
  "model.layers.*.linear_attn.out_proj": "w8a16",
  "model.layers.*.self_attn.v_proj": "w8a16",
  "model.layers.*.mlp.down_proj": "w8a16"
}
```

The corresponding quantization invocation is as follows:

```
from rknn.quantization.api import RKQuantizer

QuantTool = RKQuantizer(verbose=True)

QuantTool.load_model(...)

quant_model = QuantTool.quantize(quantized_dtype="w4a16",
                                quantized_method="group32",
                                quantized_algorithm='grq',
                                dataset='./dataset.json',
                                awq_config=None,
                                layer_quant_config="layer_quant_config.json",
                                auto_hybrid_rate=0.)
```

Method 2: Automatic proportional selection (auto_hybrid_rate)

No per-layer specification is needed. `RKQuantizer` automatically evaluates each layer's sensitivity to quantization from the calibration data and keeps the most sensitive Linear layers at w8a16 by the specified ratio. `auto_hybrid_rate` is a float in [0, 1) indicating the proportion of Linear layers set to w8a16. For example, 0.2 means about 20% of Linear layers are set to w8a16, and the rest are quantized with `quantized_dtype`. This parameter must not be used together with `layer_quant_config`. For example:

```

from rknn.quantization.api import RKQuantizer

QuantTool = RKQuantizer(verbose=True)

QuantTool.load_model(...)

## auto_hybrid_rate=0.2 means 20% of Linear layers are set to w8a16
model = QuantTool.quantize(quantized_dtype="w4a16",
                           quantized_method="group32",
                           quantized_algorithm="grq",
                           dataset='./dataset.json',
                           auto_hybrid_rate=0.2,
                           awq_config=None)

```

Exporting per-layer quantization types (export_op_quantized_dtype)

With either mixed-precision method above, invoke `export_op_quantized_dtype` after quantization to export the quantization type actually used by each layer into a JSON file, which is used for alignment in the subsequent `rknn.build()` stage. This interface takes the path of the exported ONNX model to build the mapping between torch layer names and ONNX operators. The exported JSON is in `{onnx_op_name: dtype}` format, where the key is the name of an operator such as MatMul/Gemm in the ONNX model and the value is that operator's quantization type. Taking Qwen3.5 as an example, the exported `layer_bit.json` is as follows:

```

{
  "onnx::MatMul_8968": "w8a16",
  "onnx::MatMul_8973": "w8a16",
  "onnx::MatMul_9657": "w8a16",
  ...
}

```

For example:

```

# Invoke after exporting the ONNX model, passing the ONNX model path
QuantTool.export_op_quantized_dtype(onnx_path='./Qwen3.5-0.8B-llm.onnx',
                                    op_dtype_path='layer_bit.json')

```

Corresponding changes when exporting the RKNN model

Because mixed-precision quantization is already done in the RKQuantizer stage (the ONNX weights are fake-quantized per layer type), when exporting the RKNN model via `rknn.config()` + `rknn.load_llm()` + `rknn.build()`, pass `layer_bit.json` through the `op_quantized_dtype` parameter of `rknn.config()`. This makes the RKNN conversion stage restore each layer at its specified quantization type, avoiding accuracy regression caused by re-quantizing sensitive layers to low bit widths during conversion. `op_quantized_dtype` accepts a dictionary or a JSON file path. It currently takes effect only for MatMul / Gemm / Linear operators, with supported values w6a16 or w8a16.

```
rknn.config(
    target_platform='rk1820',
    quantized_dtype='w4a16',
    quantized_algorithm='normal',
    quantized_method='group32',
    llm_config=llm_config,
    op_quantized_dtype='./layer_bit.json', # Pass the per-layer quantization
types for mixed precision
)
```

Note:

1. `layer_quant_config` and `auto_hybrid_rate` must not be used together.
2. `export_op_quantized_dtype` must be invoked after the ONNX export is complete, and the passed `onnx_path` must be the same ONNX that is later loaded by `rknn.load_llm()`. Otherwise, the mapping between torch layer names and ONNX operators fails.
3. When exporting RKNN, the JSON specified by `op_quantized_dtype` must correspond to the file exported by `export_op_quantized_dtype` in the quantization stage. Missing either one causes the mixed-precision configuration to not take effect or a conversion error.
4. If mixed-precision quantization is not enabled, there is no need to invoke `export_op_quantized_dtype` or configure `op_quantized_dtype` in `rknn.config()`.

2.3.4.5 SPECIFIC EXAMPLES

Example 1: qwen3 RKQuantizer quantization flow example. Fake-quantized ONNX model export:

```

from transformers import AutoModelForCausalLM, AutoTokenizer, AutoConfig
from rknn.quantization.api import RKQuantizer
import torch

model_path = "Qwen3-1.7B"

## Load the model
kwargs = {
    'trust_remote_code': True,
    'torch_dtype': torch.bfloat16
}
config = AutoConfig.from_pretrained(model_path, **kwargs)
update_config(config, ['use_cache'], False)
kwargs['config'] = config
model = AutoModelForCausalLM.from_pretrained(model_path, **kwargs)
tokenizer = AutoTokenizer.from_pretrained(model_path, trust_remote_code=True)

## Initialize the quantization tool
QuantTool = RKQuantizer(verbose=True)

## The quantization tool loads the model
ret = QuantTool.load_model(model=model, tokenizer=tokenizer, device='cuda',
                           system_prompt=None, tools=None, system_role='system',
                           user_role='user')
if ret != 0:
    print('Load model failed!')
    exit(ret)

## Run the quantization algorithm
dataset = args.cali_dataset
model = QuantTool.quantize(quantized_dtype="w4a16", quantized_method="group32",
                           quantized_algorithm='grq', dataset=dataset)
## Note: the weights in the model returned by QuantTool.quantize have been replaced
with fake-quantized weights
model = model.cpu()

'''
    Then export the ONNX model to obtain the fake-quantized ONNX model
'''

```

Converting the fake-quantized ONNX model to rknn:

```

import numpy as np
from rknn.api import RKNN
from rknn.api import RKNN, DEFAULT_RKNN_LLM_CONFIG

ONNX_MODEL = 'Qwen3-1.7B.onnx'
LLM_CONFIG = 'Qwen3-1.7B.config.pkl'
RKNN_MODEL = 'Qwen3-1.7B.rknn'

# Create RKNN object
rknn = RKNN(verbose=True)

llm_config = DEFAULT_RKNN_LLM_CONFIG.copy()
llm_config['attention_config'][0]['kvcache_buffer_len'] = 4*1024
llm_config['attention_config'][0]['max_position_embeddings'] = 4*1024
# llm_config['attention_config'][0]['kvcache_dtype'] = 'Int4_to_F16'

# pre-process config
print('--> config model')
## Note: quantized_algorithm must be set to 'normal' here; quantized_dtype and
quantized_method must match those in QuantTool.quantize exactly.
rknn.config(target_platform="rk1820",
            quantized_dtype='w4a16', quantized_algorithm='normal',
            quantized_method='group32', llm_config=llm_config
            )
print('done')

# Load model
print('--> Loading model')
ret = rknn.load_llm(model=ONNX_MODEL, config=LLM_CONFIG)
if ret != 0:
    print('Load model failed!')
    exit(ret)
print('done')

# Build model
print('--> Building model')
ret = rknn.build(do_quantization=True)
if ret != 0:
    print('Build model failed!')
    exit(ret)
print('done')

#Export rknn model
print('--> Export RKNN model')
ret = rknn.export_rknn(RKNN_MODEL)
if ret != 0:
    print('Export rknn failed!')
    exit(ret)
print('done')

rknn.release()

```

2.3.5 Other Quantized Model Conversion

For LLM / MLLM models, when users obtain a quantized model that meets scenario requirements with other quantization schemes (such as GPTQ / QAT), they can first save it as a Huggingface-format fake-quantized model and then export ONNX directly, without using RKQuantizer again. Note that with other quantization schemes, the supported quantization types are still `w4a16`,

w6a16 , and w8a16 , and group_size=32 is required. Likewise, in the subsequent ONNX-to-RKNN conversion, rknn.config() must set quantized_algorithm='normal' and correctly set quantized_dtype (w4a16 , w6a16 , or w8a16) and quantized_method='group32' .

Other quantized model conversion flow:

```
Floating-point PyTorch model → AutoGPTQ/AutoAWQ/LLM-Compressor → fake-quant  
PyTorch model → export fake-quant ONNX → rknn.config() →  
rknn.load_llm()/rknn.load_onnx() → rknn.build() → rknn.export_rknn()
```

2.4 Model Evaluation

After model conversion, a series of evaluations is required to verify the model's validity and performance on the target platform. This chapter describes evaluation methods based on the RKNN3-Toolkit Python API, covering the following four core dimensions:

- **Inference verification:** Verifies that the model's inference works correctly on the target device and that the output results match expectations.
- **Accuracy evaluation:** Compares the model's outputs between the RKNN backend and the original framework to quantify the accuracy loss.
- **Performance evaluation:** Measures key performance metrics such as inference latency and throughput on the NPU.
- **Memory evaluation:** Profiles memory usage during model runtime to support resource planning.

These evaluation tasks support the following two execution modes, and developers can choose flexibly based on the stage:

Execution Mode	Applicable Scenarios	Supported Capabilities
Simulator	No hardware connection required; suits quick verification in early development or CI pipelines	Inference verification, accuracy evaluation
Connected board	Connects real hardware; suits final acceptance and performance tuning before release	Inference verification, accuracy evaluation, performance evaluation, memory evaluation

Note:

1. Performance evaluation and memory evaluation depend on a real NPU environment and are available only in connected-board mode.
2. Connected-board accuracy analysis and performance evaluation require building the model with profile_mode=True configured in rknn.config() ; otherwise the corresponding data cannot be obtained.

2.4.1 Model Inference Verification

After model conversion, invoke `rknn.inference()` for inference verification to check the conversion result and functional correctness. Inference verification supports both simulator and connected-board modes. The two differ only in the parameter configuration of `init_runtime()`; the inference interface is the same:

- **Simulator inference:** Does not specify `target`. It simulates NPU execution on the PC and requires no connected development board.
- **Connected-board inference:** Specifies `target` (target platform), `device_id` (device ID), and `core_mask` (NPU core mask), and executes on a real NPU.

It is recommended to run simulator inference verification first (no hardware needed; quickly confirms the conversion result), and then connected-board inference verification (obtains actual inference results on a real NPU):

```
# 1. Simulator inference (no target specified; no development board required)
rknn.init_runtime()
outputs = rknn.inference(inputs=[img])

# 2. Connected-board inference (specify target and device_id; runs on the real NPU)
rknn.init_runtime(target=platform, device_id='515e9b401c060c0b', core_mask=0xff)
outputs = rknn.inference(inputs=[img])
```

Note: Simulator inference requires no hardware and suits quick verification of conversion results in early development; connected-board inference executes on a real NPU and obtains actual inference results and performance. Simulator performance differs from actual on-device performance; performance evaluation must be done on a connected board (see 2.4.3).

2.4.2 Accuracy Evaluation

Accuracy evaluation is enabled with `rknn.inference(accuracy_analysis=True)`. It compares the differences between the inference results of the quantized model and the floating-point model, and analyzes the source of quantization errors. Like inference verification, accuracy analysis has two modes: simulator and connected-board:

Error Type	Meaning	Sub-items
<code>simulator_error</code>	Simulator error (compared against golden)	<code>entire</code> (accumulated error), <code>single</code> (single-layer independent error)
<code>runtime_error</code>	On-device Runtime error	<code>entire</code> , <code>single_sim</code> (single-layer comparison between Runtime and simulator)

Both cosine distance and Euclidean distance are reported; abnormally low cosine or high Euclidean values are highlighted in yellow or red.

Simulator accuracy evaluation:

```
# Simulator accuracy analysis (no target specified)
rknn.init_runtime()
outputs = rknn.inference(inputs=[img], accuracy_analysis=True)
```

The analysis result is as follows:

```
I GraphPreparing : 100%|██████████████████████| 187/187
[00:00<00:00, 6481.46it/s]
I AccuracyAnalysing : 100%|██████████████████████| 187/187
[00:05<00:00, 34.62it/s]
```

simulator_error: calculate the output error of each layer of the simulator
(compared to the 'golden' value).
entire: output error of each layer between 'golden' and 'simulator',
these errors will accumulate layer by layer.
single: single-layer output error between 'golden' and 'simulator',
can better reflect the single-layer accuracy of the simulator.

layer_name	simulator_error		
	entire		single
	cos	euc	cos
euc			

[Input] images	1.00000	0.09873	1.00000
0.09873			
[Conv] /model.0/conv/Conv_output_0	0.99998	4.97621	0.99998
4.97621			
[exSwish] /model.0/act/Mul_output_0	1.00000	3.92585	1.00000
1.27413			
[Conv] /model.1/conv/Conv_output_0	1.00000	5.41606	1.00000
2.07959			
...			

Connected-board accuracy evaluation:

```
# Connected-board accuracy analysis (specify target / device_id / core_mask)
rknn.init_runtime(target=platform, device_id='515e9b401c060c0b', core_mask=0xff)
outputs = rknn.inference(inputs=[img], accuracy_analysis=True)
```

The analysis result is as follows:

```
I GraphPreparing : 100%|███████████████████████| 187/187
[00:00<00:00, 5128.55it/s]
I AccuracyAnalysing : 100%|███████████████████████| 187/187
[00:25<00:00, 7.33it/s]
```

simulator_error: calculate the output error of each layer of the simulator
(compared to the 'golden' value).

entire: output error of each layer between 'golden' and 'simulator',
these errors will accumulate layer by layer.

single: single-layer output error between 'golden' and 'simulator',
can better reflect the single-layer accuracy of the simulator.

runtime_error: calculate the output error of each layer of the runtime.

entire: output error of each layer between 'golden' and 'runtime',
these errors will accumulate layer by layer.

single_sim: single-layer output error between 'simulator' and
'runtime', can better reflect the single-layer accuracy of runtime.

layer_name					simulator_error		
runtime_error					entire		single
		single_sim			cos	euc	cos
entire	cos	euc	cos	euc			

[Input] images					1.00000	0.09873	1.00000
0.09873	1.00000	504.714	1.00000	0.70923			
[Conv] /model.0/conv/Conv_output_0					0.99998	4.97621	0.99998
4.97621	1.00000	5.45544	1.00000	1.46544			
[exSwish] /model.0/act/Mul_output_0					1.00000	3.92585	1.00000
1.27413	0.99998	7.01712	0.99998	5.80349			
...							

2.4.3 Performance Evaluation

Invoke `rknn.eval_perf()` to evaluate the model's performance on the NPU. It outputs the latency, cycles, and bandwidth of each operator.

```
rknn.init_runtime(target=platform, device_id='515e9b401c060c0b', core_mask=0xff)
rknn.eval_perf()
```

After execution, the performance evaluation result is output as follows:

```

-----
---
Op Time Ranking Table (Core 0)
-----
---
OpType           Calls   CPU(us)   NPU(us)   Total(us)   Ratio(%)
-----
---
ConvExSwish      56      0         20385     20385       47.49%
Reshape          3      0         9245      9245        21.54%
InputProc        1      0         4293      4293        10.00%
Split            6      0         2174      2174         5.06%
Concat           21      0         1878      1878         4.37%
ConvSigmoid       3      0          905       905          2.11%
exYoloPostProcess 1     784         0         784          1.83%
Mul              10      0          572       572          1.33%
Add              13      0          562       562          1.31%
MaxPool          3      0          465       465          1.08%
ConvAdd           1      0          426       426          0.99%
exSwish           1      0          416       416          0.97%
Resize           2      0          412       412          0.96%
Conv              1      0          252       252          0.59%
Sub               3      0           80        80           0.19%
Pow               3      0           75        75           0.17%
InputOperator     1      0           3          3           0.01%
OutputProc        1      0           0          0           0.00%
OutputOperator    1      0           0          0           0.00%
-----
---
Total              784      42143     42927     100.00%
-----
---
```

The performance evaluation result outputs an Op Time Ranking Table of all operators. The fields are described as follows:

Field	Description
OpType	Operator type
Calls	Number of times the operator runs within a single frame
CPU(us)	Total time of the operator on the CPU within a single frame
NPU(us)	Total time of the operator on the NPU within a single frame
Total(us)	Total time of the operator within a single frame
Ratio(%)	Ratio of the operator's total time to the total time of a single frame

2.4.4 Memory Evaluation

Invoke `rknn.eval_memory()` to obtain the model's memory consumption evaluation, including Device Memory and Per-Core Memory Allocation:

```
rknn.init_runtime(target=platform, device_id='515e9b401c060c0b', core_mask=0xff)
memory_detail = rknn.eval_memory()
```

After execution, the memory evaluation result is output as follows:

```
===== Memory Usage Information =====
Device Memory:
System :      18.98 MB total,      18.83 MB free,      0.16 MB used (  0.8%)
Node 0  :     639.50 MB total,    590.57 MB free,    48.93 MB used (  7.7%)
Node 1  :     639.50 MB total,    639.42 MB free,     0.08 MB used (  0.0%)
Node 2  :     619.90 MB total,    619.81 MB free,     0.08 MB used (  0.0%)
Node 3  :     639.50 MB total,    639.42 MB free,     0.08 MB used (  0.0%)
Node 4  :     619.90 MB total,    619.81 MB free,     0.08 MB used (  0.0%)
Node 5  :     619.90 MB total,    619.81 MB free,     0.08 MB used (  0.0%)
Node 6  :     639.50 MB total,    639.42 MB free,     0.08 MB used (  0.0%)
Node 7  :     639.50 MB total,    639.42 MB free,     0.08 MB used (  0.0%)

Per-Core Memory Allocation (MB):
Core   Command   Weight   Internal   KVCache   Total
-----
0      0.22        15.42    33.20      0.00      48.84
-----
Total  0.22        15.42    33.20      0.00      48.84
=====
```

The fields of the per-core memory allocation are described as follows:

Field	Description
Core	Core in use
Command	Memory usage of register configuration and related control information (MB)
Weight	Memory usage of model weight tensors (MB)
Internal	Memory usage of model intermediate tensors (MB)
KVCache	Memory usage of model KVCache (MB)
Total	Total memory usage of the model (MB)

3. On-device Deployment

3.1 Runtime Deployment Basics

This section describes the fundamental concepts of RKNN3 Runtime and is a prerequisite for on-device deployment (using the C API or Toolkit Lite). It covers model and context lifecycles, tensors and data layout formats, input and output memory management, synchronous and asynchronous inference mechanisms, and error handling and resource release conventions. These mechanisms apply to both general models and LLM / MLLM models.

3.1.1 Model and Context Lifecycle

`rknn3_context` is the core handle of RKNN3 Runtime and represents a runtime context instance. All model loading, inference configuration, and resource management operations are based on this handle. For on-device deployment, strictly follow the "create → use → destroy" lifecycle management convention to avoid resource leaks or runtime exceptions.

General models (CNN / ViT, etc.)

These models do not involve the session layer. Invoke `rknn3_run()` directly through the context handle to execute inference. After the context is initialized, `rknn3_run()` can be invoked multiple times, and the handle remains unchanged until the context is destroyed.

LLM / MLLM models

On top of general models, these models introduce a **Session** layer to manage the context state of text generation tasks. Create a session with `rknn3_session_init()` and configure LLM-related parameters (such as vocabulary, sampling strategy, and context window length), then invoke `rknn3_session_run()` to execute inference. One context can create multiple sessions. All sessions share model weights and internal memory, but their KVCache and LoRA enable states are independent of each other. Sessions cannot execute concurrently; only one session can run at a time. After the sessions are used, invoke `rknn3_session_destroy()` to destroy each session first, and finally invoke `rknn3_destroy()` to destroy the context.

3.1.2 Tensors and Data Layout Formats

The inputs and outputs of RKNN3 models are managed through **tensors**. This section describes the tensor structure hierarchy, data types and layouts, and the NPU-specific `NC1HWC2` data layout format.

Tensor Structure

General models and LLM / MLLM models use different tensor structure hierarchies:

General models (CNN / ViT, etc.) use `rknn3_tensor` as the input and output tensors for inference. It binds two sub-structures:

- `rknn3_tensor` : The tensor structure used for inference. It contains two core members: `mem` (pointing to `rknn3_tensor_mem`) and `attr` (pointing to `rknn3_tensor_attr`). Before inference, users must allocate memory for each input and output tensor and bind the attributes. The Runtime reads inputs and writes outputs through this structure.

- **rknn3_tensor_attr** : The tensor attribute structure, queried through `rknn3_query()` . It contains the tensor name, shape, data type (dtype), data layout (layout), quantization information (qnt_type/qnt_info), core ID, and more. For on-device deployment, query the attributes first, then allocate memory and organize data accordingly.
- **rknn3_tensor_mem** : The tensor data memory structure, allocated through interfaces such as `rknn3_create_mem()` . It represents the actual data storage of the tensor on the device. Users must synchronize data between the CPU and the device through `rknn3_mem_sync()` .

LLM / MLLM models use `rknn3_llm_input` as the input for Session inference. It distinguishes different input types through the `input_type` field and uses the corresponding sub-structure:

- **rknn3_llm_input** : The LLM inference input structure, containing `input_type` (input type) and the input data of the corresponding type. Main members:
 - `role` : The message role, such as "user" (user input) or "tool" (function call result).
 - `input_type` : The input type. Options include `RKNN3_LLM_INPUT_PROMPT` (text), `RKNN3_LLM_INPUT_TOKEN` (token id), `RKNN3_LLM_INPUT_EMBED` (embedding vector), `RKNN3_LLM_INPUT_MULTIMODAL` (multimodal), and `RKNN3_LLM_INPUT_AUX` (auxiliary input).
 - `llm_input` : A `rknn3_llm_tensor` structure when the input type is PROMPT/TOKEN/EMBED.
 - `multimodal_input` : A `rknn3_llm_multimodal_tensor` structure when the input type is MULTIMODAL (members described below).
 - `aux_input` : An auxiliary input tensor when the input type is AUX (currently `rknn3_aux_tensor` is the same type as `rknn3_tensor`).
- **rknn3_llm_tensor** : A sub-structure of `rknn3_llm_input` that represents the LLM input content. Main members:
 - `name` : The tensor name.
 - `prompt` : The text prompt input, used when the input type is `RKNN3_LLM_INPUT_PROMPT` .
 - `embed` : The embedding vector pointer, used when the input type is `RKNN3_LLM_INPUT_EMBED` (size is `n_tokens * hidden_size`).
 - `tokens` : The token id array, used when the input type is `RKNN3_LLM_INPUT_TOKEN` .
 - `n_tokens` : The number of tokens.
 - `enable_thinking` : Whether to enable thinking mode.
- **rknn3_llm_multimodal_tensor** : A sub-structure of `rknn3_llm_input` , used when the input type is `RKNN3_LLM_INPUT_MULTIMODAL` . It represents multimodal input (text, image, audio, video). Top-level members include `name` , `prompt / tokens` (text input; provide one of the two — if both are provided, prompt is used for inference by default), `n_tokens` , `enable_thinking` , and three modality sub-structures:
 - **image sub-structure**: Image input. `image_embed` (FP16 embedding pointer, size is `n_image × n_image_tokens × embedding_dim`), `n_image_tokens` (number of tokens per image), `n_image` (number of images), `image_start / image_end / image_content` (start/end/placeholder tags of the image in the text), `image_width / image_height` (image width and height).

- **audio sub-structure:** Audio input. `audio_embed` (FP16 embedding pointer, size is `n_audio × n_audio_tokens × embedding_dim`), `n_audio_tokens` (number of tokens per audio segment), `n_audio` (number of audio segments), `audio_start / audio_end / audio_content` (audio tags).
- **video sub-structure:** Video input. `video_embed` (FP16 embedding pointer, size is `n_video × n_frame_per_video × n_frame_tokens × embedding_dim`), `n_frame_tokens` (number of tokens per frame), `n_frame_per_video` (number of frames per video), `n_video` (number of videos), `video_start / video_end / video_content` (video tags).

The `*_embed` of each modality is the inference output of the corresponding vision/audio/video encoder model and is passed into the LLM through this structure. The `*_start / *_end / *_content` tags must match the special tokens of the model tokenizer. For details on how to populate and use the structure, see 3.3.6 Multimodal Input.

Data Types and Layouts

The data type (dtype) and data layout (layout) of a tensor can be queried through the `dtype` and `layout` fields of `rknn3_tensor_attr`.

- **Data type (dtype):** Includes FP32 / FP16 / INT8 / UINT8, etc., determined by the quantization configuration during model conversion. For non-quantized models, both weights and activations are FP16. For quantized models, weights can be INT8, etc.
- **Data layout (layout):** There are three main layouts: `NCHW`, `NHWC`, and `NC1HWC2`. `NCHW` and `NHWC` are common data layouts in deep learning. User inputs typically use these two formats, and the Runtime handles the conversion internally. `NC1HWC2` is a memory layout format optimized for NPU hardware. To directly access NPU memory (for example, in zero-copy scenarios), organize the data in the `NC1HWC2` layout.

NC1HWC2 Data Layout Format

`NC1HWC2` stores the channel dimension in blocks, where C1 is the number of channel blocks ($\text{ceil}(C/C2)$) and C2 is the number of sub-channels, determined by the platform and data type:

	RK1820/RK3572
INT8	32
FP16	16

The data layout and storage of `NC1HWC2` are shown in the figure below. The number 0 represents one data element; C2 data elements are stored at a time, and the storage order follows the increasing order of the values in the figure (0-15 are stored first, then 16-31). The figure below uses $C2 = 8$ as an example (to illustrate the layout principle; on current platforms $C2 = 16$ for FP16 and $C2 = 32$ for INT8, with the same layout). When the feature is (1, 13, 4, 4), the corresponding `NC1HWC2` is (1, 2, 4, 4, 8). Only the first 5 data elements in each C2 block are valid; the rest are alignment padding.

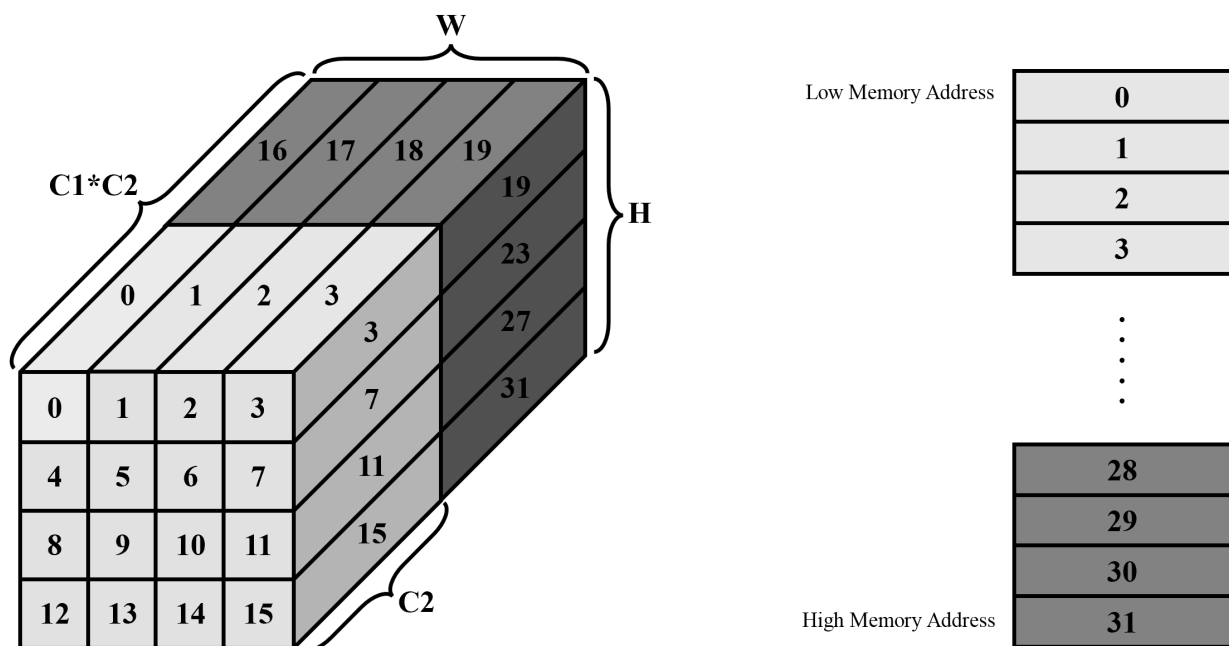


Figure 3-1 RKNPU NC1HWC2 data layout and storage

Taking feature (1, 13, 2, 2) with $C2 = 8$ as an example, the NC1HWC2 is (1, 2, 2, 2, 8). Its storage is expanded below; the red parts are invalid data used for alignment padding:

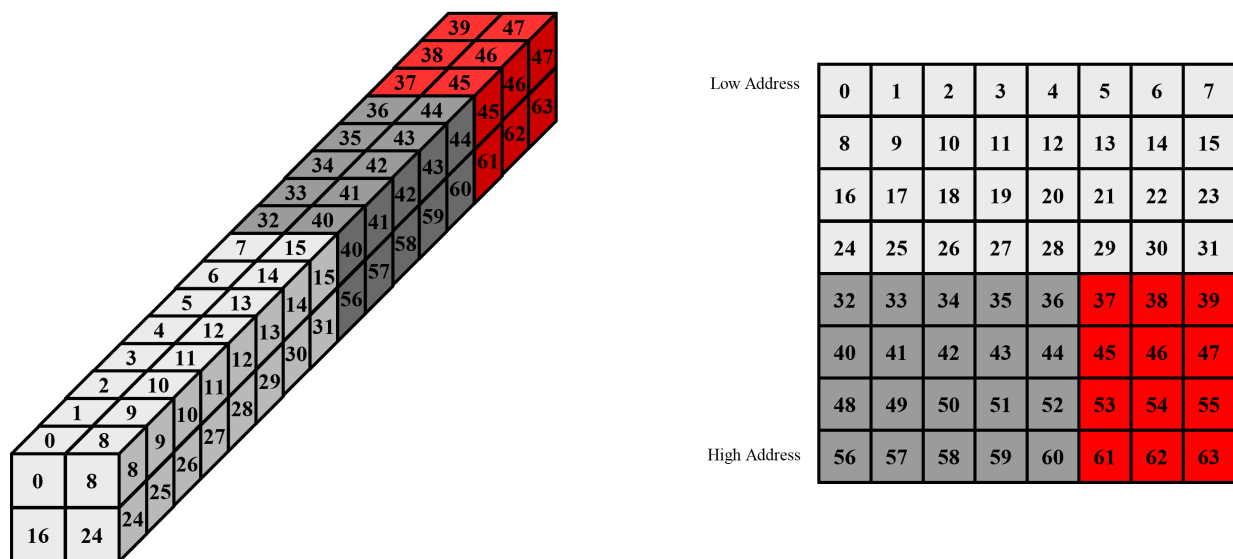


Figure 3-2 NC1HWC2 data layout expanded

After removing the invalid data, it becomes NCHW (1, 13, 2, 2):

Low Address	0	8	16	24
	1	9	17	25
	2	10	18	26
	3	11	19	27
	4	12	20	28
	5	13	21	29
	6	14	22	30
	7	15	23	31
High Address	32	40	48	56
	33	41	49	57
	34	42	50	58
	35	43	51	59
	36	44	52	60

Figure 3-3 NCHW data layout

After removing the invalid data, it becomes NHWC (1, 2, 2, 13):

Low Address	0	1	2	3	4	5	6	7	32	33	34	35	36
	8	9	10	11	12	13	14	15	40	41	42	43	44
	16	17	18	19	20	21	22	23	48	49	50	51	52
	24	25	26	27	28	29	30	31	56	57	58	59	60
High Address													

Figure 3-4 NHWC data layout

Layout Conversion

When you need to directly access NPU memory (for example, in zero-copy scenarios), convert the data between NC1HWC2 and NCHW / NHWC. The following are common conversion examples:

NC1HWC2 to NCHW (INT8):

```

#define ALIGNED_SIZE(x, bytes) (((x) + ((bytes) - 1)) & ~(bytes) - 1))
int NC1HWC2_to_NCHW(const int8_t* src, int8_t* dst, int* dims, int channel)
{
    int batch      = dims[0];
    int height     = dims[2];
    int width      = dims[3];
    int cn         = dims[4];
    int align_surf  = width * height == 1 ? 1 : ALIGNED_SIZE(width * height, 4);

    int src_batch_elems = dims[1] * align_surf * dims[4];
    int dst_batch_elems = channel * width * height;

    for (int i = 0; i < batch; ++i) {
        int idx = 0;
        for (int c = 0; c < channel; ++c) {
            int plane = c / cn;
            const int8_t* src_c = plane * align_surf * cn + src;
            int offset = c % cn;

            for (int cur_h = 0; cur_h < height; ++cur_h) {
                for (int cur_w = 0; cur_w < width; ++cur_w) {
                    int cur_hw = cur_h * width + cur_w;
                    dst[idx++] = src_c[cn * cur_hw + offset];
                }
            }
        }

        src += src_batch_elems;
        dst += dst_batch_elems;
    }
    return 0;
}

```

NCHW to NC1HWC2 (INT8):

```

#define ALIGNED_SIZE(x, bytes) (((x) + ((bytes) - 1)) & ~(bytes) - 1))
int NCHW_to_NC1HWC2(const int8_t* src, int8_t* dst, int* dims, int channel)
{
    int batch      = dims[0];
    int height     = dims[2];
    int width      = dims[3];
    int cn         = dims[4];
    int align_surf = width * height == 1 ? 1 : ALIGNED_SIZE(width * height, 4);
    int hw = height * width;

    int src_batch_elems = channel * width * height;
    int dst_batch_elems = dims[1] * align_surf * cn;

    for (int i = 0; i < batch; ++i) {
        for (int c = 0; c < channel; ++c) {
            int plane = c / cn;
            int offset = c % cn;
            int8_t* dst_c = dst + (size_t)plane * align_surf * cn;

            for (int cur_h = 0; cur_h < height; ++cur_h) {
                for (int cur_w = 0; cur_w < width; ++cur_w) {
                    int cur_hw = cur_h * width + cur_w;
                    dst_c[cn * cur_hw + offset] = src[c * hw + cur_h * width + cur_w];
                }
            }
        }

        src += src_batch_elems;
        dst += dst_batch_elems;
    }

    return 0;
}

```

Note: For the complete conversion code (including `NC1HWC2` to `NHWC`), refer to the utility functions in `rknn3-model-zoo`.

3.1.3 Input and Output Memory

In RKNN3 Runtime, the input and output data of a model are managed through `rknn3_tensor_mem` memory objects. For on-device deployment, allocate memory for each input and output tensor, bind it to the tensor attributes, synchronize the data between the CPU and the NPU, and finally release the memory. This section describes memory allocation methods, the synchronization mechanism, and release conventions.

Memory Allocation Methods

RKNN3 provides multiple memory allocation methods for different scenarios:

- `rknn3_create_mem()` : The Runtime allocates device memory of the specified size. This is the most commonly used allocation method. When allocating, specify the memory size, core ID (the core the memory belongs to in multi-core scenarios), and memory allocation flags (`rknn3_mem_alloc_flags`). The flags control the cache attributes of the memory: `RKNN3_FLAG_MEMORY_CACHEABLE` (default; creates cacheable memory with better

performance but requires synchronization with `rknn3_mem_sync()` and `RKNN3_FLAG_MEMORY_NON_CACHEABLE` (creates non-cacheable memory; no synchronization required but lower access performance). The allocated memory object is bound to the tensor through the `rknn3_tensor.mem` field, and the Runtime reads inputs and writes outputs through this binding during inference. This method suits most regular deployment scenarios.

- `rknn3_create_mem_from_fd()` : Creates a memory object from an existing file descriptor (fd) and reuses externally allocated memory (such as DMA-BUF). It suits zero-copy scenarios where memory must be shared with other hardware modules (such as cameras or displays). It avoids data copy overhead but requires a valid fd and a matching size for the external memory.
- `rknn3_create_mem_from_phys()` : Creates a memory object from a physical address (not implemented in the current version). It is designed for advanced scenarios that directly access physical memory.

Selection advice: For regular deployment, use `rknn3_create_mem()` first; it is simple and reliable. For zero-copy integration with modules such as cameras or RGA, use `rknn3_create_mem_from_fd()`.

Allocation example:

```
// Allocate cacheable memory for the input tensor
inputs[i].mem = rknn3_create_mem(ctx, input_attrs[i].aligned_size,
                                input_attrs[i].core_id,
                                RKNN3_FLAG_MEMORY_CACHEABLE);
```

Memory Synchronization

When memory is allocated with the `RKNN3_FLAG_MEMORY_CACHEABLE` flag (or the default `RKNN3_FLAG_MEMORY_FLAGS_DEFAULT`; both behave the same), the CPU and the device maintain separate caches. Data written directly to CPU-side memory may not be immediately visible to the device, and vice versa. Therefore, explicitly synchronize the memory data between the CPU and the device through `rknn3_mem_sync()` before and after inference to ensure data consistency. If memory is allocated with the `RKNN3_FLAG_MEMORY_NON_CACHEABLE` flag, no synchronization is required, but access performance is lower.

The synchronization direction is specified by `rknn3_mem_sync_mode` :

- `RKNN3_MEMORY_SYNC_TO_DEVICE` : Synchronizes CPU-side data to the device. Use it after the input data is ready and before inference executes, to ensure the device reads the latest input data.
- `RKNN3_MEMORY_SYNC_FROM_DEVICE` : Synchronizes device-side data to the CPU. Use it after inference completes and before reading the output results, to ensure the CPU reads the final results written by the device.
- `RKNN3_MEMORY_SYNC_BIDIRECTIONAL` : Synchronizes in both directions. It suits scenarios where both the CPU and the device read and write.

Synchronization example:

```
// Before inference: synchronize input data to the device
for (int i = 0; i < n_inputs; i++) {
    rknn3_mem_sync(ctx, inputs[i].mem, RKNN3_MEMORY_SYNC_TO_DEVICE);
}

// After inference: synchronize output data to the CPU
for (int i = 0; i < n_outputs; i++) {
    rknn3_mem_sync(ctx, outputs[i].mem, RKNN3_MEMORY_SYNC_FROM_DEVICE);
}
```

Precautions:

1. Synchronization is a blocking operation; it waits for the data transfer to complete. Perform input synchronization after all input data is written, and perform output synchronization after inference completes.
2. Each input tensor must be synchronized to the device individually, and each output tensor must be synchronized back to the CPU individually.
3. If you skip synchronization and directly read the output of cacheable memory, you may read stale cache data and get incorrect results.

Memory Release

After the memory is used, release it through `rknn3_destroy_mem()`. Release must follow these lifecycle conventions:

1. Memory release must complete before the context is destroyed (`rknn3_destroy()`); otherwise, resource leaks may occur.
2. Release the input and output memory first, then destroy the context.
3. Do not use released memory objects again; doing so causes undefined behavior.

Release example:

```
// Release input and output memory first
for (int i = 0; i < n_inputs; i++) {
    rknn3_destroy_mem(ctx, inputs[i].mem);
}
for (int i = 0; i < n_outputs; i++) {
    rknn3_destroy_mem(ctx, outputs[i].mem);
}
// Then destroy the context
rknn3_destroy(ctx);
```

3.1.4 Synchronous and Asynchronous Inference

RKNN3 Runtime provides two inference execution modes: synchronous and asynchronous.

- **Synchronous inference** (`rknn3_run()`): Blocks the current thread after invocation and returns after inference completes. It is simple to use and suits most scenarios. Before invocation, ensure the input data has been synchronized to the device (`RKNN3_MEMORY_SYNC_TO_DEVICE`); after inference completes, synchronize the output data back to the CPU (`RKNN3_MEMORY_SYNC_FROM_DEVICE`).

```

// 1. Synchronize input data to the device
for (int i = 0; i < n_inputs; i++) {
    rknn3_mem_sync(ctx, inputs[i].mem, RKNN3_MEMORY_SYNC_TO_DEVICE);
}

// 2. Run synchronous inference (blocks until completion)
int ret = rknn3_run(ctx, inputs, n_inputs, outputs, n_outputs);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_run failed! ret=%d\n", ret);
    return -1;
}

// 3. Synchronize output data to the CPU
for (int i = 0; i < n_outputs; i++) {
    rknn3_mem_sync(ctx, outputs[i].mem, RKNN3_MEMORY_SYNC_FROM_DEVICE);
}

```

- **Asynchronous inference** (`rknn3_run_async()` + `rknn3_wait()`): First initiates non-blocking inference through `rknn3_run_async()` . The main thread can continue with other tasks, then waits for inference to complete through `rknn3_wait()` . It suits scenarios that process other work in parallel during inference (such as data pre-processing) and can improve overall throughput. **Note: `rknn3_run_async()` and `rknn3_wait()` are not implemented in the current version.**

Selection advice: The current version supports only synchronous inference (`rknn3_run()`); the asynchronous inference interfaces are not implemented yet. Once supported in later versions, use the asynchronous mode if parallel tasks are needed during inference.

3.1.5 Error Handling and Resource Release

Error Handling

Interfaces of the RKNN3 C API return an `int` status code. `RKNN3_SUCCESS` (0) indicates success, and negative values indicate errors. For on-device deployment, check the return value of every interface invocation. On error, do not continue with subsequent operations; jump to the resource cleanup routine and exit, to avoid undefined behavior caused by executing in an abnormal state.

Common status codes and their meanings:

Status code	Value	Description
<code>RKNN3_SUCCESS</code>	0	Execution succeeded
<code>RKNN3_ERR_FAIL</code>	-1	Execution failed
<code>RKNN3_ERR_ARGUMENT_INVALID</code>	-2	Invalid argument
<code>RKNN3_ERR_MODEL_INVALID</code>	-3	Invalid model
<code>RKNN3_ERR_CTX_INVALID</code>	-4	Invalid context
<code>RKNN3_ERR_OUT_OF_MEMORY</code>	-6	Out of memory
<code>RKNN3_ERR_TIMEOUT</code>	-7	Execution timed out
<code>RKNN3_ERR_DEVICE_UNAVAILABLE</code>	-10	Device unavailable
<code>RKNN3_ERR_COMMUNICATION</code>	-13	Communication error
<code>RKNN3_ERR_INCOMPATIBLE_VERSION</code>	-18	Incompatible component versions

For the complete error code list, see the *RKNN3_Runtime_C_API_Reference*.

Error handling example for the initialization stage (jumps to the cleanup routine on error):

```
int ret = rknn3_init(&ctx, NULL);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_init failed! ret=%d\n", ret);
    goto out;
}
```

Resource Release

In on-device deployment, release all types of resources in the correct order to avoid resource leaks. It is recommended to use `goto` to jump to a unified cleanup routine, ensuring correct resource release on both the normal and error paths.

- **General models:** Release order: release input and output memory -> destroy the context.

```

out:
    // Release input and output memory
    for (int i = 0; i < n_inputs; i++) {
        if (inputs[i].mem) rknn3_destroy_mem(ctx, inputs[i].mem);
    }
    for (int i = 0; i < n_outputs; i++) {
        if (outputs[i].mem) rknn3_destroy_mem(ctx, outputs[i].mem);
    }
    // Destroy the context
    if (ctx) rknn3_destroy(ctx);

```

- **LLM / MLLM models:** Release order: destroy the session -> release input and output memory -> destroy the context.

```

out:
    // Destroy the session
    if (session) rknn3_session_destroy(session);
    // Release input and output memory
    for (int i = 0; i < n_inputs; i++) {
        if (inputs[i].mem) rknn3_destroy_mem(ctx, inputs[i].mem);
    }
    for (int i = 0; i < n_outputs; i++) {
        if (outputs[i].mem) rknn3_destroy_mem(ctx, outputs[i].mem);
    }
    // Destroy the context
    if (ctx) rknn3_destroy(ctx);

```

Note: After the context is destroyed, all resources created from it (memory, sessions, etc.) are no longer accessible. Ensure all resources are released before the context is destroyed.

3.2 General Model C API Deployment

This section describes how to deploy general models (CNN / ViT, etc.) with the RKNN3 Runtime C API, covering the complete flow from model initialization through inference execution to resource release. It suits production deployment and performance-sensitive scenarios.

3.2.1 Complete Invocation Flow

The typical invocation sequence for general model C API deployment is: initialize context -> load model -> initialize model -> query attributes -> create memory -> prepare input -> execute inference -> process output -> release resources. The overall flow is shown in the figure below:

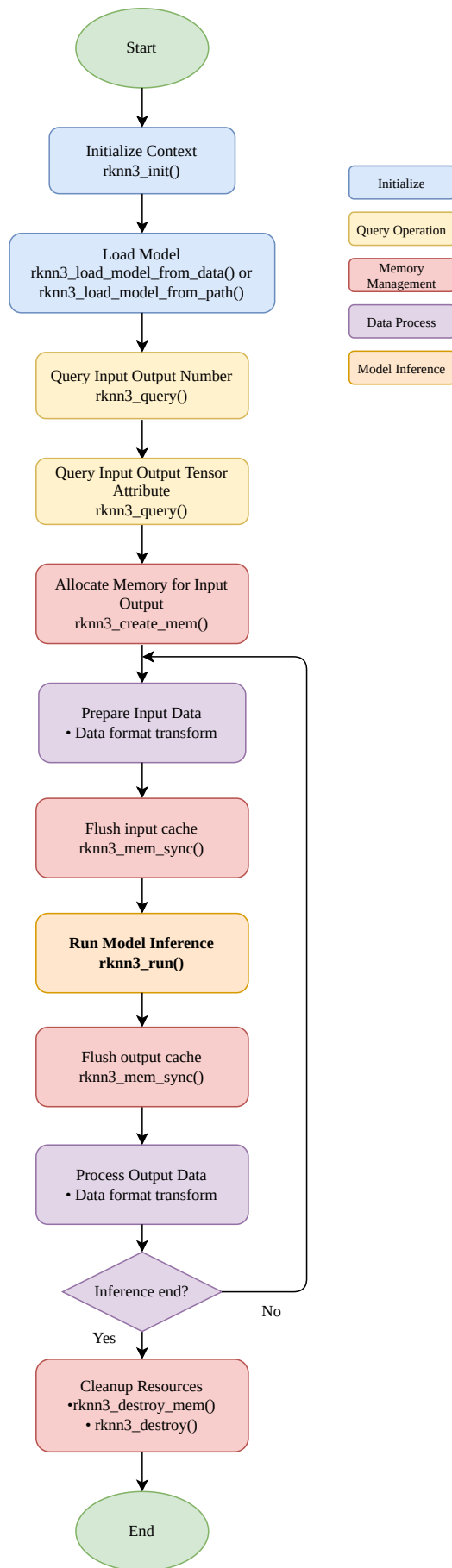


Figure 3-5 RKNN3 C API inference flow

The following subsections describe the operations and precautions for each step in this flow.

3.2.2 Model Initialization

Model initialization consists of three steps: create the context, load the model, and configure runtime parameters.

1. Create the context

Create the runtime context through `rknn3_init()`. This is the prerequisite for all subsequent operations. In multi-device scenarios, specify the device ID through `init_extend`; in single-device scenarios, pass `NULL`:

```
rknn3_context ctx;
int ret = rknn3_init(&ctx, NULL);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_init failed! ret=%d\n", ret);
    goto out;
}
```

2. Load the model

Load the `.rknn` model and `.weight` weights from a file through `rknn3_load_model_from_path()`, or load from memory through `rknn3_load_model_from_data()` (suits scenarios where the model is already in memory, such as loading an encrypted model after decryption):

```
// Load from file
ret = rknn3_load_model_from_path(ctx, model_path, weight_path);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_load_model_from_path failed! ret=%d\n", ret);
    goto out;
}
```

Note: Passing `NULL` for `weight_path` enables streaming weight loading mode. After `rknn3_model_init()`, upload the weights in chunks through `rknn3_load_weight_chunk()`. This suits memory-constrained scenarios.

3. Configure runtime parameters

Configure the model runtime parameters and complete initialization through `rknn3_model_init()`. The `rknn3_config` structure contains the core mask (`run_core_mask`), run priority (`priority`), timeout (`run_timeout`), and other configurations. The core mask controls which NPU cores the model runs on. It must match the `core_mask` configured during model conversion; otherwise, an error is reported:

```
rknn3_config config;
memset(&config, 0, sizeof(rknn3_config));
config.run_core_mask = 0xff; // Use all NPU cores
ret = rknn3_model_init(ctx, &config);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_model_init failed! ret=%d\n", ret);
    goto out;
}
```

3.2.3 Input and Output Preparation

After model initialization, query the model input and output attributes, then allocate memory and prepare the input data accordingly.

1. Query input and output attributes

Query the number and attributes of the model input and output tensors through `rknn3_query()`. First query the numbers (`RKNN3_QUERY_IN_OUT_NUM`), then query the attributes of each tensor (`RKNN3_QUERY_INPUT_ATTR` / `RKNN3_QUERY_OUTPUT_ATTR`) to obtain information such as shape / dtype / layout, quantization parameters, `aligned_size` (aligned memory size), and `core_id` (the core the tensor belongs to):

```
// Query the number of inputs and outputs
rknn3_input_output_num io_num;
ret = rknn3_query(ctx, RKNN3_QUERY_IN_OUT_NUM, &io_num, sizeof(io_num));
if (ret != RKNN3_SUCCESS) {
    printf("query io_num failed! ret=%d\n", ret);
    goto out;
}

// Query the attributes of each input tensor
rknn3_tensor_attr input_attrs[io_num.n_input];
for (uint32_t i = 0; i < io_num.n_input; i++) {
    input_attrs[i].index = i;
    ret = rknn3_query(ctx, RKNN3_QUERY_INPUT_ATTR, &input_attrs[i],
sizeof(rknn3_tensor_attr));
    if (ret != RKNN3_SUCCESS) {
        printf("query input_attr %d failed! ret=%d\n", i, ret);
        goto out;
    }
}

// Query the attributes of each output tensor
rknn3_tensor_attr output_attrs[io_num.n_output];
for (uint32_t i = 0; i < io_num.n_output; i++) {
    output_attrs[i].index = i;
    ret = rknn3_query(ctx, RKNN3_QUERY_OUTPUT_ATTR, &output_attrs[i],
sizeof(rknn3_tensor_attr));
    if (ret != RKNN3_SUCCESS) {
        printf("query output_attr %d failed! ret=%d\n", i, ret);
        goto out;
    }
}
```

2. Allocate input and output memory

Based on the queried `aligned_size` and `core_id`, allocate memory for each input and output tensor through `rknn3_create_mem()` and bind it to the `mem` field of `rknn3_tensor`:

```
rknn3_tensor inputs[io_num.n_input];

// Create memory for the input tensor
for (uint32_t i = 0; i < io_num.n_input; i++) {
    inputs[i].mem = rknn3_create_mem(ctx, input_attrs[i].aligned_size,
                                     input_attrs[i].core_id,
                                     RKNN3_FLAG_MEMORY_CACHEABLE);

    if (inputs[i].mem == NULL) {
        printf("create input mem %d failed!\n", i);
        goto out;
    }
}

// Create memory for the output tensor
rknn3_tensor outputs[io_num.n_output];
for (uint32_t i = 0; i < io_num.n_output; i++) {
    outputs[i].mem = rknn3_create_mem(ctx, output_attrs[i].aligned_size,
                                       output_attrs[i].core_id,
                                       RKNN3_FLAG_MEMORY_CACHEABLE);

    if (outputs[i].mem == NULL) {
        printf("create output mem %d failed!\n", i);
        goto out;
    }
}
```

3. Prepare input data

Before inference, write the preprocessed input data (after resize, color space conversion, etc.) into the corresponding input tensor memory (`inputs[i].mem`). After writing, invoke `rknn3_mem_sync()` to synchronize the data to the NPU device side.

Note:

- **Data attribute consistency:** The `dtype` (data type) and `shape` of the input data must strictly match the queried tensor attributes; otherwise, inference errors occur.
- **Mean and normalization processing:** If `mean_values` and `std_values` were configured in `rknn.config()` during model conversion, the C API inference stage **does not require manual mean normalization**, which effectively reduces latency in the deployment stage.
- **Data layout requirements:** The input data layout must match the `layout` field in the tensor attributes. RKNN3 Runtime internally supports the common `NHWC` and `NCHW` layouts and automatically handles the necessary layout conversions during inference.

```
// Write pre-processed data into the input memory
// memcpy(inputs[0].mem->virt_addr, input_data, input_attrs[0].aligned_size);

// Synchronize input data to the device
for (uint32_t i = 0; i < io_num.n_input; i++) {
    ret = rknn3_mem_sync(ctx, inputs[i].mem, RKNN3_MEMORY_SYNC_TO_DEVICE);
    if (ret != RKNN3_SUCCESS) {
        printf("mem_sync input %d failed! ret=%d\n", i, ret);
        goto out;
    }
}
```

3.2.4 Inference Execution

After the input data is synchronized to the device, execute synchronous inference through `rknn3_run()`, passing the context, the input tensor array and count, and the output tensor array and count. This interface blocks until inference completes:

```
ret = rknn3_run(ctx, inputs, io_num.n_input, outputs, io_num.n_output);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_run failed! ret=%d\n", ret);
    goto out;
}
```

After inference completes, the output data has been written to the output tensor memory but remains on the device side. Synchronize it back to the CPU through `rknn3_mem_sync()` before reading it (see 3.2.5 Output Processing).

3.2.5 Output Processing

After inference completes, the output data is on the device side. Synchronize it back to the CPU first, then read and process the results according to the output tensor attributes.

1. Synchronize output data

Synchronize the memory of each output tensor from the device back to the CPU:

```
for (uint32_t i = 0; i < io_num.n_output; i++) {
    ret = rknn3_mem_sync(ctx, outputs[i].mem, RKNN3_MEMORY_SYNC_FROM_DEVICE);
    if (ret != RKNN3_SUCCESS) {
        printf("mem_sync output %d failed! ret=%d\n", i, ret);
        goto out;
    }
}
```

2. Read output data

After synchronization, read the output data through the virtual address of `outputs[i].mem`. Note the following when reading:

- **Data type:** Determine the data type to read based on `output_attrs[i].dtype` (such as FP16 / INT8).

- **Data layout:** The output layout may be `NC1HWC2`. Convert it to `NCHW` or `NHWC` before post-processing (see 3.1.2 for the conversion method).
- **Quantization parameters:** If the output is quantized (INT8, etc.), dequantize it with the `scale` and `zero_point` in `output_attrs[i].qnt_info` to restore floating-point values.

```
// Read output data (FP16 in this example)
float16* output_data = (float16*)outputs[0].mem->virt_addr;
// Post-process according to output_attrs[0].shape and layout
// ...
```

3. Post-processing

Post-processing depends on the specific model and application scenario, for example NMS for object detection models or argmax for classification models. Refer to the example code for each model in rknn3-model-zoo.

3.2.6 Complete Example

The following is a complete example of general model C API inference, integrating model initialization, input and output preparation, inference execution, output processing, and resource release:

```

#include "rknn3_api.h"

int main() {
    int ret;
    rknn3_context ctx = NULL;

    // 1. Initialize the context
    ret = rknn3_init(&ctx, NULL);
    if (ret != RKNN3_SUCCESS) {
        printf("rknn3_init failed! ret=%d\n", ret);
        goto out;
    }

    // 2. Load the model
    ret = rknn3_load_model_from_path(ctx, "model.rknn", "model.weight");
    if (ret != RKNN3_SUCCESS) {
        printf("rknn3_load_model_from_path failed! ret=%d\n", ret);
        goto out;
    }

    // 3. Initialize the model
    rknn3_config config = {0};
    config.run_core_mask = 0xff;
    ret = rknn3_model_init(ctx, &config);
    if (ret != RKNN3_SUCCESS) {
        printf("rknn3_model_init failed! ret=%d\n", ret);
        goto out;
    }

    // 4. Query the number of inputs and outputs
    rknn3_input_output_num io_num;
    ret = rknn3_query(ctx, RKNN3_QUERY_IN_OUT_NUM, &io_num, sizeof(io_num));
    if (ret != RKNN3_SUCCESS) {
        printf("query io_num failed! ret=%d\n", ret);
        goto out;
    }

    // 5. Query attributes and allocate memory
    rknn3_tensor_attr input_attrs[io_num.n_input];
    rknn3_tensor_attr output_attrs[io_num.n_output];
    rknn3_tensor inputs[io_num.n_input];
    rknn3_tensor outputs[io_num.n_output];

    for (uint32_t i = 0; i < io_num.n_input; i++) {
        input_attrs[i].index = i;
        ret = rknn3_query(ctx, RKNN3_QUERY_INPUT_ATTR, &input_attrs[i],
sizeof(rknn3_tensor_attr));
        if (ret != RKNN3_SUCCESS) {
            printf("query input_attr %d failed! ret=%d\n", i, ret);
            goto out;
        }
        inputs[i].mem = rknn3_create_mem(ctx, input_attrs[i].aligned_size,
input_attrs[i].core_id,
RKNN3_FLAG_MEMORY_CACHEABLE);
        if (inputs[i].mem == NULL) {
            printf("create input mem %d failed!\n", i);
            goto out;
        }
    }
    for (uint32_t i = 0; i < io_num.n_output; i++) {

```

```

        output_attrs[i].index = i;
        ret = rknn3_query(ctx, RKNN3_QUERY_OUTPUT_ATTR, &output_attrs[i],
sizeof(rknn3_tensor_attr));
        if (ret != RKNN3_SUCCESS) {
            printf("query output_attr %d failed! ret=%d\n", i, ret);
            goto out;
        }
        outputs[i].mem = rknn3_create_mem(ctx, output_attrs[i].aligned_size,
                                         output_attrs[i].core_id,
RKNN3_FLAG_MEMORY_CACHEABLE);
        if (outputs[i].mem == NULL) {
            printf("create output mem %d failed!\n", i);
            goto out;
        }
    }

    // 6. Prepare input data and synchronize it to the device
    // ... Fill inputs[0].mem->virt_addr ...
    for (uint32_t i = 0; i < io_num.n_input; i++) {
        ret = rknn3_mem_sync(ctx, inputs[i].mem, RKNN3_MEMORY_SYNC_TO_DEVICE);
        if (ret != RKNN3_SUCCESS) {
            printf("mem_sync input %d failed! ret=%d\n", i, ret);
            goto out;
        }
    }

    // 7. Run inference
    ret = rknn3_run(ctx, inputs, io_num.n_input, outputs, io_num.n_output);
    if (ret != RKNN3_SUCCESS) {
        printf("rknn3_run failed! ret=%d\n", ret);
        goto out;
    }

    // 8. Synchronize outputs and process them
    for (uint32_t i = 0; i < io_num.n_output; i++) {
        ret = rknn3_mem_sync(ctx, outputs[i].mem, RKNN3_MEMORY_SYNC_FROM_DEVICE);
        if (ret != RKNN3_SUCCESS) {
            printf("mem_sync output %d failed! ret=%d\n", i, ret);
            goto out;
        }
    }

    // ... Read outputs[i].mem->virt_addr and post-process ...

out:
    // 9. Release resources
    for (uint32_t i = 0; i < io_num.n_input; i++) {
        if (inputs[i].mem) rknn3_destroy_mem(ctx, inputs[i].mem);
    }
    for (uint32_t i = 0; i < io_num.n_output; i++) {
        if (outputs[i].mem) rknn3_destroy_mem(ctx, outputs[i].mem);
    }
    if (ctx) rknn3_destroy(ctx);
    return 0;
}

```

Note: For complete example code of more models, refer to rknn3-model-zoo (such as `examples/yolov6/cpp/`).

3.2.7 Debugging Methods

If accuracy or performance issues arise during on-device deployment, use the following debugging tools for troubleshooting:

Exporting intermediate features

`rknn3_dump_features()` exports the model intermediate features layer by layer to a specified directory as `.npy` files, making it easy to compare with a floating-point model layer by layer and analyze accuracy issues. When exporting, specify the core mask (`core_mask`) and export range (`dump_selector` ; for example, `"*"` or `"all"` to export all operators):

```
ret = rknn3_dump_features(ctx, inputs, n_inputs, NULL, 0,
                          "./dump/", 0, "all");
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_dump_features failed! ret=%d\n", ret);
}
```

Per-layer operator performance profiling

`rknn3_profile_ops()` prints the per-layer operator performance information, including statistics such as latency, cycles, and bandwidth, and outputs a summary table. `log_level` controls the output verbosity: 0 prints only the latency summary, 1 adds the per-operator latency table, and 2 further adds cycle and bandwidth information:

```
ret = rknn3_profile_ops(ctx, inputs, n_inputs, NULL, 0, 2);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_profile_ops failed! ret=%d\n", ret);
}
```

Troubleshooting common issues

1. **Memory alignment:** Use `aligned_size` instead of `n_elems * sizeof(dtype)` when allocating memory; otherwise, out-of-bounds memory access may occur.
2. **Core ID:** Use `attr->core_id` when creating memory to ensure the memory is allocated on the correct core.
3. **Cache synchronization:** When using cacheable memory, invoke `rknn3_mem_sync()` correctly; otherwise, data inconsistency may occur.
4. **Layout conversion:** The `NC1HWC2` layout requires special handling. Note that FP16 and INT8 have different sub-channel counts (see 3.1.2).
5. **Quantization parameters:** INT8 models require `qnt_info.scale` and `zero_point` for quantization/dequantization.
6. **core_mask consistency:** The runtime `core_mask` must match the configuration used during model conversion; otherwise, an error is reported.

3.3 LLM and MLLM Session Deployment

This section describes how to deploy LLM / MLLM models with the RKNN3 Runtime Session API, covering the session lifecycle, input and callbacks, inference execution, chat templates,

multimodal input, and more. The Session API suits LLM / MLLM deployment scenarios that require advanced features such as session management, KVCache, and sampling control.

3.3.1 Complete Invocation Flow

The typical invocation sequence for LLM / MLLM Session deployment is: initialize context -> load model -> initialize model -> initialize Session -> set callbacks -> prepare input -> execute inference -> process results -> destroy Session -> destroy context. The overall flow is shown in the figure below:

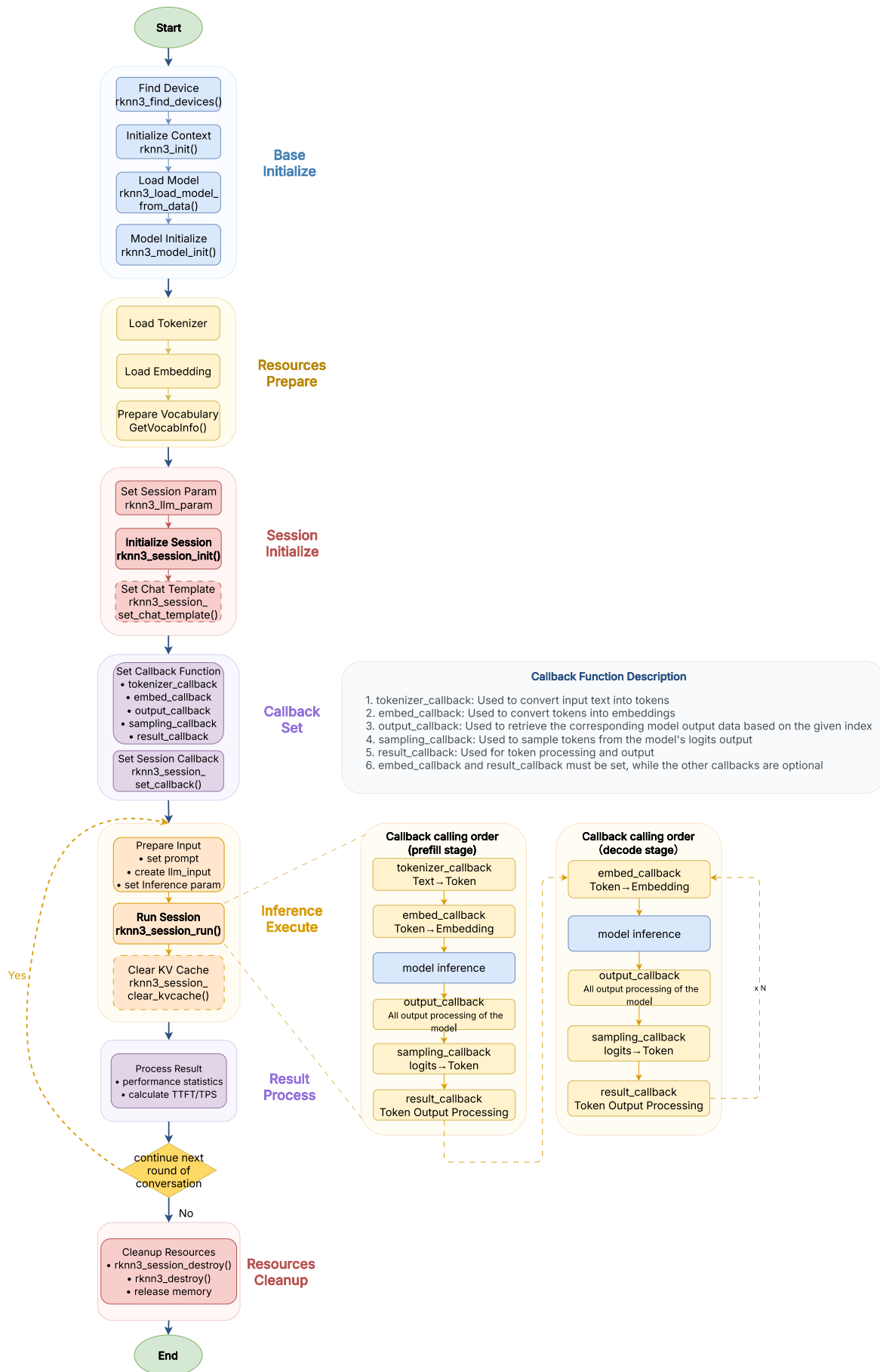


Figure 3-6 LLM model deployment flow based on Session management

The following subsections describe the operations and precautions for each step in this flow.

3.3.2 Session Lifecycle

After the model is loaded and initialized (`rknn3_init()` -> `rknn3_load_model_from_path()` -> `rknn3_model_init()`), create a session through `rknn3_session_init()` . When creating the session, configure the `rknn3_llm_param` structure, which contains the following key parameters:

- `logits_name` : The name of the inference output node.
- `max_context_len` : The maximum context length (in tokens).
- `sampling_param` : The sampling parameters, which control the randomness and quality of generation, including:
 - `temperature` : The sampling temperature. A larger value produces more random generation (higher diversity); a smaller value produces more deterministic generation (favoring high-probability tokens).
 - `top_k` : Top-K sampling. Samples only from the K highest-probability tokens. A smaller value produces more conservative generation.
 - `top_p` : Top-P sampling. Samples from the smallest set of tokens whose cumulative probability reaches P, dynamically adjusting the candidate range.
- `repeat_penalty` : The repeat penalty. Penalizes already generated tokens to reduce the probability of generating repeated content.
- `frequency_penalty` : The frequency penalty. Applies an additional penalty to high-frequency tokens.
- `presence_penalty` : The presence penalty. Penalizes tokens that have already appeared, encouraging the generation of new content.
- `vocab_info` : The vocabulary information, usually obtained from the tokenizer's `GetVocabInfo()` , including:
 - `vocab_size` : The vocabulary size.
- `special_bos_id` : The list of special beginning-of-sequence (BOS) token IDs; `n_special_bos_id` is the count.
- `special_eos_id` : The list of special end-of-sequence (EOS) token IDs; `n_special_eos_id` is the count. It controls when inference ends.
- `linefeed_id` : The token ID of the line feed.
- `ignore_eos_token` : Whether to ignore the EOS token during generation. When set to True, generation does not stop even if an end token is encountered.

```
rknn3_llm_param params;
memset(&params, 0, sizeof(rknn3_llm_param));

params.logits_name = "logits";
params.max_context_len = 1024;
params.sampling_param.temperature = 1.0f;
params.sampling_param.top_k = 1;
params.sampling_param.top_p = 0.9;
params.sampling_param.repeat_penalty = 1.1f;
params.vocab_info.vocab_size = vocab_info.vocab_size;
params.vocab_info.n_special_eos_id = vocab_info.n_special_eos_id;
memcpy(params.vocab_info.special_eos_id, vocab_info.special_eos_id,
sizeof(vocab_info.special_eos_id));

rknn3_session *session = rknn3_session_init(ctx, &params, 1); /* The third
parameter n_params is the number of parameter sets */
if (session == NULL) {
    printf("rknn3_session_init failed!\n");
    goto out;
}
```

After the session is used, destroy it through `rknn3_session_destroy()` to release session-related resources (KVCache, etc.). Do not use the session pointer after destruction:

```
ret = rknn3_session_destroy(session);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_session_destroy failed! ret=%d\n", ret);
}
```

Note: One context can create multiple sessions. The sessions share model weights and internal memory, but their KVCache and LoRA enable states are independent of each other. Sessions cannot execute concurrently; only one session can run at a time.

3.3.3 Input and Callbacks

LLM inference handles tokenization, embedding, sampling, result output, and other steps through the callback mechanism. After creating a session, register the callback functions through `rknn3_session_set_callback()`. The `RKLLMCallback` structure contains the following callbacks:

- **result_callback** : The result callback. Triggered each time a token is generated, and used to process the model output (for example, converting tokens to text output). Typically used with the tokenizer's `TokenToPiece()` or `Decode()`.
- **tokenizer_callback** (optional): The tokenizer callback. Converts input text into a token id sequence. Typically implemented by invoking the tokenizer's `Tokenize()`.
- **embed_callback** (optional): The embedding callback. Converts token ids into embedding vectors. Typically looked up by token id from the `embed.bin` file.
- **sampling_callback** (optional): The sampling callback. Samples token ids from the logits output by the model. If not set, the Runtime uses the default sampling strategy (configured

through `sampling_param`).

- **output_callback** (optional): The output callback. Obtains the model output tensors. Use it with the `output_tensors` and `n_output_tensors` fields to specify the output tensor array. It suits scenarios that directly process output tensors (for example, using the output as input to other models).
- **input_callback** (optional): The input callback. Provides custom input tensors. Use it with the `input_tensors_index` field to specify the input tensor indexes. It suits scenarios that directly pass in preprocessed tensors.

```
RKLLMCallback callback;
memset(&callback, 0, sizeof(RKLLMCallback));

callback.result_callback    = result_callback;
callback.result_userdata    = tokenizer;
callback.tokenizer_callback = tokenizer_callback;
callback.tokenizer_userdata = tokenizer;
callback.embed_callback     = embed_callback;
callback.embed_userdata     = &embedding_info;
callback.sampling_callback  = sampling_callback;
callback.sampling_userdata  = &embedding_info;

ret = rknn3_session_set_callback(session, &callback);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_session_set_callback failed! ret=%d\n", ret);
    goto out;
}
```

Note: For implementations of `tokenizer_callback` and `embed_callback`, refer to 2.2.4 Model and Tokenizer Adaptation. If Tie Word mode is used (see 4.2.8 Tie Word Embedding), `embed_callback` does not need to be set.

3.3.4 Inference Execution and Interruption

Executing inference

After the session is created and the callbacks are set, execute inference through `rknn3_session_run()`. Prepare the `rknn3_llm_input` input structure and the `rknn3_llm_infer_param` inference parameters. Specify the input type through `input_type`, commonly `RKNN3_LLM_INPUT_PROMPT` (text input). The inference parameters include `keep_history` (whether to keep the history context) and `max_new_tokens` (the maximum number of tokens to generate):

```

char *prompt = "Please explain the basic concept of relativity";
rknn3_llm_infer_param param = {.keep_history = 0, .max_new_tokens = 128};
rknn3_llm_tensor input_tensor = {.name = NULL, .prompt = prompt,
                                  .embed = NULL, .tokens = NULL,
                                  .n_tokens = 0, .enable_thinking = false};

rknn3_llm_input input;
input.input_type = RKNN3_LLM_INPUT_PROMPT;
input.llm_input = input_tensor;

rknn3_llm_input inputs[1] = {input};
ret = rknn3_session_run(session, inputs, 1, &param);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_session_run failed! ret=%d\n", ret);
    goto out;
}

```

During inference, each generated token is returned through the `result_callback` callback, and users process the output in the callback (for example, printing text). Inference continues until `max_new_tokens` is reached or an EOS token is encountered.

The Runtime also provides the asynchronous inference interface `rknn3_session_run_async()`. Its usage is the same as the synchronous interface, but it does not block the current thread.

Note: For general models, `rknn3_run_async()` / `rknn3_wait()` are not implemented in the current version (see 3.1.4); for LLM Sessions, `rknn3_session_run_async()` is available.

Interrupting inference

During inference, interrupt the current session inference through `rknn3_session_stop()`. The Runtime stops generation and returns:

```

ret = rknn3_session_stop(session);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_session_stop failed! ret=%d\n", ret);
}

```

3.3.5 Chat Template

Applicable Scenarios

Different LLMs use different conversation formats (such as ChatML / Llama-3 / Qwen). Models learn specific system prompts, role markers, and delimiters during training. Feeding raw user text directly into the model degrades generation quality or produces irrelevant answers due to missing format markers. The Chat Template wraps the user input with the conversation format required by the model so that the input meets the model's expectations.

Usage

Set the chat template through `rknn3_session_set_chat_template()`. It consists of three parts:

- `system_prompt` : The system prompt. It defines the model's role and behavioral context and is always prepended at the beginning of each conversation round.

- `prompt_prefix` : The user input prefix. It marks the beginning of a user message (such as `<|im_start|>user\n`).
- `prompt_postfix` : The user input postfix. It marks the end of a user message and prompts the model to generate (such as `<|im_end|>\n<|im_start|>assistant\n`).

Once set, whenever a PROMPT input is passed to `rknn3_session_run()`, the Runtime automatically concatenates `system_prompt + prompt_prefix + user input + prompt_postfix` into the complete conversation and feeds it to the model. Invoke this interface after `rknn3_session_init()` and before `rknn3_session_run()`. Usually it only needs to be set once.

```
const char* system_prompt = "<|im_start|>system\nYou are a helpful assistant.
<|im_end|>\n";
const char* prompt_prefix = "<|im_start|>user\n";
const char* prompt_postfix = "<|im_end|>\n<|im_start|>assistant\n";
ret = rknn3_session_set_chat_template(session, system_prompt, prompt_prefix,
prompt_postfix);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_session_set_chat_template failed! ret=%d\n", ret);
    goto out;
}
```

Note: The template strings must strictly match the conversation format used during model training. They can be derived from the `chat_template` field in the model's `tokenizer_config.json`. An incorrectly set template causes chaotic or repetitive model output. If a chat template was already configured through the config file during model conversion (see 2.2.2), this setting overrides it at runtime; if not set, the conversion-time configuration is used.

3.3.6 Multimodal Input

Applicable Scenarios

In addition to text, an MLLM (multimodal large language model) also accepts inputs such as images, audio, and video. The typical flow is: first use the corresponding encoder model (vision/audio/video) to encode the raw modality data into embedding vectors (embeds), then feed the text prompt and the modality embeddings together into the LLM, which fuses the multimodal information to generate an answer. RKNN3 supports this flow through the `RKNN3_LLM_INPUT_MULTIMODAL` input type.

Input Structure

Multimodal input uses the `rknn3_llm_multimodal_tensor` structure (`rknn3_llm_input.multimodal_input`). Its members fall into two parts:

- **Text part:** Provide either `prompt` (text prompt) or `tokens` (token id array); if both are provided, `prompt` is used for inference by default. `enable_thinking` controls whether thinking mode is enabled.
- **Modality sub-structures:** The three sub-structures `image`, `audio`, and `video` are structurally symmetric. Each contains the `*_embed` embedding pointer from the

corresponding encoder model output, the token count (`n*_tokens`), the modality count (`n_image / n_audio / n_video`), and the placeholder tags in the text (`*_start / *_end / *_content` , which must match the special tokens of the model tokenizer). Populate one or more of them based on the actual input modalities. For the complete member definitions, see the *RKNN3_Runtime_C_API_Reference*.

Usage (VLM example)

In VLM (vision-language model) deployment, the vision model and the LLM model are two separate `.rknn` files that must be initialized and run separately. The vision model uses the general RKNN3 C API (see 3.2) to infer the image embeddings. On the LLM side, set the input type to `RKNN3_LLM_INPUT_MULTIMODAL` , populate the `image` sub-structure and the text prompt, and then invoke `rknn3_session_run()` :

```
/* img_embeds is the output of vision model inference */
rknn3_llm_multimodal_tensor tensor;
memset(&tensor, 0, sizeof(rknn3_llm_multimodal_tensor));
tensor.name = "input_embeds";
tensor.prompt = prompt; // Text prompt

tensor.image.image_embed = img_embeds; // Image embeddings output by the
vision model
if (rknn_app_ctx.vision.embeds_ndims == 2) {
    tensor.image.n_image_tokens = rknn_app_ctx.vision.embeds_shape[0];
    tensor.image.n_image = 1;
} else {
    tensor.image.n_image_tokens = rknn_app_ctx.vision.embeds_shape[1];
    tensor.image.n_image = rknn_app_ctx.vision.embeds_shape[0];
}
tensor.image.image_width = rknn_app_ctx.vision.model_width;
tensor.image.image_height = rknn_app_ctx.vision.model_height;
tensor.image.image_start = "<|vision_start|>"; // Must match the model
tokenizer
tensor.image.image_end = "<|vision_end|>";
tensor.image.image_content = "<|image_pad|>";
tensor.enable_thinking = false;

rknn3_llm_input inputs[1];
memset(inputs, 0, sizeof(inputs));
inputs[0].input_type = RKNN3_LLM_INPUT_MULTIMODAL;
inputs[0].multimodal_input = tensor;

ret = rknn3_session_run(session, inputs, 1, &param);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_session_run failed! ret=%d\n", ret);
    goto out;
}
```

Note: The audio / video modalities are populated exactly like the image modality; simply replace the `image_*` members with the corresponding `audio_*` / `video_*` members. For complete multimodal deployment examples (vision + LLM integration), refer to the Qwen2.5-VL / FastVLM example projects in `rknn3-model-zoo` .

3.3.7 Complete Example

This section provides a complete invocation example of LLM Session deployment, chaining together the lifecycle, callback, and inference steps described above. For the complete project code, refer to the SDK example directory `rknn3-runtime/examples/rknn3_session_test_demo/`.

The complete example covers the following steps:

1. **Initialize the context and load the model:** `rknn3_init()` -> `rknn3_load_model_from_path()` -> `rknn3_model_init()`.
2. **Create the session:** Configure `rknn3_llm_param` (`logits_name` / `max_context_len` / `sampling_param` / `vocab_info`), and invoke `rknn3_session_init()`.
3. **Set callbacks:** Populate `RKLLMCallback` (`result_callback` / `tokenizer_callback` / `embed_callback` / `sampling_callback`), and invoke `rknn3_session_set_callback()`.
4. **Set the chat template** (optional): Invoke `rknn3_session_set_chat_template()`.
5. **Execute inference:** Populate `rknn3_llm_input` (`input_type=PROMPT`) and `rknn3_llm_infer_param`, and invoke `rknn3_session_run()`. Results are returned in streaming mode through `result_callback`.
6. **Destroy resources:** `rknn3_session_destroy()` -> `rknn3_destroy()`.

```

#include "rknn3_runtime.h"
#include "Tokenizer.h"

/* Result callback: invoked after each token is generated; handle different states
by LLMCallState */
static int result_callback(void* userdata, RKLLMResult* result, LLMCallState state)
{
    Tokenizer* tokenizer = (Tokenizer*)userdata;

    if (state == RKLLM_RUN_NORMAL) {
        std::string piece;
        if (result->num_tokens == 1) {
            piece = tokenizer->TokenToPiece(result->token_ids[0]);
        } else {
            piece = tokenizer->Decode(result->token_ids, result->num_tokens);
        }
        printf("%s", piece.c_str());
    } else if (state == RKLLM_RUN_FINISH) {
        printf("\n[generation finished]\n");
    } else if (state == RKLLM_RUN_ERROR) {
        printf("\n[inference error]\n");
    }
    fflush(stdout);
    return 0;
}

/* Tokenizer callback: converts text into a token id sequence */
int tokenizer_callback(void *userdata, const char *text, int32_t text_len, int32_t
*tokens, int32_t n_tokens_max)
{
    int n_tokens = 0;
    Tokenizer *tokenizer = (Tokenizer *)userdata;
    n_tokens = tokenizer->Tokenize(text, text_len, tokens, n_tokens_max);

    if (n_tokens <= 0)
    {
        printf("tokenizer failed for %s\n", text);
        return n_tokens;
    }
    return n_tokens;
}

/* Embedding callback: looks up embeddings in embed.bin by token id */
int embed_callback(void* userdata, int32_t* tokens, uint64_t num_tokens, void*
embed, uint64_t len)
{
    struct embedding_info* embed_info = (struct embedding_info*)userdata;

    if (len != num_tokens * embed_info->embedding_dim * sizeof(float16)) {
        printf("invalid embed buffer\n");
        return -1;
    }

    for (int n = 0; n < num_tokens; n++) {
        memcpy((unsigned char*)embed +
            n * embed_info->embedding_dim * sizeof(float16),
            embed_info->embedding_data + tokens[n] * embed_info->embedding_dim,
            embed_info->embedding_dim * sizeof(float16));
    }
    return 0;
}

```

```

}

int main(int argc, char** argv)
{
    int ret = 0;
    rknn3_context ctx = NULL;
    rknn3_session* session = NULL;

    /* Creation/loading of tokenizer and embedding_info is omitted; see the SDK
    example for the complete implementation */
    Tokenizer* tokenizer = NULL;
    struct embedding_info embedding_info;

    /* 1. Initialize the context and load the model */
    ret = rknn3_init(&ctx, NULL);
    if (ret != RKNN3_SUCCESS) {
        printf("rknn3_init failed! ret=%d\n", ret);
        goto out;
    }
    ret = rknn3_load_model_from_path(ctx, "model.rknn", NULL);
    if (ret != RKNN3_SUCCESS) {
        printf("rknn3_load_model_from_path failed! ret=%d\n", ret);
        goto out;
    }
    rknn3_config config = {0};
    config.run_core_mask = 0xff; /* Configure according to the actual core count
    */
    ret = rknn3_model_init(ctx, &config);
    if (ret != RKNN3_SUCCESS) {
        printf("rknn3_model_init failed! ret=%d\n", ret);
        goto out;
    }

    /* 2. Create the session */
    rknn3_llm_param params;
    memset(&params, 0, sizeof(rknn3_llm_param));
    params.logits_name = "logits";
    params.max_context_len = 1024;
    params.sampling_param.temperature = 1.0f;
    params.sampling_param.top_k = 1;
    params.sampling_param.top_p = 0.9;
    params.sampling_param.repeat_penalty = 1.1f;
    /* vocab_info is populated by tokenizer->GetVocabInfo(); omitted here */

    session = rknn3_session_init(ctx, &params, 1);
    if (session == NULL) {
        printf("rknn3_session_init failed!\n");
        goto out;
    }

    /* 3. Set callbacks */
    RKLLMCallback callback;
    memset(&callback, 0, sizeof(RKLLMCallback));
    callback.result_callback = result_callback;
    callback.result_userdata = tokenizer;
    callback.tokenizer_callback = tokenizer_callback;
    callback.tokenizer_userdata = tokenizer;
    callback.embed_callback = embed_callback;
    callback.embed_userdata = &embedding_info;
    ret = rknn3_session_set_callback(session, &callback);
    if (ret != RKNN3_SUCCESS) {

```

```

        printf("rknn3_session_set_callback failed! ret=%d\n", ret);
        goto out;
    }

    /* 4. Set the chat template */
    const char* system_prompt = "<|im_start|>system\nYou are a helpful assistant.
<|im_end|>\n";
    const char* prompt_prefix = "<|im_start|>user\n";
    const char* prompt_postfix = "<|im_end|>\n<|im_start|>assistant\n";
    ret = rknn3_session_set_chat_template(session, system_prompt, prompt_prefix,
    prompt_postfix);
    if (ret != RKNN3_SUCCESS) {
        printf("rknn3_session_set_chat_template failed! ret=%d\n", ret);
        goto out;
    }

    /* 5. Run inference */
    char* prompt = "Please explain the basic concept of relativity";
    rknn3_llm_infer_param param = {.keep_history = 0, .max_new_tokens = 128};
    rknn3_llm_tensor input_tensor = {.name = NULL, .prompt = prompt,
        .embed = NULL, .tokens = NULL,
        .n_tokens = 0, .enable_thinking = false};

    rknn3_llm_input inputs[1];
    inputs[0].input_type = RKNN3_LLM_INPUT_PROMPT;
    inputs[0].llm_input = input_tensor;

    ret = rknn3_session_run(session, inputs, 1, &param);
    if (ret != RKNN3_SUCCESS) {
        printf("rknn3_session_run failed! ret=%d\n", ret);
        goto out;
    }

out:
    /* 6. Release resources */
    if (session) {
        rknn3_session_destroy(session);
    }
    rknn3_destroy(ctx);
    return ret;
}

```

Note: The example above omits preparation steps such as tokenizer initialization, the `GetVocabInfo()` invocation, and `embed.bin` loading. For the complete implementation, refer to the SDK example `rknn3-runtime/examples/rknn3_session_test_demo/`. The `state` parameter of `result_callback` is an `LLMCallState` enum. Common values include `RKLLM_RUN_NORMAL` (normal generation), `RKLLM_RUN_FINISH` (inference finished), `RKLLM_RUN_ERROR` (error), and `RKLLM_RUN_MAX_NEW_TOKEN_REACHED` (maximum generation count reached). For the complete state definitions, see the *RKNN3_Runtime_C_API_Reference*.

3.4 Toolkit Lite Python Deployment

Prerequisites: Before using Toolkit Lite, convert the model to `.rknn` + `.weight` with RKNN3-Toolkit on the PC side.

Positioning and Applicable Scenarios

RKNN3-Toolkit Lite is the on-device Python inference interface. It wraps the RKNN3 Runtime (C API) in Python and provides a high-level inference experience consistent with the traditional `RKNN-Toolkit Lite`, while supporting LLM / MLLM capabilities such as Session, Callback, and KVCache. Compared with the C API:

Item	Toolkit Lite (Python)	Runtime (C API)
Development efficiency	High; rapid Python iteration	Low; requires compilation and deployment
Runtime performance	Slightly lower (wrapping overhead)	Optimal
Typical scenarios	Prototype validation, algorithm debugging, application development	Production deployment, maximum performance
Supported platforms	RK1820 / RK1828 / RK3572	Same as left

Toolkit Lite suits rapid on-device Python validation and upper-layer application development. When performance and resources are critical and production is planned, migrate to the C API.

3.4.1 Python Runtime Environment

Toolkit Lite runs on the device side. The currently supported platforms are as follows:

Item	Supported range
Hardware platform	RK1820 / RK1828 / RK3572
Operating system	Debian 10 / 11 / 12 (aarch64)
Python	3.9 / 3.10 / 3.11 / 3.12
Main Python dependencies	<code>numpy</code> , <code>transformers</code> , <code>jinja2</code>

For the detailed version matrix, installation steps, and import methods, see Chapter 1 of the *RKNN3_Toolkit_Lite_Python_API_Reference*.

3.4.2 RKNN3Lite Object Initialization and Release

`RKNN3Lite` is the main Python entry point of Toolkit Lite, and all operations revolve around this object. Construct the object before use and release it after use. The entire lifecycle follows the order: create object -> load model -> initialize runtime -> inference -> release.

Constructing the object

Constructing `RKNN3Lite()` only builds the Python-side wrapper. It does not immediately create a Runtime or Session; the underlying resources are created by the subsequent `init_runtime()` / `init_llm_session()`. Main parameters:

- `llm_mode` : Whether to use LLM / MLLM mode. Defaults to `False`. Set it to `True` for LLM / MLLM to initialize the Session-related code; leave it unset for general models.
- `verbose` : Whether to print detailed logs to the terminal. Defaults to `False`.
- `verbose_file` : The log file path. When specified, logs are also written to this file. Defaults to `None`.

```

from rknn3lite.api import RKNN3Lite

# General model
rknn_lite = RKNN3Lite(verbose=True)
# LLM / MLLM model
rknn_lite = RKNN3Lite(llm_mode=True, verbose=True, verbose_file='./inference.log')

```

Releasing the object

`release()` releases resources. In LLM / MLLM multi-Session scenarios, specify `session_index` to release only one Session while keeping the shared Runtime; if unspecified, all Sessions and the Runtime are released:

```

# Release all resources
rknn_lite.release()
# Release only the specified LLM session (keep the Runtime for other sessions)
rknn_lite.release(session_index=0)

```

Note: Invoke `release()` after all inference is complete, and invoke it exactly once for each `RKNN3Lite` object to avoid resource leaks.

3.4.3 Loading and Initializing the Model

Loading the model

`load_rknn()` loads the converted `.rknn` and `.weight` files. The two paths must be paired; otherwise, loading fails:

```

ret = rknn_lite.load_rknn(model_path='mobilenetv2-12.rknn',
                          weight_path='mobilenetv2-12.weight')
if ret != 0:
    raise RuntimeError(f"load_rknn failed, ret={ret}")

```

Initializing the runtime

`init_runtime()` initializes the runtime environment. The parameters differ between general models and LLM / MLLM models:

- **General models:** Specify `target`, `core_mask`, and `device_id` (obtained through `get_devices_id()` for multiple devices).
- **LLM / MLLM:** Additionally pass sampling and other parameters through `llm_args`, and pass the callback object through `llm_callback`.

```
# General model
device_id = rknn_lite.get_devices_id()
ret = rknn_lite.init_runtime(target='rk1820', core_mask=0x01,
device_id=device_id[0])

# LLM / MLLM
ret = rknn_lite.init_runtime(target='rk1820', core_mask=0xff,
                             llm_args=ARGS, llm_callback=callback)
```

Initialization key points

- `core_mask` must match the number of cores configured in `rknn.config(core_num=...)` during model conversion; otherwise, an error is reported. RK1820 / RK1828 support 0x1~0xff (1-8 cores), and RK3572 supports only 0x1 (single core).
- For LLM / MLLM, the recommended step-by-step flow is: invoke `init_runtime` first, then query the model configuration with `rknn3_query(RKNN3_QUERY_LLM_CONFIG)`, and finally invoke `init_llm_session()`. This reads the model's actual `vocab_size`, `embedding_dim`, `max_ctx_len`, KVCache configuration, and more before creating the Session, avoiding hardcoding. If the application has already determined `llm_args` and the Callback, they can also be passed directly at `init_runtime()` time, and the initialization flow creates the first Session.
- One LLM object can create multiple Sessions on the same Runtime (managed by `session_index`). Each Session independently maintains its conversation state, KVCache, and per-Session configuration, but only one Session can run at a time.
- The sampling parameters of `llm_args` (`top_k` / `top_p` / `temperature` / `repeat_penalty` / `max_new_tokens`, etc.) correspond in meaning to the C API's `rknn3_sampling_params` / `rknn3_llm_param`. See 3.3.2.

3.4.4 Tensor and Memory Management

Toolkit Lite wraps tensor and memory management at the Python layer. By ownership, there are three categories:

- **High-level inputs and outputs:** numpy ndarray inputs and outputs for traditional models. The high-level interfaces automatically handle data preparation and layout conversion, so users do not need to care about the underlying memory.
- **Tensor Memory created by the Runtime:** Device memory allocated through interfaces such as `create_mem()`. The Runtime manages its lifecycle, and users synchronize the data between the CPU and the device through `mem_sync()`.
- **User-provided model runtime memory:** Pre-allocated user memory bound to the model through `set_internal_mem()` / `set_weight_mem()`. It suits advanced scenarios such as memory reuse and cross-process sharing.

Organizing multimodal embeddings

In MLLM deployment, the outputs of each modality encoder model (vision/audio/video) must be converted to the FP16 format expected by Toolkit Lite and then passed into the LLM through the corresponding dedicated type. The three modality types are structurally symmetric. Each

contains the embedding pointer from the encoder model output, the token count, the modality count, and the text placeholder tags:

Modality	Type	Embedding pointer	Token count	Modality count	Placeholder tags
Image	RKNN3Image	image_embed	n_image_tokens	n_image	image_start / image_end / image_content
Audio	RKNN3Audio	audio_embed	n_audio_tokens	n_audio	audio_start / audio_end / audio_content
Video	RKNN3Video	video_embed	n_frame_tokens / n_frame_per_video	n_video	video_start / video_end / video_content

Taking an image as an example, convert the vision model output to FP16, populate `RKNN3Image`, and pass it into the LLM:

```
import numpy as np
import ctypes
from rknn3lite.api import RKNN3Image, Float16

# Convert the vision model output to FP16
outputs = np.float16(np.expand_dims(rknn_vision.inference(inputs=[feature])[0], 0))

llm_input = RKNN3Image()
llm_input.image_embed = outputs.ctypes.data_as(ctypes.POINTER(Float16))
llm_input.n_image_tokens = outputs.shape[1]
llm_input.n_image = outputs.shape[0]
llm_input.image_width = 392
llm_input.image_height = 392
llm_input.image_start = "<|vision_start|>".encode('utf-8')
llm_input.image_end = "<|vision_end|>".encode('utf-8')
llm_input.image_content = "<|image_pad|>".encode('utf-8')

ret, perf = rknn_llm.session_run(inputs=[llm_input], prompt="<image>Describe the content of this image")
```

Audio and video are populated exactly like images: convert the encoder model output to FP16 and populate the members of the corresponding type (`audio_embed` / `video_embed` , etc.). Multiple modalities can be passed in at the same time; construct the corresponding `RKNN3Image` / `RKNN3Audio` / `RKNN3Video` lists based on the actual inputs and pass them to `session_run(inputs=...)` .

Note: The placeholder tags of each modality must match the special tokens of the model tokenizer. The underlying structure corresponds to the C API's `rknn3_llm_multimodal_tensor` (see 3.3.6). For the complete member definitions, see section 5.5.25 of the *RKNN3_Runtime_C_API_Reference*.

3.4.5 Executing Inference

General models

Use `inference()` for synchronous inference. Both inputs and outputs are numpy ndarrays:

```

outputs = rknn_lite.inference(inputs=[img])
if outputs is None:
    raise RuntimeError("inference failed")

```

Asynchronous `inference_async()` -> `wait()` -> `get_outputs()` is also supported. It suits scenarios that execute other tasks while waiting for the results.

LLM / MLLM

Use `session_run()` for inference. It supports streaming output, and the results are returned asynchronously through `result_callback`:

```

prompt = "Describe how LLM models work."
ret, [n_decode_tokens, n_prefill_tokens, llm_start_time, llm_end_time] = \
    rknn_llm.session_run(prompt=prompt)
if ret != 0:
    raise RuntimeError(f"session_run failed, ret={ret}")

```

- `prompt` (text) and `embeds` (embedding vectors, 3D: batch × seq × hidden) are mutually exclusive; the latter skips tokenization.
- Multimodal input passes dedicated types such as `RKNN3Image` / `RKNN3Audio` / `RKNN3Video` / `RKNN3AuxTensor` through `inputs`, populated in correspondence with the C API's `rknn3_llm_multimodal_tensor`.
- The return value contains performance statistics `[n_decode_tokens, n_prefill_tokens, llm_start_time, llm_end_time]`, which can be used for performance analysis.
- Asynchronous inference uses `session_run_async()`, combined with `session_stop()` / `session_pause()` / `session_resume()` to interrupt, pause, and resume.

3.4.6 Complete Example

This section provides two minimal runnable examples, one for the general model flow and one for the LLM flow. For the complete code, see the `rknn3-toolkit-lite/examples` directory in the SDK.

General model example:

Using MobileNetV2 image classification as an example, this demonstrates the complete flow: create object -> load model -> initialize runtime -> query tensor attributes -> inference -> post-processing -> release:

```

from rknn3lite.api import RKNN3Lite

rknn_lite = RKNN3Lite()

# 1. Load the model
ret = rknn_lite.load_rknn(model_path='mobilenetv2-12.rknn',
                          weight_path='mobilenetv2-12.weight')

# 2. Initialize the runtime (core_mask must match the one used for conversion)
device_id = rknn_lite.get_devices_id()
ret = rknn_lite.init_runtime(target='rk1820', core_mask=0x01,
                             device_id=device_id[0])

# 3. Query input and output tensor attributes (shape/dtype/layout, etc.)
for attr in rknn_lite.get_inputs_tensor_attr():
    print("input:", attr)
for attr in rknn_lite.get_outputs_tensor_attr():
    print("output:", attr)

# 4. Run inference (input is a pre-processed numpy ndarray)
outputs = rknn_lite.inference(inputs=[img])

# 5. Post-process (Top5 classification) and release
show_top5(outputs[0])
rknn_lite.release()

```

LLM example:

Using Qwen2.5-0.5B Instruct as an example, this demonstrates the complete flow in LLM mode:
 create object -> load model -> construct callbacks -> initialize runtime -> set chat template ->
 multi-round inference -> release:

```

import ctypes, numpy as np
from transformers import AutoTokenizer
from rknn3lite.api import (RKNN3Lite, RKLLMCallback,
                           LLMResultCallback, LLMGetEmbedCallback,
                           LLMTOKENIZERCallback)

# Preparation: load the tokenizer and embed.bin (vocabulary vectors, shape =
vocab_size × embedding_dim)
tokenizer = AutoTokenizer.from_pretrained(TOKENIZER_PATH)
embeds_data = np.fromfile(EMBED_PATH, dtype=np.float16).reshape(VOCAB_SIZE, -1)

# 1. Create the LLM-mode object and load the model
rknn = RKNN3Lite(llm_mode=True, verbose=True)
rknn.load_rknn(RKNN_MODEL, WEIGHT_MODEL)

# 2. Build callbacks: result (streaming output), tokenizer (text to token), embed
(vocabulary lookup by token)
callback = RKLLMCallback()
callback.result_callback = LLMResultCallback(result_callback)
callback.tokenizer_callback = LLMTOKENIZERCallback(tokenizer_callback)
callback.embed_callback = LLMGetEmbedCallback(embed_callback)
# userdata passes Python object pointers via ctypes.py_object + ctypes.cast
userdata = ctypes.py_object(tokenizer)
callback.tokenizer_userdata = ctypes.cast(ctypes.pointer(userdata),
ctypes.c_void_p)
userdata = ctypes.py_object(embeds_data)
callback.embed_userdata = ctypes.cast(ctypes.pointer(userdata), ctypes.c_void_p)

# 3. Initialize the runtime (llm_args contains sampling parameters; llm_callback
passes the callback object)
ARGS = [{"top_k": 1,
        "top_p": 0.8,
        "temperature": 0.8,
        "repeat_penalty": 1.1,
        "vocab_size": VOCAB_SIZE,
        "special_eos_id": 151645,
        "max_context_len": 1024,
        "keep_history": 0,
        "max_new_tokens": 1024}]
rknn.init_runtime(target='rk1820', core_mask=0xff, llm_args=ARGS,
llm_callback=callback)

# 4. Set the chat template
rknn.set_chat_template(system_prompt, prompt_prefix, prompt_postfix)

# 5. Multi-round inference; the return value contains performance statistics
for prompt in prompts:
    ret, [n_decode_tokens, n_prefill_tokens, llm_start_time, llm_end_time] = \
        rknn.session_run(prompt=prompt)
    printf_perf(first_token, n_decode_tokens, n_prefill_tokens, llm_start_time,
llm_end_time)

rknn.release()

```

Key points of the examples

- **Callback implementation:** `result_callback` handles branches by `LLMCallState` (`RKLLM_RUN_NORMAL` for normal generation, `RKLLM_RUN_FINISH` for completion,

`RKLLM_RUN_ERROR` for errors, `RKLLM_RUN_WAITING` for waiting for UTF-8 characters, `RKLLM_RUN_STOP` for stop, `RKLLM_RUN_MAX_NEW_TOKEN_REACHED` for reaching the maximum token count, `RKLLM_RUN_PAUSE` / `RKLLM_RUN_RESUME` for pause/resume). In the normal state, it accumulates tokens and prints them with incremental decoding, solving the issue of UTF-8 characters being truncated across tokens. `tokenizer_callback` invokes the HuggingFace tokenizer for tokenization; `embed_callback` looks up the vocabulary vectors from `embeds_data` by token id and copies them to the buffer.

- **userdata passing:** Python objects are wrapped with `ctypes.py_object`, cast to `c_void_p`, and passed into the callback. The callback restores and uses them, preventing them from being collected by the Python GC.
- **Performance statistics:** Combined with the first-token time, `[n_decode_tokens, n_prefill_tokens, llm_start_time, llm_end_time]` returned by `session_run` can be used to calculate the per-token latency and throughput (tokens/s) of the prefill and generate stages separately.
- **Multi-round conversation:** With `keep_history=1`, the Session automatically keeps the history KVCache for multi-round context; with `keep_history=0`, each round is independent.

Note: The specified RKNN model path must match the paired weight file path. Due to differences in model version, input image, or runtime environment, actual outputs may differ slightly from the examples. For the complete example code, see the `rknn3-toolkit-lite/examples` directory in the SDK.

3.5 RKLLM3 Server Integration

rkllm3-server is a lightweight LLM HTTP Server built on RKNPU3. It provides inference services compatible with the OpenAI API, with the underlying inference acceleration implemented by RKLLM3 Runtime (which wraps RKNN3 Runtime). This section describes its applicable scenarios, deployment methods, and client invocation.

3.5.1 Server Applicable Scenarios

rkllm3-server suits the following scenarios:

- **Service-oriented deployment:** Wraps LLM / MLLM models as an HTTP service that multiple clients invoke through the standard OpenAI API, integrating with the existing application ecosystem.
- **Multimodal inference:** Supports text, image, and audio inputs (video input is not supported yet), covering various model types such as LLMs, VLMs, speech models, and multimodal fusion models.
- **Rapid validation and integration:** No C/Python inference code development required. Invoke it through `curl`, the OpenAI SDK, or any HTTP client. It suits model quality validation and prototype integration.
- **Multiple models and multiple sessions:** Supports loading multiple models in a single process with concurrent sessions (up to 16). It suits scenarios such as multi-model routing and multi-user services.

Positioning difference from native interfaces: rkllm3-server trades some peak performance and flexibility (limited by HTTP and the OpenAI protocol) for ease of use and ecosystem compatibility. Use the C API or Toolkit Lite when peak performance, fine-grained control (such as KVCache/LoRA/Callback), or embedded integration is required.

3.5.2 Service Deployment Overview

rkllm3-server is deployed as an on-device executable, with the model, vocabulary, sampling configuration, and more specified through command-line parameters. The typical deployment flow is: prepare the model artifacts (`.rknn` / `.weight` / `tokenizer.gguf` / `embed.bin` , etc.) -> start the service process -> client access.

Startup examples

```
# LLM model (standard mode; embed file required)
./rkllm3-server -m qwen2.5-3b.rknn --vocab qwen2.5-3b.tokenizer.gguf \
--embed qwen2.5-3b.embed.bin --host 0.0.0.0 --port 8080 \
-c 768 --n_predict 512 --top-k 1 --top-p 0.8 --temp 0.8

# Multimodal model (LLM + Vision)
./rkllm3-server -m Qwen2.5-VL-3B-llm.rknn --model2 Qwen2.5-VL-3B-vision.rknn \
--vocab Qwen2.5-VL-3B-llm.tokenizer.gguf --embed Qwen2.5-VL-3B-llm.embed.bin \
--host 0.0.0.0 --port 8080 -c 768 --n_predict 512 \
--img-start "<|vision_start|>" --img-end "<|vision_end|>" --img-content "
<|image_pad|>" \
--img-width 392 --img-height 392

# Multimodal model (LLM + Vision + Audio)
./rkllm3-server -m Qwen2.5-Omni-3B-llm.rknn \
--model2 Qwen2.5-Omni-3B-vision.rknn --model3 Qwen2.5-Omni-3B-audio.rknn \
--vocab Qwen2.5-Omni-3B-llm.tokenizer.gguf --embed Qwen2.5-Omni-3B-llm.embed.bin \
--mel-filter mel_128_filters.txt --host 0.0.0.0 --port 8080 \
--img-start "<|vision_bos|>" --img-end "<|vision_eos|>" --img-content "<|IMAGE|>" \
--audio-start "<|audio_bos|>" --audio-end "<|audio_eos|>" --audio-content "
<|AUDIO|>"
```

Key parameter categories

Category	Common parameters	Description
Model paths	<code>-m</code> / <code>--weight</code> / <code>--model2</code> / <code>--model3</code>	LLM/vision/audio model and weight paths; for multimodal, use <code>--model2</code> / <code>--model3</code> to specify the vision/audio models
Vocabulary and embedding	<code>--vocab</code> / <code>--embed</code> / <code>--mel-filter</code>	tokenizer (<code>.gguf</code>), vocabulary embedding (<code>.bin</code>), audio mel filter
Context and generation length	<code>-c</code> / <code>--n_predict</code>	Context length (0 = loaded from model), number of tokens to generate (-1 = context size)
Sampling parameters	<code>--temp</code> / <code>--top-k</code> / <code>--top-p</code> / <code>-repeat-penalty</code> / <code>--presence-penalty</code> / <code>--frequency-penalty</code>	Same meaning as the C API sampling parameters; see 3.3.2

Category	Common parameters	Description
Multimodal placeholder tags	<code>--img-start / --img-end / --img-content</code> (and audio/video counterparts)	Must match the special tokens of the model tokenizer
Service configuration	<code>--host / --port / --timeout / --n-session</code>	Listen address and port, timeout, concurrent session count (up to 16)
NPU cores	<code>--core-mask / --core-mask2 / --core-mask3</code>	NPU core masks for the LLM/vision/audio models; RK182X supports 0x1~0xff, and RK3572 supports only 0x1
Advanced features	<code>--lora-weight / --tie-word-embedding / --checkpoint-* / --embedding</code>	LoRA, Tie Word, KVCache Checkpoint, embedding model (<code>--embedding</code> identifies an embedding model, which differs from <code>--embed</code> specifying the embed.bin path)

Note: The multimodal placeholder tags (such as `--img-start`) must strictly match the special tokens of the model tokenizer; otherwise, the model cannot correctly identify the positions of the modality inputs. `--core-mask` must match the `core_mask` used during model conversion. For the complete parameter list and the multi-model `--params-file` JSON configuration format, see the *RKLLM3_Server_User_Guide_and_API_Reference*.

Library-based integration

In addition to the executable, `rkllm3-server` also provides `librkllm3-server.so` , which integrates the service capabilities into your own application as a library. The configuration is passed in as a JSON string. It suits embedded integration scenarios that do not want to deploy a standalone service process:

```
#include "rkllm3-server.h"
// Start the server with JSON configuration; parameters are identical to the
command line / params.json
rkllm3_server_start(json_config_str);
```

3.5.3 Client Invocation Methods

`rkllm3-server` is compatible with the OpenAI API. Clients can invoke it directly with the OpenAI SDK, `curl` , or any HTTP client.

Invoking with the OpenAI SDK

Point `base_url` to the service address to reuse existing OpenAI application code:

```

from openai import OpenAI

client = OpenAI(base_url="http://<board_ip>:8080/v1", api_key="rkllm3")
response = client.chat.completions.create(
    model="Qwen2.5-3B",
    messages=[{"role": "user", "content": "Please explain the basic concept of
    relativity"}],
    stream=True, # Streaming output supported
)
for chunk in response:
    print(chunk.choices[0].delta.content or "", end="", flush=True)

```

Multimodal invocation

Image input is passed through `image_url` in `messages`, and audio input is passed through the corresponding field. The format follows the OpenAI multimodal specification:

```

response = client.chat.completions.create(
    model="Qwen2.5-VL-3B",
    messages=[{"role": "user", "content": [
        {"type": "text", "text": "Describe the content of this image"},
        {"type": "image_url", "image_url": {"url": "file:///path/to/image.jpg"}},
    ]}],
)

```

Invoking with curl

```

curl http://<board_ip>:8080/v1/chat/completions \
-H "Content-Type: application/json" \
-d '{"model": "Qwen2.5-0.5B", "messages": [{"role": "user", "content": "Who are
you?"}], "stream": false}'

```

Note: The `model` field must match the `-a/--alias` or model name used at startup. Both streaming (`stream:true`) and non-streaming modes are supported. For supported endpoints and the complete request/response fields, see the [RKLLM3_Server_User_Guide_and_API_Reference](#).

3.5.4 Choosing Between Native C/Python Interfaces

The three LLM / MLLM deployment methods are compared below. Choose based on the scenario:

Item	rkllm3-server	C API (Runtime)	Toolkit Lite (Python)
Invocation method	HTTP/OpenAI API	C inference interface	Python inference interface
Ease of use	Highest; standard protocol	Low; requires compilation	High; rapid iteration
Performance	Slightly lower (HTTP overhead)	Optimal	Slightly lower (wrapping overhead)

Item	rkllm3-server	C API (Runtime)	Toolkit Lite (Python)
Feature completeness	Within OpenAI protocol coverage	Most complete	Slightly lower
Typical scenarios	Service-oriented, multi-client, ecosystem integration	Production deployment, peak performance, fine-grained control	Prototype validation, algorithm debugging

Selection advice:

- **Service-oriented / multi-client / rapid integration:** Choose rkllm3-server.
- **Production / peak performance / fine-grained control:** Choose the C API.
- **On-device Python validation / upper-layer applications:** Choose Toolkit Lite.

3.5.5 Server Documentation Entry

For the complete rkllm3-server content (all command-line parameters, multi-model JSON configuration, OpenAI-compatible endpoints and request/response fields, error codes, `librkllm3-server.so` integration API, etc.), see the standalone document *RKLLM3_Server_User_Guide_and_API_Reference*. This guide only covers the core deployment and invocation flow.

4. Advanced Features and Extensions

This chapter describes the advanced features and extension capabilities of RKNN3, including multi-core NPU, dynamic shape, and memory management at the Runtime layer; multiple input types, callbacks, Function Calling, KVCache / LoRA, and more at the LLM / MLLM layer; and extension and security capabilities such as post-process plugins, custom operators, and model encryption. These features target advanced deployment and optimization scenarios; you can skip this chapter for basic deployment.

4.1 Runtime Advanced Features

4.1.1 NPU Multi-Core

Applicable Scenarios

The RK1820 / RK1828 coprocessor NPU contains 8 cores (RK3572 has a single core). Multi-core operation splits computation across multiple cores for parallel execution. It suits models with computationally heavy layers (such as large-parameter LLMs and high-resolution vision models). For small networks, operator switching between cores requires CPU involvement and may degrade performance; single-core operation is recommended.

Usage

Multi-core operation requires coordinated configuration on both the model conversion side and the on-device inference side:

- **Conversion side:** `rknn.config(core_num=N)` specifies the number of cores the model occupies. `core_num=0` is automatic mode (models loaded with `load_llm` use 8 cores; all others use 1 core).
- **Inference side:** The core mask must match the `core_num` used at conversion time. Set it through `rknn3_config.run_core_mask` (C) or `init_runtime(core_mask=)` (Python). The range is 0x1~0xff.

```
/* Query the core count and build the mask */
uint32_t core_num = 0;
rknn3_query(ctx, RKNN3_QUERY_CORE_NUMBER, &core_num, sizeof(core_num));
uint32_t core_mask = 0;
for (uint32_t i = 0; i < core_num; i++) core_mask |= (1 << i);

rknn3_config config;
memset(&config, 0, sizeof(rknn3_config));
config.run_core_mask = core_mask;
rknn3_model_init(ctx, &config);
```

Note: For CNN models, `core_num=1` is recommended; set Transformer-class models as needed. Invoke `RKNN3_QUERY_CORE_MEM_SIZE` to query per-core memory usage and assist memory planning. RK3572 supports single core only (`core_mask=0x1`).

4.1.2 Dynamic Shape

Applicable Scenarios

Dynamic shape allows the same model to switch input sizes at runtime. One model replaces multiple static models, saving Flash/DDR footprint. It suits scenarios with varying input sizes (such as variable-resolution detection and multi-tier image classification).

Prerequisites

- **The model must support dynamic shape:** If the original model is not dynamic-shape, RKNN3-Toolkit can extend it into a dynamic-shape model. However, the model must not contain operators or subgraph structures that restrict dynamic shape (for example, constant shapes cannot change); otherwise the conversion fails with an error. If extension is not supported, modify the model structure according to the error message and retrain. Using a model that is natively dynamic-shape is recommended.
- **Limited number of shapes:** Due to NPU hardware characteristics, a dynamic-shape model does not support arbitrary input shape changes. Users must explicitly enumerate a finite set of input shapes at conversion time. For multi-input models, each input must have the same number of shapes, and the shapes must correspond one-to-one in order; cross combinations are not supported.

Usage

- **Conversion side:** `rknn.config(dynamic_input=[...])` configures multiple sets of input shapes, each described by `rknn3_shape_info`. During quantization, the shape set with the largest total size is used. All shapes share one set of quantization parameters.

```
dynamic_shapes = [
    [[1, 3, 224, 224]],
    [[1, 3, 192, 192]],
    [[1, 3, 160, 160]],
]
rknn.config(mean_values=[[0,0,0]], std_values=[[255,255,255]],
            dynamic_input=dynamic_shapes)
```

- **Inference side:** The model uses Shape ID 0 by default. To switch, release the old memory first, then set the new ID, reallocate, and synchronize:

```
rknn3_destroy_mem(ctx, old_mem);           /* Release the old shape memory */
rknn3_set_shape(ctx, shape_id);            /* Switch the shape */
rknn3_create_mem(ctx, ...);                /* Allocate memory for the new shape */
rknn3_mem_sync(ctx, mem, RKNN3_MEMORY_SYNC_TO_DEVICE);
```

Invoke `RKNN3_QUERY_DYNAMIC_SHAPE_CONFIG` / `RKNN3_QUERY_DYNAMIC_SHAPE_INFO` to query the list of supported shapes.

Limitation: Runtime shape switching must follow the order "release old memory -> `rknn3_set_shape()` -> reallocate memory -> synchronize". Dynamic-shape models do not support weight sharing or context reuse (`rknn3_dup_context()`). Extreme size combinations may affect accuracy because all shapes share one set of quantization parameters.

4.1.3 Cacheable Memory Coherence

Applicable Scenarios

The CPU and NPU have independent caches. When they exchange data through shared memory with cache attributes, missing synchronization causes data incoherence and inference errors. The cacheable memory coherence mechanism uses `rknn3_mem_sync()` to explicitly flush the cache when the data flow direction changes, ensuring coherence.

Usage

Specify memory attributes through the flags of `rknn3_create_mem()` : `RKNN3_FLAG_MEMORY_CACHEABLE` (with cache) or `RKNN3_FLAG_MEMORY_NON_CACHEABLE` (without cache). For cacheable memory, synchronize according to the data flow direction when data passes between the CPU and NPU:

Data Flow Direction	Sync Mode	Timing
CPU write -> NPU read	<code>RKNN3_MEMORY_SYNC_TO_DEVICE</code>	Before inference (input)
NPU write -> CPU read	<code>RKNN3_MEMORY_SYNC_FROM_DEVICE</code>	After inference (output)

Usage Recommendations:

1. When the data flow direction is fixed, use one-way synchronization. This reduces redundant cache flush operations and improves inference performance.
2. When sending input data to the device, invoke `rknn3_mem_sync(..., RKNN3_MEMORY_SYNC_TO_DEVICE)`.
3. When reading output data from the device, invoke `rknn3_mem_sync(..., RKNN3_MEMORY_SYNC_FROM_DEVICE)`.

```
rknn3_tensor_mem* mem = rknn3_create_mem(ctx, size, 0,
RKNN3_FLAG_MEMORY_CACHEABLE);
/* After filling in input data, synchronize to the device before inference */
rknn3_mem_sync(ctx, mem, RKNN3_MEMORY_SYNC_TO_DEVICE);
rknn3_run(ctx, ...);
/* After inference, synchronize output back to the CPU for reading */
rknn3_mem_sync(ctx, mem, RKNN3_MEMORY_SYNC_FROM_DEVICE);
```

4.1.4 Internal Memory Reuse

Applicable Scenarios

When deploying multimodal models (such as VLMs), the vision encoder and the LLM often execute serially. Reusing the same Internal memory between them significantly reduces overall on-device memory usage and effectively eases the bottleneck of limited memory resources.

The core idea is "time slicing + spatial reuse": after the vision model finishes inference and passes image features to the LLM, the Internal memory it occupied is released and reused for the LLM's intermediate computation. This avoids allocating separate memory spaces for both models.

This strategy is especially suitable for the following scenarios:

- **Multi-model pipeline deployment:** Combined inference tasks such as VLM and speech recognition + large language model (LLM).
- **Memory-constrained edge devices:** Devices that need to run multi-stage models within limited memory.
- **Multimodal interaction applications:** Applications that must maximize memory utilization while maintaining response speed.

Usage

Prerequisite: set `rknn3_config.user_mem_internal=1` so that internal memory is allocated by the user. Procedure (taking two serially executed models A and B as an example):

1. Invoke `RKNN3_QUERY_CORE_NUMBER` on both models to query the core counts, then invoke `RKNN3_QUERY_CORE_MEM_SIZE` to query the internal memory size of each core (allocate a `rknn3_core_mem_size` array sized by the core count to receive the results).
2. Iterate over the cores of both models, match them by `core_id` to find the parts running on the same physical core, and use `max(A_internal, B_internal)` as the shared size.
3. Total memory blocks = A core count + B core count - overlapping core count.
4. For reused cores, B's internal_mems pointer points directly to the memory already allocated by A; for non-reused cores, allocate separately (allocate uniformly with A's ctx for later unified release and to avoid double release).
5. Invoke `rknn3_set_internal_mem()` to set the memory for both models. The temporary pointer arrays can then be freed; the internal memory blocks themselves are released by `rknn3_destroy_mem()`.

The following is sample code implementing Internal memory reuse in a VLM:

```

...
int ret = -1;

uint32_t core_num_vision = 0;
uint32_t core_num_llm = 0;
// Get the core_num of the vision model
ret = rknn3_query(app_ctx->vision.rknn_ctx, RKNN3_QUERY_CORE_NUMBER,
&core_num_vision, sizeof(core_num_vision));
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_query failed! ret=%d\n", ret);
    return ret;
}
// Get the core_num of the LLM model
ret = rknn3_query(app_ctx->llm.rknn_ctx, RKNN3_QUERY_CORE_NUMBER, &core_num_llm,
sizeof(core_num_llm));
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_query failed! ret=%d\n", ret);
    return ret;
}

rknn3_core_mem_size* core_mem_sizes_vision =
(rknn3_core_mem_size*)malloc(sizeof(rknn3_core_mem_size) * core_num_vision);
if (!core_mem_sizes_vision) {
    printf("Failed to allocate memory for core_mem_sizes_vision\n");
    return ret;
}
rknn3_core_mem_size* core_mem_sizes_llm =
(rknn3_core_mem_size*)malloc(sizeof(rknn3_core_mem_size) * core_num_llm);
if (!core_mem_sizes_llm) {
    printf("Failed to allocate memory for core_mem_sizes_llm\n");
    return ret;
}
// Get the weight and internal memory usage of the vision model on each NPU core
ret = rknn3_query(app_ctx->vision.rknn_ctx, RKNN3_QUERY_CORE_MEM_SIZE,
core_mem_sizes_vision, sizeof(rknn3_core_mem_size) * core_num_vision);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_query core memory size failed! ret=%d\n", ret);
    return ret;
}
// Get the weight and internal memory usage of the LLM model on each NPU core
ret = rknn3_query(app_ctx->llm.rknn_ctx, RKNN3_QUERY_CORE_MEM_SIZE,
core_mem_sizes_llm, sizeof(rknn3_core_mem_size) * core_num_llm);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_query core memory size failed! ret=%d\n", ret);
    return ret;
}

int llm_to_vision[core_num_llm];
for (int i = 0; i < core_num_llm; i++) { llm_to_vision[i] = -1; }

// Iterate over vision and LLM cores to find cores running on the same physical NPU
core
int core_num_same = 0;
for (int i = 0; i < core_num_vision; i++) {
    for (int j = 0; j < core_num_llm; j++) {
        if (core_mem_sizes_vision[i].core_id == core_mem_sizes_llm[j].core_id) {
            uint64_t internal_size =
std::max(core_mem_sizes_vision[i].internal_size,
core_mem_sizes_llm[j].internal_size);
            core_mem_sizes_vision[i].internal_size = internal_size;

```

```

        core_mem_sizes_llm[j].internal_size = internal_size;
        core_num_same ++;
        llm_to_vision[j] = i;
        break;
    }
}

// Allocate storage for all internal memory block pointers to be assigned; total
blocks = vision core count + LLM core count - overlapping core count (the reused
part)
app_ctx->n_internal_mems = core_num_vision + core_num_llm - core_num_same;
app_ctx->internal_mems = (rknn3_tensor_mem**)calloc(app_ctx->n_internal_mems,
sizeof(rknn3_tensor_mem*));
if (!app_ctx->internal_mems) {
    printf("Failed to allocate memory for app_ctx->internal_mems array\n");
    return -1;
}

// Allocate a pointer array for each vision model core (used later to set the model
internal memory)
rknn3_tensor_mem** internal_mems_vision =
(rknn3_tensor_mem**)calloc(core_num_vision, sizeof(rknn3_tensor_mem*));
if (!internal_mems_vision) {
    printf("Failed to allocate memory for internal_mems_vision array\n");
    return -1;
}

// Allocate a pointer array for each LLM core (used later to set the model internal
memory)
rknn3_tensor_mem** internal_mems_llm = (rknn3_tensor_mem**)calloc(core_num_llm,
sizeof(rknn3_tensor_mem*));
if (!internal_mems_llm) {
    printf("Failed to allocate memory for internal_mems_llm array\n");
    return -1;
}

// Allocate and initialize the internal memory blocks for the vision model
int idx = 0;
for (uint32_t i = 0; i < core_num_vision; i++) {
    internal_mems_vision[i] = rknn3_create_mem(app_ctx->vision.rknn_ctx,
core_mem_sizes_vision[i].internal_size, core_mem_sizes_vision[i].core_id,
RKNN3_FLAG_MEMORY_CACHEABLE);
    if (!internal_mems_vision[i]) {
        return -1;
    }
    printf("Created user internal memory for core %d: size=%lu, virt_addr=%p,
phys_addr=0x%lx\n", core_mem_sizes_vision[i].core_id,
        internal_mems_vision[i]->size, internal_mems_vision[i]->virt_addr,
internal_mems_vision[i]->phys_addr);
    app_ctx->internal_mems[idx++] = internal_mems_vision[i];
}

// Allocate internal memory for the LLM (reused or newly allocated)
for (uint32_t i = 0; i < core_num_llm; i++) {
    if (llm_to_vision[i] != -1) {
        // Reuse: the LLM core and a vision model core are on the same physical
core; point directly to the allocated memory
        internal_mems_llm[i] = internal_mems_vision[llm_to_vision[i]];
        continue;
    }
    // Otherwise allocate separate memory for this LLM core; allocation uses
vision.rknn_ctx for unified release later (to avoid double release)

```

```

        internal_mems_llm[i] = rknn3_create_mem(app_ctx->vision.rknn_ctx,
        core_mem_sizes_llm[i].internal_size, core_mem_sizes_llm[i].core_id,
        RKNN3_FLAG_MEMORY_CACHEABLE);
        if (!internal_mems_llm[i]) {    // All allocated via vision.rknn_ctx for easy
        release later
            return -1;
        }
        printf("Created user internal memory for core %d: size=%lu, virt_addr=%p,
        phys_addr=0x%lx\n", core_mem_sizes_llm[i].core_id,
            internal_mems_llm[i]->size, internal_mems_llm[i]->virt_addr,
            internal_mems_llm[i]->phys_addr);
        // Add the newly allocated memory blocks to the global management array
        app_ctx->internal_mems[idx++] = internal_mems_llm[i];
    }

    // Set the allocated internal memory for the vision model
    ret = rknn3_set_internal_mem(app_ctx->vision.rknn_ctx, internal_mems_vision,
    core_num_vision);
    if (ret != RKNN3_SUCCESS) {
        printf("rknn3_set_internal_mem failed! ret=%d\n", ret);
        return ret;
    }
    // Set the allocated internal memory (including the reused part) for the LLM model
    ret = rknn3_set_internal_mem(app_ctx->llm.rknn_ctx, internal_mems_llm,
    core_num_llm);
    if (ret != RKNN3_SUCCESS) {
        printf("rknn3_set_internal_mem failed! ret=%d\n", ret);
        return ret;
    }

    // Release the temporary pointer arrays (the internal memory blocks themselves are
    allocated by rknn3_create_mem and must be released later with rknn3_destroy_mem)
    free(internal_mems_vision);
    free(internal_mems_llm);
    free(core_mem_sizes_vision);
    free(core_mem_sizes_llm);

    return ret;

```

Note: Reuse is safe only when the two models execute serially (only one runs at any moment); reuse is not allowed with concurrent execution.

4.2 LLM and MLLM Advanced Features

4.2.1 Prompt, Token, and Embedding Inputs

Applicable Scenarios

LLM / MLLM inference supports three types of text input, specified through `rknn3_llm_input.input_type`, each corresponding to a different data preparation stage. Choosing the type based on the actual data form reduces pre-processing overhead:

Input Type	Applicable Scenarios	Data Preparation Stage
RKNN3_LLM_INPUT_PROMPT	General text input; the SDK performs tokenizer segmentation	Natural language text
RKNN3_LLM_INPUT_TOKEN	Tokenization already done; skips the text->token conversion	token id sequence
RKNN3_LLM_INPUT_EMBED	Embedding already done; skips text->token->embedding	embedding vectors

Usage

All three types pass through `llm_input ( rknn3_llm_tensor )` uniformly. Fill the `rknn3_llm_input` array and pass it to `rknn3_session_run()`, populating the corresponding fields by type. The three types are mutually exclusive; each `rknn3_llm_input` can carry only one.

- **PROMPT input:** Enter natural language text directly; the SDK performs tokenizer conversion. Recommended for most general scenarios.

```
rknn3_llm_tensor input_tensor = {.name = NULL, .prompt = prompt,
                                .embed = NULL, .tokens = NULL,
                                .n_tokens = 0, .enable_thinking = false};
input.input_type = RKNN3_LLM_INPUT_PROMPT;
input.llm_input = input_tensor;
```

- **TOKEN input:** Enter a token id sequence directly. It suits scenarios where tokenization is already done and the text->token conversion should be skipped (such as reusing historical tokens or custom segmentation).

```
int32_t tokens[6] = {151628, 151629, 151630, 151631, 151632, 151633};
rknn3_llm_tensor input_tensor = {.name = NULL, .prompt = NULL,
                                .embed = NULL, .tokens = tokens,
                                .n_tokens = 6, .enable_thinking = false};
input.input_type = RKNN3_LLM_INPUT_TOKEN;
input.llm_input = input_tensor;
```

- **EMBED input:** Enter embedding vectors directly (size: `n_tokens * embedding_dim * sizeof(float16)`). It skips the text->token->embedding pipeline and suits models without a traditional tokenization flow or with custom embeddings.

```
float16* embed = (float16*)malloc(n_tokens * embedding_dim * sizeof(float16));
/* Fill in the embed data */
rknn3_llm_tensor input_tensor = {.name = NULL, .prompt = NULL,
                                .embed = embed, .tokens = NULL,
                                .n_tokens = n_tokens, .enable_thinking = false};
input.input_type = RKNN3_LLM_INPUT_EMBED;
input.llm_input = input_tensor;
```

Note: For structure members, see 3.1.2. PROMPT/TOKEN/EMBED are mutually exclusive; each `rknn3_llm_input` can carry only one.

4.2.2 Multimodal and Aux Inputs

Applicable Scenarios

Besides text inputs (see 4.2.1), MLLM inference also supports multimodal inputs and auxiliary inputs, covering multimodal fusion and the extended input needs of special models:

Input Type	Applicable Scenarios	Data Preparation Stage
<code>RKNN3_LLM_INPUT_T_MULTIMODAL</code>	Multimodal input; supports image/audio/video individually or combined	Embeddings output by per-modality encoding models
<code>RKNN3_LLM_INPUT_T_AUX</code>	Other input types, used with the main input for special models (such as the 3 deepstack auxiliary inputs of Qwen3-VL)	Auxiliary tensors

Usage

- **MULTIMODAL input:** Use the `rknn3_llm_multimodal_tensor` structure (`rknn3_llm_input.multimodal_input`). Populate the `image / audio / video` substructures for the actual modalities and leave unused modalities unset. Each substructure contains the pointer to the embeddings output by the encoding model, the token count, the modality count, and text placeholder markers. Placeholders in the prompt use the `<image> / <audio> / <video>` tags, and their counts must match `n_image / n_audio / n_video`. Placeholder markers (such as `<|vision_start|>` or `<|vision_bos|>`) vary by model tokenizer; follow the special tokens of the actual model tokenizer.

```
rknn3_llm_multimodal_tensor tensor;
memset(&tensor, 0, sizeof(tensor));
tensor.prompt = "1.<image> 2.<audio> 3.<video> Please analyze the above content";
tensor.image.image_embed = vision_output; /* n_image x n_image_tokens x
embedding_dim */
tensor.image.n_image_tokens = 196;
tensor.image.n_image = 1;
tensor.image.image_start = "<|vision_bos|>";
tensor.image.image_end = "<|vision_eos|>";
tensor.image.image_content = "<|IMAGE|>";
/* Fill the audio/video sub-structures in the same way */
input.input_type = RKNN3_LLM_INPUT_MULTIMODAL;
input.multimodal_input = tensor;
```

- **AUX input:** Other input types that provide auxiliary tensors together with the main input, serving as extended inputs for special models. Take Qwen3-VL-2B as an example: besides 1 MULTIMODAL main input, it needs 3 deepstack auxiliary inputs, for 4 `rknn3_llm_input` entries in total. Each AUX carries auxiliary data through `rknn3_aux_tensor.mem` (allocated by `rknn3_create_mem()`):

```

/* Main input: MULTIMODAL, occupies llm_inputs[0] */
/* Auxiliary inputs: 3 AUX, occupy llm_inputs[1..3] */
rknn3_aux_tensor aux_input_tensor0;
rknn3_llm_input aux_input0;

aux_input_tensor0.mem = rknn3_create_mem(ctx, n * embedding_dim * sizeof(float16),
                                         0, RKNN3_FLAG_MEMORY_CACHEABLE);
aux_input0.input_type = RKNN3_LLM_INPUT_AUX;
aux_input0.aux_input = aux_input_tensor0;
llm_inputs[1] = aux_input0;
/* aux_input1 and aux_input2 follow the same pattern */

```

Note: The placeholder markers of each modality must match the special tokens of the model tokenizer. For MULTIMODAL structure members, see 3.1.2; for the full definition, see *RKNN3_Runtime_C_API_Reference* 5.5.25; for an example showing how to populate the structure, see 3.3.6. AUX cannot be used alone; it must be combined with the main input in the same `llm_inputs` array and passed to `rknn3_session_run()`.

4.2.3 Callback Mechanism

Applicable Scenarios

In LLM inference, stages such as tokenization, embedding lookup, sampling, and result output are delegated to user implementations through the callback mechanism, providing flexibility. Six types of callbacks are configured uniformly through the `RKLLMCallback` structure and registered with `rknn3_session_set_callback()`:

Callback	Purpose	Required
<code>result_callback</code>	Invoked after each token is generated; processes output (such as converting to text and printing)	Required
<code>tokenizer_callback</code>	Converts text to a token id sequence	Optional (default is used if unset)
<code>embed_callback</code>	Looks up embeddings by token id and fills the buffer	Optional (not needed in Tie Word mode)
<code>sampling_callback</code>	Custom sampling strategy (argmax/top-k/top-p/temperature)	Optional (default sampling is used if unset)
<code>output_callback</code>	Retrieves output tensors (such as logits); requires <code>output_tensors</code> to be preconfigured	Optional
<code>input_callback</code>	Modifies/fills input tensors before inference (such as Per-Layer Embeddings)	Optional

Each callback pairs with a `*_userdata` field to pass in context (such as a tokenizer instance or embedding_info).

```

RKLLMCallback callback;
memset(&callback, 0, sizeof(RKLLMCallback));
callback.result_callback = result_callback;
callback.tokenizer_callback = tokenizer_callback;
callback.embed_callback = embed_callback;
callback.sampling_callback = sampling_callback;
rknn3_session_set_callback(session, &callback);

```

`result_callback` handles the different stages according to `LLMCallState` (`RKLLM_RUN_NORMAL` for normal generation, `RKLLM_RUN_FINISH` for completion, `RKLLM_RUN_ERROR` for errors, `RKLLM_RUN_PAUSE/RESUME` for pause/resume, and so on).

Note: `output_callback` additionally requires `callback.output_tensors` and `n_output_tensors` to be set. For complete callback signatures and parameters, see *RKNN3_Runtime_C_API_Reference*.

4.2.4 Function Calling

Applicable Scenarios

Function Calling allows model output during inference to trigger preset function tools, enabling data interaction with external systems (such as weather queries, schedule reminders, and database retrieval) and extending the capability boundary of the LLM.

Usage

Invoke `rknn3_session_set_function_tools()` to set a group of tools (JSON describes function names and parameter definitions). During inference, the model decides whether to initiate a call according to the session template and provides the call request in the output. The calling flow involves two rounds of inference:

1. **Round 1** (role="user"): The model generates a function call request. Parse the output to obtain `tool_call_list`, perform the actual function invocation, and store the result as `tool_prompt`.
2. **Round 2** (role="tool"): Feed the function call result back to the model; the model processes it and returns the final result.

```

/* Set the function tools */
char* function_tools = R"([{"type":"function","function":{"
    "name":"get_current_temperature","description":"...",
    "parameters":{"type":"object","properties":{"..."},"required":[...]}}}]);
rknn3_session_set_function_tools(session, function_tools);

/* Round 1: the user asks a question */
input.role = "user";
rknn3_session_run(session, inputs, 1, &param);
/* Parse the output -> invoke the actual function -> build tool_prompt */

/* Round 2: the tool returns the result */
input.role = "tool";
input_tensor.prompt = tool_prompt.c_str();
rknn3_session_run(session, inputs, 1, &param);

```

Note: For a complete example, see `rknn3-runtime/examples/rknn_session_test_demo/src/rknn_session_test_function_call.c` .

4.2.5 KVCache Management

Applicable Scenarios

KVCache caches the key-value states of attention to accelerate autoregressive generation. RKNN3 provides three KVCache policies plus import/export capabilities, covering scenarios such as long conversations, resuming interrupted computation, and cross-session context saving.

Three Policies

Set through `rknn3_session_set_kvcache_policy()` (`rknn3_kvcache_policy`):

Policy	Behavior	Applicable Scenarios
<code>RKNN3_KVCACHE_POLICY_RECURRENT</code> (default)	KVCache is overwritten automatically; <code>n_keep</code> / <code>n_keep_aligned</code> keeps the system prompt from being overwritten	Long multi-turn conversations
<code>RKNN3_KVCACHE_POLICY_NORMAL</code>	Uses only <code>max_context_len</code> of KVCache; generation stops after that	Fixed-length context
<code>RKNN3_KVCACHE_POLICY_SAVE_CHECKPOINT</code>	Enables Checkpoint save mode (see 4.2.6)	Acceleration for Linear/Sliding Attention models

```
rknn3_kvcache_policy_param param;  
memset(&param, 0, sizeof(param));  
param.recurrent.n_keep = 21; /* Actual token count of the system prompt */  
param.recurrent.n_keep_aligned = 32; /* Aligned to 32 */  
rknn3_session_set_kvcache_policy(session, RKNN3_KVCACHE_POLICY_RECURRENT, &param);
```

Import and Export

`rknn3_session_save_kvcache()` (save to file), `rknn3_session_load_kvcache_from_path()` (load from file), and `rknn3_session_load_kvcache_from_data()` (load from memory) apply to resuming interrupted computation and cross-session context saving. Typical flow: save after the first round of inference, load before the second round.

Note: KVCache state is strongly tied to the model and runtime library version; do not mix it across different models/versions.

4.2.6 KVCache Checkpoint

Applicable Scenarios

The Checkpoint policy saves KV Cache snapshots at fixed intervals. Generation can later resume from a checkpoint, avoiding re-prefilling the entire prefix. It takes effect only for Linear Attention

/ Sliding Attention models (such as the Qwen3.5 and Gemma4 series). Full Attention follows a separate path, and the setting has no effect on other models.

Usage

With the `RKNN3_KVCACHE_POLICY_SAVE_CHECKPOINT` policy, configure four parameters in `rknn3_kvcache_policy_param.save_checkpoint` (all must be aligned to 128):

Parameter	Description
<code>checkpoint_start_pos</code>	Base position, not saved itself; actual save positions = start + k×interval (k=1..max_count)
<code>checkpoint_interval</code>	Save interval; common values are 128/256/512/1024
<code>max_checkpoint_count</code>	Maximum number of saves; upper bound = floor((max_context_len - start)/interval)
<code>checkpoint_tail_overwrite</code>	false by default; when set to true, the last slot is overwritten in a rolling manner — suitable when memory is limited but the most recent restore point must be kept

```
rknn3_kvcache_policy_param param;
memset(&param, 0, sizeof(param));
param.save_checkpoint.checkpoint_start_pos    = 0;
param.save_checkpoint.checkpoint_interval     = 128;
param.save_checkpoint.max_checkpoint_count    = 2;
param.save_checkpoint.checkpoint_tail_overwrite = false;
rknn3_session_set_kvcache_policy(session, RKNN3_KVCACHE_POLICY_SAVE_CHECKPOINT,
&param);
```

Note: When reusing only a fixed system prompt, `max_checkpoint_count≤2` is recommended. Validation rules: an error is raised directly if `start_pos > max_context_len - interval`; unaligned parameters are forced up to alignment; a `max_count` above the upper bound is forced down. Verify the effect through the log `load LINEAR_ATTN_CACHE kvcache_checkpoint pos: ...`.

4.2.7 LoRA

Applicable Scenarios

LoRA (Low-Rank Adaptation) allows adapters to be loaded/switched dynamically without modifying base model weights. It suits multi-task fine-tuned deployment: one base model + multiple LoRA adapters, switched per task, saving storage and memory. LoRA usage involves two stages: in the conversion stage, RKNN3-Toolkit converts the LoRA weights together with the base model into an RKNN model; in the on-device deployment stage, the Runtime interfaces load adapters and enable them on demand.

Conversion Stage

RKNN3-Toolkit provides the `load_lora`, `enable_lora`, and `disable_lora` interfaces for loading and controlling LoRA weights during model conversion. LoRA weights must be loaded after the model is loaded (`load_onnx` / `load_llm`) and before the model is built (`build`):

```
rknn.load_lora(lora_paths=[["./adapter_model.safetensors"]],
              lora_config_paths=[["./adapter_config.json"]],
              lora_patterns=["lora0_pattern0"],
              lora_distribute_strategy='best_perf')
rknn.enable_lora(enable_patterns=["lora0_pattern0"])
rknn.disable_lora(disable_patterns=["lora0_pattern0"])
```

On-Device Deployment

On the device side, load and control LoRA through the Runtime interfaces. `rknn3_lora_init()` initializes context-level LoRA support from a weight file path and loads the metadata. Invoke it after `rknn3_model_init()` and before any session uses LoRA. Afterwards, invoke `RKNN3_QUERY_LORA_NUM` and `RKNN3_QUERY_LORA_INFO` to query the loaded LoRA information, then invoke `rknn3_lora_load()` to load the specified adapter.

LoRA weights are loaded and managed at the context level and shared by multiple sessions under the same context. Whether they take effect for a session is controlled at two levels:

- **Context level** (takes effect for all inference under the context; updated once at invocation time):

Interface	Function
<code>rknn3_lora_init()</code> / <code>rknn3_lora_init_from_data()</code>	Initializes LoRA metadata from file/memory
<code>rknn3_lora_load()</code> / <code>rknn3_lora_unload()</code>	Loads/unloads adapter weights
<code>rknn3_lora_enable()</code> / <code>rknn3_lora_disable()</code>	Enables/disables at the context level

- **Session level** (controls only that session; updated on every `session_run`, without affecting other sessions):

Interface	Function
<code>rknn3_session_enable_lora()</code> / <code>rknn3_session_disable_lora()</code>	Enables/disables at the session level without reloading weights

On-device invocation example:

```
rknn3_lora* lora = NULL;
rknn3_lora_init(ctx, "lora_weight_path");           /* Initialize the metadata;
must be called after model_init */
rknn3_query(ctx, RKNN3_QUERY_LORA_INFO, &lora, sizeof(rknn3_lora*) *
RKNN3_MAX_LORA_NUM); /* Query the valid lora descriptors */
rknn3_lora_load(ctx, lora);                         /* Load the adapter weights */
rknn3_session_enable_lora(session, lora);           /* Enable at the session level
*/
```

Note: Mixing the two levels of interfaces is not recommended: session-level settings override context-level settings, making the effective order indeterminate. LoRA weights support int4 / int6 / int8 / float16 quantization, with multi-core splitting strategies `best_perf` / `less_mem`.

4.2.8 Tie Word Embedding

Applicable Scenarios

Tie Word Embeddings is a weight-tying technique: the input embedding layer and the output lm_head layer share the same weight matrix. When enabled, the device side reuses the lm_head weights inside the model, and **the Host side does not need to load `embed.bin`**, saving Host memory and disk space.

Usage

- **Confirm support:** Check whether the `tie_word_embeddings` field in the HuggingFace `config.json` is true (common in the Qwen series).
- **Conversion configuration:** Set `llm_config['vocab'][0]['tie_embeddings'] = True`. After conversion, verify with `strings xxx.rknn | grep tie_word_embeddings`.

```
llm_config = DEFAULT_RKNN_LLM_CONFIG.copy()
llm_config['vocab'][0]['tie_embeddings'] = True
rknn.config(..., llm_config=llm_config)
```

- **Deployment differences:** In Tie mode, `embed.bin` is not loaded (no open/mmap) and `embed_callback` is not set; in non-Tie mode, both are required.

Mode	embed.bin	embed_callback	Host Memory
Tie	Not required	Not required	Lower
Non-Tie	Required	Required	Higher

Note: Even if the model contains `tie_word_embeddings:true`, you can still set the environment variable `RKNN_ENABLE_TIE_WORD_EMBEDDINGS=0` to force non-Tie mode. However, `embed.bin` must then be loaded and `embed_callback` must be set, losing the memory advantage.

4.2.9 Pause, Resume, and State Query

Applicable Scenarios

A session can be paused/resumed during inference to insert other tasks during generation (such as handling user interruption or higher-priority tasks). State query monitors inference progress, performance statistics, and LoRA/KVCache status.

Usage

- **Pause:** `rknn3_session_pause()`.
- **Resume:** `rknn3_session_resume()`.

On pause, `result_callback` is invoked with `RKLLM_RUN_PAUSE`; on resume, with `RKLLM_RUN_RESUME`. Handle these states in the callback to implement business logic.

```
rknn3_session_pause(session);
/* Handle other tasks while paused */
rknn3_session_resume(session);
```

- **State query:** `rknn3_session_query_state()` queries the current state and stores the result in `RKLLMRunState`. Main fields:

Field	Description
<code>n_max_tokens</code>	Maximum number of inference tokens
<code>n_total_tokens</code>	Number of tokens already inferred
<code>n_prefill_tokens</code>	Number of tokens processed in the prefill stage
<code>n_decode_tokens</code>	Number of tokens generated in the generate stage
<code>n_output_tokens</code>	Number of model output tokens (= <code>n_decode_tokens</code> + 1)
<code>input_tokens</code>	Pointer to the input token array (users must not free it)
<code>output_tokens</code>	Pointer to the output token array (users must not free it)
<code>kvcache_policy</code>	Current KVCache policy
<code>n_loras_enabled</code>	Number of enabled LoRAs

Note: For the complete field list, see *RKNN3_Runtime_C_API_Reference* 5.5.31.

4.3 Extensions and Security

4.3.1 Custom Operators

Overview

RKNN3 provides a custom operator extension mechanism that allows users to implement custom operators executing on the CPU cores of the RK182X coprocessor. This extends operator implementations, reduces data transfer back to the host, lowers the host SoC load, and simplifies application development. Typical scenarios include post-processing operations such as bounding box decoding and NMS for object detection models (YOLOv5/v6/v8), and intermediate operators in the model graph that RKNN3 does not yet support.

Two Operator Types

Based on position and purpose in the model, custom operators fall into two types:

Type	plugin_type	Position	Inputs and Outputs
Post-process operator	<code>RKNN3_OP_PLUGIN_TYPE_POSTPROCESS</code>	End of the model	Customized by the plugin's <code>get_output_num</code> / <code>get_attrs</code>
Custom CPU operator	<code>RKNN3_OP_PLUGIN_TYPE_CUSTOM_OP</code>	Any position in the compute graph	Determined by the model graph

Both types share the same plugin interface (`rknn3_custom_op` structure), compilation method, and loading API (`rknn3_register_custom_ops_plugins()`). They differ in the `plugin_type` field and the corresponding callback responsibilities.

Plugin Interface Definition

A custom operator implements the interface defined by the `rknn3_custom_op` structure, which is defined in `rknn3_api.h`:

```

typedef struct _rknn3_custom_op
{
    const char *op_type;                // Operator type name
    rknn3_op_plugin_type plugin_type;    // Plugin type
    rknn3_op_target_type target;         // Execution backend (CPU)
    uint32_t version;                   // Version number
    const char *author;                  // Author information
    const char *description;             // Description

    // Lifecycle callbacks
    int (*init)(rknn3_custom_op_context *op_ctx);    // [Optional] Initialization
    int (*prepare)(rknn3_custom_op_context *op_ctx,  // [Optional] Preparation
                  rknn3_tensor *inputs, uint32_t n_inputs,
                  rknn3_tensor *outputs, uint32_t n_outputs);
    int (*compute)(rknn3_custom_op_context *op_ctx,  // [Required] Compute
                  rknn3_tensor *inputs, uint32_t n_inputs,
                  rknn3_tensor *outputs, uint32_t n_outputs);
    int (*deinit)(rknn3_custom_op_context *op_ctx);  // [Optional] De-
initialization

    // Post-process plugin specific interfaces
    int (*get_output_num)(rknn3_custom_op_context *op_ctx); // [Optional] Get the
output count
    int (*get_attrs)(rknn3_custom_op_context *op_ctx,        // [Optional] Get
input and output attributes
                    rknn3_tensor_attr *input_attrs, uint32_t n_inputs,
                    rknn3_tensor_attr *output_attrs, uint32_t n_outputs);
} rknn3_custom_op;

```

Key Data Structures:

- **rknn3_custom_op_context** : Operator context, containing the RKNN3 context handle, a private data pointer (**priv_data** , managed by the framework; users must not modify it), a user data pointer (**user_data** , managed by the user), and more. Custom CPU operators can also read operator attributes through its **get_param** interface.
- **rknn3_tensor** : Tensor structure, containing memory information (**rknn3_tensor_mem**) and attribute information (**rknn3_tensor_attr**). Plugins read inputs from **inputs[i].mem->virt_addr** and write outputs to **outputs[j].mem->virt_addr** .
- **rknn3_tensor_attr** : Tensor attributes, including shape / dtype / layout and quantization parameters.

Plugin Type and Execution Backend Enums:

```

typedef enum _rknn3_op_plugin_type {
    RKNN3_OP_PLUGIN_TYPE_POSTPROCESS = 0, // Post-process operator (at the end of
the model)
    RKNN3_OP_PLUGIN_TYPE_CUSTOM_OP = 1,   // Custom CPU operator (any position in
the graph)
} rknn3_op_plugin_type;

typedef enum _rknn3_op_target_type {
    RKNN3_OP_TARGET_TYPE_CPU = 1,         // CPU backend
} rknn3_op_target_type;

```

The rest of this section describes implementation steps, compilation/loading, and examples by operator type: 4.3.1.1 post-process operators and 4.3.1.2 custom CPU operators.

4.3.1.1 POST-PROCESS OPERATORS

Post-process operators sit at the end of the model. They move post-processing computation (such as YOLO bounding box decoding and NMS) from the host SoC to the coprocessor, reducing data transfer back to the host and lowering the host load. The plugin customizes the output count and attributes through `get_output_num / get_attrs`.

4.3.1.1.1 Plugin Implementation Steps

The custom operator plugin development flow is shown in the figure below:

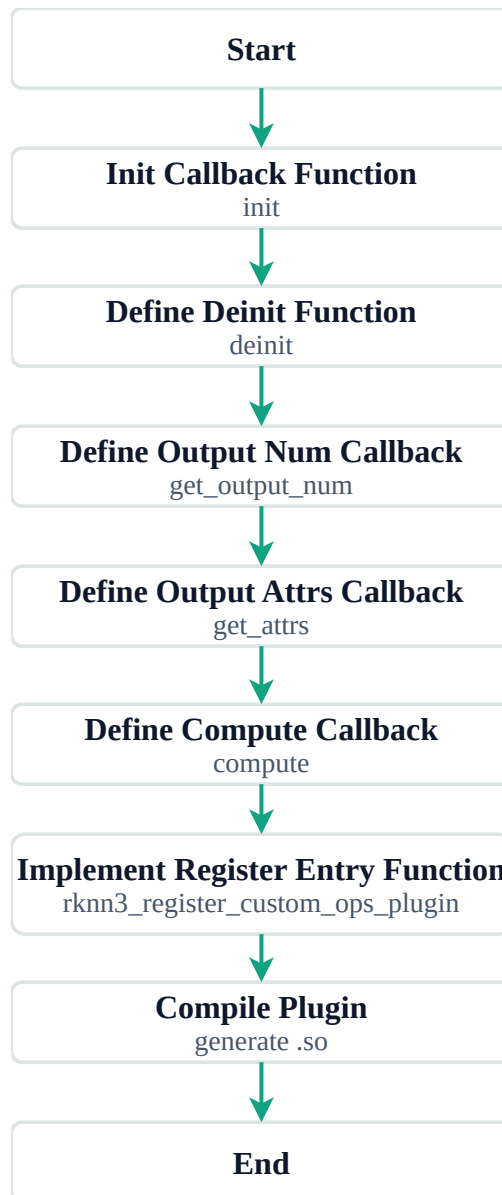


Figure 4-1 Custom operator plugin development flow

As shown, plugin implementation completes each callback in turn, assembles them into the operator structure for registration, and finally compiles them into a `.so`. The steps below use a YOLO post-process plugin (`YoLoPostprocess`) as an example:

- **Implement the initialization callback:** `init` is invoked at operator initialization. It allocates private data (such as thresholds and model dimensions) and stores it in `op_ctx->priv_data` for later callbacks.

```
static int yolo_postprocess_plugin_init(rknn3_custom_op_context *op_ctx)
{
    rknn3_yolo_postprocess_context *pp_ctx = malloc(sizeof(...));
    memset(pp_ctx, 0, sizeof(...));
    pp_ctx->conf_threshold = 0.25;
    pp_ctx->nms_threshold = 0.45;
    op_ctx->priv_data = pp_ctx;
    return 0;
}
```

- **Define the deinitialization function:** `deinit` is invoked at operator destruction. It releases private data and other resources allocated in `init`.

```
static int yolo_postprocess_plugin_deinit(rknn3_custom_op_context *op_ctx)
{
    if (op_ctx->priv_data) {
        free(op_ctx->priv_data);
        op_ctx->priv_data = NULL;
    }
    return 0;
}
```

- **Define the output tensor callback:** `get_output_num` returns the number of output tensors of the operator. The output count of a post-process plugin is defined by the plugin itself.

```
static int yolo_postprocess_plugin_get_output_num(rknn3_custom_op_context *op_ctx)
{
    return 1; /* Output 1 tensor: detection results */
}
```

- **Define the output attribute callback:** `get_attrs` defines attributes such as shape, dtype, and aligned_size of the output tensors.

```
static int yolo_postprocess_plugin_get_attrs(rknn3_custom_op_context *op_ctx,
                                             rknn3_tensor_attr *input_attrs,
                                             uint32_t n_inputs,
                                             rknn3_tensor_attr *output_attrs,
                                             uint32_t n_outputs)
{
    output_attrs[0].n_dims = 3;
    output_attrs[0].shape[0] = input_attrs[0].shape[0]; /* batch */
    output_attrs[0].shape[1] = 256; /* Maximum number of detections */
    output_attrs[0].shape[2] = 6; /* score, class_id, x1, y1, x2, y2 */
    output_attrs[0].dtype = RKNN3_TENSOR_FLOAT32;
    return 0;
}
```

- **Define the compute callback:** `compute` is the required callback. It executes the operator's actual computation logic, reads inputs from `inputs[i].mem->virt_addr`, and writes the processed results to `outputs[j].mem->virt_addr`.

```
static int yolo_postprocess_plugin_compute(rknn3_custom_op_context *op_ctx,
                                          rknn3_tensor *inputs, uint32_t n_inputs,
                                          rknn3_tensor *outputs, uint32_t
n_outputs)
{
    int result_count = post_process(op_ctx->priv_data, inputs, n_inputs, outputs,
n_outputs);
    return (result_count >= 0) ? 0 : -1;
}
```

- **Implement the registration entry function:** Assemble the callbacks above into an `rknn3_custom_op` structure (specifying `op_type`, `plugin_type`, `target`, and the callback pointers), then register through the fixed-name entry function `rknn3_register_custom_ops_plugin()`. The framework invokes it repeatedly by `op_index` to retrieve the registered operators. One `.so` can register multiple operators.

```
static rknn3_custom_op yolo_postprocess_op = {
    .op_type      = "YoloPostprocess",
    .plugin_type   = RKNN3_OP_PLUGIN_TYPE_POSTPROCESS,
    .target       = RKNN3_OP_TARGET_TYPE_CPU,
    .init         = yolo_postprocess_plugin_init,
    .compute      = yolo_postprocess_plugin_compute,
    .deinit       = yolo_postprocess_plugin_deinit,
    .get_output_num = yolo_postprocess_plugin_get_output_num,
    .get_attrs    = yolo_postprocess_plugin_get_attrs,
};

static rknn3_custom_op* registered_ops[] = { &yolo_postprocess_op, NULL };

rknn3_custom_op* rknn3_register_custom_ops_plugin(int op_index)
{
    if (op_index < 0 || registered_ops[op_index] == NULL) {
        return NULL;
    }
    return registered_ops[op_index];
}
```

- **Compile the plugin:** Compile the C source above into a `.so` shared library. For compilation requirements and scripts, see 4.3.1.1.2.

4.3.1.1.2 Plugin Compilation, Loading, and Usage

Compilation Requirements

- Pure C only (C++ classes/templates/namespaces/exceptions are not supported).
- The compiler is the RISC-V toolchain `riscv64-unknown-elf-gcc` (dedicated to the RK182X CPU cores).

- Key compilation options: `-mcpu=c908v -mrvv-v0p10-compatible -march=rv64gcv -mabi=lp64d -O2 -g2 -mcmmodel=medany -fpic -fno-plt`.
- Link options: `-nostartfiles -nostdlib -static-libgcc -e 0 -Wl,-r,-z,max-page-size=1024`. The output is a `.so`, trimmed with `riscv64-unknown-elf-strip`.

```
CC=riscv64-unknown-elf-gcc
STRIP=riscv64-unknown-elf-strip
CFLAGS="-mcpu=c908v -mrvv-v0p10-compatible -march=rv64gcv -mabi=lp64d -O2 -g2 \
        -mcmmodel=medany -fpic -fno-plt -D_POSIX_C_SOURCE=1 -I${RKNPU3_INCLUDE}"
LDFLAGS="-mcpu=c908v -O2 -g2 -Wl,-r,-z,max-page-size=1024 -fpic \
        -nostartfiles -nostdlib -static-libgcc -e 0"
${CC} ${LDFLAGS} -o libpostprocess_yolov5_rk182x.so *.o
${STRIP} --strip-unneeded -R .hash libpostprocess_yolov5_rk182x.so
```

Loading and Usage

Load plugins with `rknn3_register_custom_ops_plugins()`, which must be invoked after `rknn3_model_init()` and before `rknn3_run()`. The third parameter `size` is a reserved field; pass `0` for now. After loading, the framework rebuilds the operator dispatch table automatically, and `rknn3_run()` invokes the plugin compute automatically.

API interface:

```
int rknn3_register_custom_ops_plugins(rknn3_context ctx,
                                     const char* plugin_path,
                                     int64_t size);
```

Usage flow (using the YOLOv5 post-process plugin as an example):

```

/* 1. Initialize and load the model */
rknn3_init(&ctx, NULL);
rknn3_load_model_from_path(ctx, "yolov5.rknn", "yolov5.weight");
rknn3_model_init(ctx, &config);

/* 2. Load the post-process plugin (must be called after model_init) */
const char* plugin_path = "./libpostprocess_yolov5_rk182x.so";
ret = rknn3_register_custom_ops_plugins(ctx, plugin_path, 0);
if (ret != RKNN3_SUCCESS) {
    printf("Plugin load failed! ret=%d\n", ret);
    return -1;
}

/* 3. Query the plugin output information */
rknn3_input_output_num io_num;
rknn3_query(ctx, RKNN3_QUERY_POSTPROCESS_IN_OUT_NUM, &io_num, sizeof(io_num));
rknn3_tensor_attr output_attrs[io_num.n_output];
for (int i = 0; i < io_num.n_output; i++) {
    output_attrs[i].index = i;
    rknn3_query(ctx, RKNN3_QUERY_POSTPROCESS_OUTPUT_ATTR,
                &output_attrs[i], sizeof(rknn3_tensor_attr));
}

/* 4. Run inference (the framework invokes the plugin compute automatically) */
rknn3_run(ctx, inputs, n_inputs, outputs, n_outputs);

/* 5. Read the plugin output (already contains the post-processed detection boxes)
*/
float* result = (float*)outputs[0].mem->virt_addr;

```

Note: The same context can load plugins multiple times; a later load replaces an earlier one with the same `op_type` (the old plugin is dlclosed first, invoking `deinit`). After a post-process plugin is loaded, use `RKNN3_QUERY_POSTPROCESS_IN_OUT_NUM` / `RKNN3_QUERY_POSTPROCESS_OUTPUT_ATTR` to query the output count and attributes, and allocate output memory accordingly. This compilation and loading method applies to all custom operators, including custom CPU operators.

4.3.1.1.3 Examples

The YOLOv5 post-processing example is located at `rknn3-model-zoo/examples/yolov5/cpp/libpostprocess_rk182x/` and includes both Anchor-based (yolov5) and Anchor-free (yolov8) implementations. Typical configuration: `MAX_OBJ_NUM=256`, `OBJ_CLASS_NUM=80`, `NMS_THRESH=0.45`, `BOX_THRESH=0.25`; output format `[N, 256, 6]` (score, class_id, x1, y1, x2, y2).

4.3.1.2 CUSTOM CPU OPERATORS

Custom CPU operators (`RKNN3_OP_PLUGIN_TYPE_CUSTOM_OP`) can sit anywhere in the model compute graph, replacing or supplementing operators that RKNN3 does not support. During toolchain conversion, nodes that cannot be mapped to the NPU are preserved as `CustomOperator` and executed at runtime on the coprocessor CPU cores. Main differences from post-process operators:

Aspect	Post-Process Operator	Custom CPU Operator
Position	End of the model	Any position in the compute graph
Inputs and outputs	Customized by the plugin's <code>get_output_num / get_attrs</code>	Determined by the model graph; the plugin does not implement these two callbacks
Operator attributes	None	From the original framework (ONNX), read through <code>get_param</code>

4.3.1.2.1 Plugin Implementation Steps

The implementation steps of a custom CPU operator are similar to those of a post-process operator (init / deinit / compute / registration entry), with two key differences:

- **Do not implement `get_output_num / get_attrs`** : Inputs and outputs are determined by the model graph; no plugin customization is needed.
- **Read operator attributes through `get_param`** : Operator attributes come from the original framework (ONNX). `rknn3_custom_op_context` contains `rknn_ctx`, `priv_data` (do not modify), `user_data` (managed by the user), and `get_param`. `get_param` supports 6 dtypes: `i` (int32), `f` (float), `s` (char[]), `is` (int32[]), `fs` (float[]), and `ss` (char[] NUL-separated).

The `compute` callback typically has three steps: read attributes -> fetch inputs and convert layout/type (unpack NC1HWC2+FP16 into NCHW+FP32) -> compute and write back (pack back into NC1HWC2).

```
/* op_type must exactly match the operator type name in the model (case-sensitive)
*/
static rknn3_custom_op my_op = {
    .op_type      = "RknnCustomOpExam",
    .plugin_type   = RKNN3_OP_PLUGIN_TYPE_CUSTOM_OP,
    .target       = RKNN3_OP_TARGET_TYPE_CPU,
    .init         = my_op_init,
    .compute      = my_op_compute,
    .deinit       = my_op_deinit,
};
static rknn3_custom_op* registered_ops[] = { &my_op, NULL };
```

4.3.1.2.2 Plugin Compilation, Loading, and Usage

The compilation requirements, loading API, and usage flow of custom CPU operators are identical to those of post-process operators; see 4.3.1.1.2. The only difference is that after loading, there is no need to query `RKNN3_QUERY_POSTPROCESS_*` (the inputs and outputs of a custom CPU operator are determined by the model graph; the regular `RKNN3_QUERY_IN_OUT_NUM / RKNN3_QUERY_OUTPUT_ATTR` queries suffice).

4.3.1.2.3 Examples

For a complete example, see `rknn3-runtime/examples/rknn3_custom_op_demo/` (`RknnCustomOpExam: y=clamp(x*scale+shift, min, max)`). It demonstrates the full flow of reading attributes with `get_param` and executing computation with `compute`. Use `fp16_to_fp32 / fp32_to_fp16` from `float16.h` for FP16/FP32 conversion.

4.3.2 Encrypted Model Deployment

Applicable Scenarios

Protect model intellectual property and prevent unauthorized copying and distribution of `.rknn` / `.weight` files. A hybrid encryption architecture is used:

- **AES-128-CTR** (symmetric): Encrypts the model body. High performance and supports random-access decryption.
- **RSA-OAEP** (asymmetric): Protects the AES key for secure distribution.

Key system: a user key (AES-128, generated by the user, encrypts and decrypts the model files) plus an RK key pair (RSA, preloaded by RK; the public key encrypts the user key, and the private key built into the chip decrypts it).

Usage

- **Encryption:** Encrypt the model with `rknn3_model_encrypt.py`. It outputs the encrypted model and the key envelope:

```
python3 rknn3_model_encrypt.py \  
    --model model.rknn --weight model.weight \  
    --rk-pubkey rk_public_key.pem --output-dir ../encrypted  
# Output: model.rknn.enc, model.weight.enc, user_key_iv.enc (RSA-encrypted key  
envelope)
```

- **Loading:** Set the decryption key after `rknn3_init()` and before `rknn3_load_model_from_path()`. The runtime decrypts automatically, and the subsequent flow is identical to that of unencrypted models:

```
rknn3_init(&ctx, NULL);  
rknn3_set_decrypt_key_from_path(ctx, "user_key_iv.enc"); /* Must be called before  
load_model */  
rknn3_load_model_from_path(ctx, "model.rknn.enc", "model.weight.enc");  
rknn3_config config = {0};  
config.run_core_mask = 0xff; /* Configure according to the actual core count */  
rknn3_model_init(ctx, &config);  
/* Subsequent inference is identical to unencrypted models */
```

Note: Security features include two-layer encryption; the key can be decrypted only by the private key built into the chip; weights are transmitted as ciphertext (no plaintext is exposed on PCIe / USB / Ethernet links); and the decrypted key exists only briefly in memory and is cleared after use. For a complete example, see `rknn3-runtime/examples/rknn3_encrypt_demo/`.

5. Issue Diagnosis and Optimization

This chapter describes diagnosis methods and optimization techniques for common issues in model deployment, covering three core issue categories—accuracy, performance, and memory—as well as common error handling. The overall troubleshooting approach is: first confirm environment and version consistency, then verify model correctness on the simulator, and finally locate Runtime or deployment issues on the device.

5.1 Accuracy Issue Diagnosis

Applicable Scenarios

Inference results deviate excessively from the original model after model conversion or quantization, and you need to determine whether the cause is a configuration error, quantization error, pre-processing and post-processing error, or a Runtime implementation issue. Accuracy troubleshooting is divided by quantization workflow into internal quantization accuracy diagnosis and RKQuantizer quantization accuracy diagnosis; the two differ slightly in their troubleshooting stages.

5.1.1 Internal Quantization Accuracy Diagnosis

Accuracy troubleshooting for the internal quantization workflow follows the progressive sequence of "simulator FP16 accuracy -> simulator quantized accuracy -> connected-board accuracy -> Runtime accuracy"; correctness at each stage is a prerequisite for the next. The troubleshooting steps are shown in the figure below:

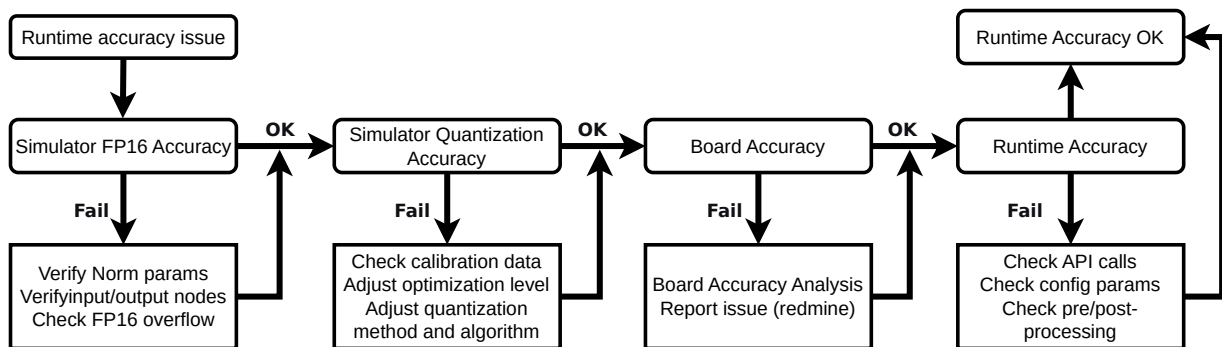


Figure 5-1 Internal quantization accuracy troubleshooting steps

Cosine distance serves as the basic criterion for judging accuracy, and final verification must be performed on a dataset. The methods for each stage are described below in order.

Stage 1: Simulator FP16 Accuracy Troubleshooting

Run inference on the simulator with quantization disabled (FP16 by default) to confirm basic correctness. Focus on the following:

- **Verify normalization parameters:** `mean_values` / `std_values` must match those used during training. Note the mismatch between the BGR order read by cv2 and the NHWC default values. For an N-channel input, use `[[c1, c2, ..., cn]]`; for multiple inputs, use `[[...], [...]]`, with each sub-list corresponding to one input:

```
rknn.config(mean_values=[[128, 128, 128, 128]], # 4-channel single input
            std_values=[[1, 1, 1, 1]],
            target_platform='rk1820', core_num=1)
```

- **Verify input and output nodes:** The input/output node names of `load_onnx()`, `input_size_list` (input sizes), and `data_format` must match the model definition.
- **Check for FP16 overflow:** If an intermediate tensor exceeds -65504~65504, use accuracy analysis to check for `inf`; this can be resolved by modifying the model structure, such as adding a BN layer. If FP16 inference results already differ significantly from the original model and it is not a configuration or overflow issue, the Runtime accuracy may be poor; go directly to Stage 4 for Runtime accuracy troubleshooting.

Stage 2: Simulator Quantized Accuracy Troubleshooting

After FP16 correctness is confirmed, enable quantization and start troubleshooting. Focus on the following:

- **Check calibration data:** The calibration set must cover the data distribution of the actual application scenario, typically 20~200 sets. Calibration data can be images (jpg/png, read with OpenCV) or npy files (read with numpy); npy is recommended for inputs that are not RGB/BGR images. Each line in `dataset.txt` is one set of inputs; for multiple inputs, separate them with spaces:

```
sampleA_in0.npy sampleA_in1.npy sampleA_in2.npy
sampleB_in0.npy sampleB_in1.npy sampleB_in2.npy
```

For 3-channel image input with quantization data in image format, confirm whether the model input is RGB or BGR, and set `quant_img_RGB2BGR` accordingly. This parameter only controls whether channels are converted when calibration set images are read during quantization; it does not affect inference: if the model was trained with RGB, set it to `False` or leave it unset, and feed RGB input during inference; if the model was trained with BGR, set it to `True` and feed BGR input during inference. If quantization data is in npy format, it is recommended to leave this parameter unset to avoid confusion.

- **Adjust the optimization level:** `optimization_level` defaults to 3 (speed-prioritized). Lowering it (for example to 0) disables optimizations that affect accuracy.
- **Adjust the quantization method and algorithm:** If accuracy is insufficient, optimize in the following order:
 - **Switch the quantization method:** For w8a8, change from `layer` to `channel`; for w4a16, change from `channel` to `group32`.
 - **Switch the quantization algorithm:** `normal` -> `kl_divergence` (effective when feature distributions are uneven) -> `mmse` (iteratively clips weights; greatly increases time and memory).
- **Hybrid quantization:** Use higher precision for sensitive layers (for example w8a16) and lower precision for the rest (for example w4a16); specify the quantization type of each layer via `op_quantized_dtype`.

Quantization Algorithms and Calibration Data:

Algorithm	Characteristics	Recommended Data Volume
normal (default)	Standard calibration, fast	20~200 sets
kl_divergence	Better improvement when feature distributions are uneven; slightly slower than normal	20~200 sets
mmse	Iteratively clips weights of intermediate layers; may improve accuracy but greatly increases time and memory	20~50 sets

Start with `normal` ; if accuracy is unsatisfactory, try `kl_divergence` or `mmse` . The quantization algorithm only affects the conversion stage and does not affect runtime performance.

If some layers in the `single` column show severe accuracy loss that cannot be improved, the numerical distribution of the weights may be unsuitable for quantization; the model structure must be modified or the model retrained.

Stage 3: Connected-Board Accuracy Troubleshooting

After the simulator works correctly, run connected-board inference (specify `target` + `core_mask`) and compare the results with the simulator results. Focus on the following:

- **Connected-board accuracy analysis:** If differences are large, perform connected-board accuracy analysis (`rknn.inference(accuracy_analysis=True)`), requires `profile_mode=True`) to compare per-layer output differences between the simulator and the board. Focus on checking whether pre-processing and post-processing are consistent with the simulator (normalization, channel order, data type), and the C API input/output configuration (whether normalization is applied twice, channel order, `rknn3_create_mem()` size and data format).
- **Report the issue (redmine):** If differences persist and cannot be eliminated, report the accuracy analysis results to the NPU team (redmine).

Stage 4: Runtime Accuracy Troubleshooting

If connected-board inference is correct but on-device deployed inference is abnormal, perform on-device accuracy analysis and check the `single_sim` column in `runtime_error` (low cos or high euc values are highlighted in yellow or red). Focus on the following:

- **Check API calls:** Verify that the API call order and parameter passing are correct.
- **Check configuration parameters:** Verify that configurations such as `core_mask` , memory allocation, and data types match the model.
- **Check pre-processing and post-processing:** Verify that input pre-processing (normalization, channel order, data type) and output post-processing (dequantization, business logic) are consistent with the simulator.

If abnormal `single_sim` values cause the `entire` results to diverge further from the simulator, the Runtime implementation of that layer may be faulty; report it to the NPU team.

5.1.2 RKQuantizer Quantization Accuracy Diagnosis

The output of the RKQuantizer quantization workflow is already a fake-quantized model (weights are fake-quantized), so FP16 accuracy troubleshooting is not needed. Accuracy troubleshooting follows the progressive sequence of "quantized accuracy -> connected-board accuracy -> Runtime accuracy". The troubleshooting steps are shown in the figure below:

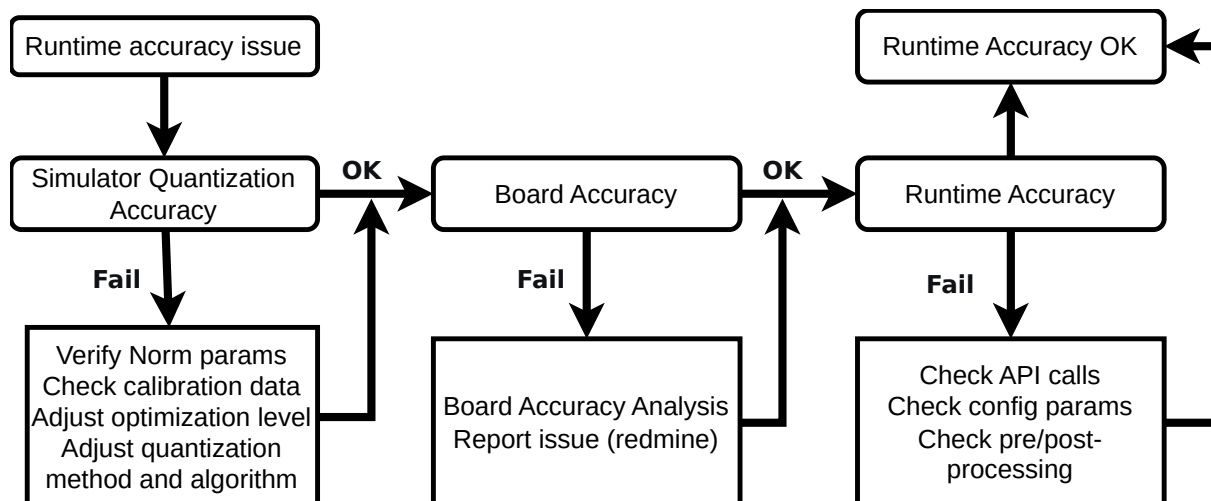


Figure 5-2 RKQuantizer quantization accuracy troubleshooting steps

Stage 1: Quantized Accuracy Troubleshooting

For models quantized by RKQuantizer, start directly with quantized accuracy troubleshooting. Focus on the following:

- **Verify normalization parameters:** `mean_values` / `std_values` must match those used during training. Note the mismatch between the BGR order read by cv2 and the NHWC default values.
- **Check calibration data:** The GRQ algorithm of RKQuantizer requires calibration data to collect the activation distribution of each layer and fine-tune the weights. Calibration data is provided as JSON files and must cover the data distribution of the actual application scenario, typically 20~200 samples. For the specific format and preparation method of calibration data, see 2.3.4 RKQuantizer Quantization Workflow.
- **Adjust the optimization level:** In the RKNN conversion stage, `optimization_level` of `rknn.config()` defaults to 3 (speed-prioritized). Lowering it (for example to 0) disables optimizations that affect accuracy. Note that when converting with the RKQuantizer quantization workflow, `quantized_algorithm` must be set to `normal`.
- **Adjust the quantization method and algorithm:** RKQuantizer supports two algorithms, GRQ and Normal, with GRQ as the first choice. If accuracy is insufficient, adjust `quantized_dtype` (w4a16 / w6a16 / w8a16) or use hybrid precision (`layer_quant_config` / `auto_hybrid_rate`); see 2.3.4 for details.

LLM / MLLM/VIT Quantization Recommendations: Use `quantized_dtype='w4a16'` and `quantized_method='group32'`; to further improve w4a16 accuracy, apply RKQuantizer GRQ quantization first.

If some layers in the `single` column show severe accuracy loss that cannot be improved, the numerical distribution of the weights may be unsuitable for quantization; the model structure must be modified or the model retrained.

Stage 2: Connected-Board Accuracy Troubleshooting

Same as the connected-board accuracy troubleshooting for internal quantization; see Stage 3 in 5.1.1.

Stage 3: Runtime Accuracy Troubleshooting

Same as the Runtime accuracy troubleshooting for internal quantization; see Stage 4 in 5.1.1.

5.2 Performance Analysis and Optimization

Applicable Scenarios

Model inference speed falls short of expectations, and you need to determine whether the cause is an NPU compute bottleneck, a bandwidth bottleneck, CPU operator overhead, or time spent on pre-processing/post-processing and data transfer, and then optimize accordingly.

Performance Analysis Tools

Tool/Interface	Function	Usage
<code>rknn.eval_perf(log_level=)</code>	Toolkit performance evaluation; outputs Op Time Ranking	Requires <code>config(profile_mode=True)</code> first
<code>rknn3_profile_ops(ctx, ..., log_level)</code>	Per-layer operator performance on device	<code>log_level</code> 0 for summary / 1 for per operator / 2 for Cycle and bandwidth
<code>rknn3_session_query_state()</code>	LLM session performance statistics	Queries <code>RKLLMRunState</code>
Lite <code>session_run</code> return value	LLM Python performance statistics	Returns <code>[n_decode_tokens, n_prefill_tokens, llm_start_time, llm_end_time]</code>

Key LLM performance metrics: TTFT (first-token latency), TPOT (time per token), Prefill TPS, and Decode TPS. Use `llm_start_time / llm_end_time` plus the first-token time to calculate the time and throughput of each stage.

Performance Optimization Techniques

- **Multi-core configuration:** Use multiple cores (`core_num>1`) for compute-intensive models, and a single core for small networks (multi-core switching requires CPU involvement and can be slower). `core_mask` must match the `core_num` used at conversion.
- **Quantization acceleration:** Quantization improves both bandwidth and compute utilization, and is the fundamental means of performance optimization.
- **Offload CPU operators to the NPU:** Most performance issues stem from CPU operators (size limits exceeded/unsupported/hardware limitations). Identify CPU operators with `eval_perf / profile_ops` , then replace them with NPU-supported equivalent operators or adjust their sizes.
- **Graph-level optimization:** Hardware fusion (Conv+Relu/Clip/Add), replacing algorithm-equivalent subgraphs with a single OP, and merging same-type operators.
- **Operator-level optimization:** DDR alignment (align Channel to Int8=32/FP16=16, and HW to 4), avoid inefficient operators (Transpose/Reshape/Resize/Sigmoid/Tanh), and optimize convolution sizes (utilization is higher when Channel<256 and output HW>=32).

- **LLM-specific optimization:** `keep_one_logit` reduces `lm_head` computation in the prefill stage, `kvcache_dtype` selects Int4/Int8 to reduce KVCache access, and multi-core configuration accelerates prefill.

Performance Analysis Baseline

Performance data is meaningful only under **fixed frequency** (data fluctuates significantly without it). On the board, use `opp_dump` to check CPU/NPU frequencies, and `srv_set_rate / npu_set_rate` to set frequencies. Use `rknn-smi info -w` to check the NPU load; a low load indicates the NPU is waiting, so optimize pre-processing and post-processing or use multi-threading.

Note: To identify NPU operator bottlenecks, compare DDR Cycles vs NPU Cycles: `DDR > NPU` indicates a bandwidth bottleneck, otherwise a compute bottleneck. Deployment time consists of three parts: the user application (pre-processing and post-processing, which can be accelerated with the Matmul API), input/output transfer (in coprocessor mode, choose appropriate tensor data types/layout), and inference (average multiple runs).

5.3 Memory Analysis and Optimization

Applicable Scenarios

On-device memory is limited; when multiple models coexist or large models are deployed, memory becomes insufficient, requiring analysis of the memory usage composition and optimization. For LLM models, memory optimization is especially critical because KVCache consumes a large amount of memory.

Memory Analysis Tools

Tool/Interface	Function
<code>rknn.eval_memory()</code>	Toolkit memory evaluation; outputs Device Memory and the Per-Core Memory Allocation table
<code>rknn3_query(ctx, RKNN3_QUERY_CORE_MEM_SIZE, ...)</code>	Queries weight/internal usage of each NPU core on device
<code>rknn3_profile_mem(ctx)</code>	Prints per-core Command/Weight/Internal/KVCache and device memory statistics on device

The Per-Core Memory Allocation output of `eval_memory` includes Command (register configuration), Weight (weights), Internal (intermediate tensors), KVCache (LLM-specific), and Total.

Memory Composition

Runtime memory = weights + internal tensors + register configuration + input/output tensors; LLMs additionally have KVCache. Memory is allocated at `rknn3_model_init() / rknn3_session_init()`.

Memory Optimization Techniques

- **Internal memory reuse:** In multimodal serial inference (vision -> LLM), internal memory on the same core can be reused (see 4.1.4); the total number of blocks = number of vision

cores + number of LLM cores - number of overlapping cores.

- **Multi-session sharing:** Multiple sessions under the same context share weights/internal/LoRA, with only KVCache being independent; this saves memory compared with multiple contexts.
- **KVCache optimization:** `kvcache_buffer_len` (32-aligned; set as needed), `kvcache_dtype` (memory usage: FP16 > Int8_to_F16 > Int4_to_F16), and `position_embeddings_host_storage` (stores on the Host side to save device memory).
- **Tie Word Embedding:** Reuses `lm_head` weights, saves `embed.bin`, and reduces Host-side memory (see 4.2.8).
- **llm_head device designation:** `llm_head.device` can designate `lm_head` to the main SoC (rk3588/rk3576) to reduce coprocessor memory.
- **core_mask control:** `core_num / core_mask` determine how many cores the weights/internal are distributed across, directly affecting Total memory; configure as needed rather than using all cores.

Note: Dynamic-shape models do not support weight sharing or context reuse (`rknn3_dup_context()`), so memory reuse is limited. Use `rknn.eval_memory()` to check Total on-device memory and assess whether it is sufficient.

5.4 FAQ and Error Handling

Version and Compatibility

- **Version inconsistency:** `rknn3-toolkit`, the communication services (`rknn3_transfer_proxy` / `rknn3-server`), and the Runtime library must be the same version. Otherwise, model conversion may succeed but the model fails to run on the board, possibly with `protocol fingerprint mismatch` or an `Invalid RKNN format` error. Version query commands:

```
pip list | grep rknn3-toolkit           # Toolkit version
strings xxx.rknn | grep rknn3-toolkit  # Version used for model
conversion
strings librknn3_api.so | grep -i "librknn3_api version" # Runtime version
strings rknn3_transfer_proxy | grep "Transfer version"  # Coprocessor mode
communication service version
rknn3-server -v                          # Non-coprocessor mode
```

- **Cross-platform incompatibility:** Models for RK1820 / RK1828 / RK3572 are mutually incompatible; re-convert for the target platform.

Tool Installation

- **Installation failure due to dependency conflicts:** When all dependencies are installed but versions do not match, the pip environment check blocks installation. Add `--no-deps` to skip it: `pip install rknn3-toolkit-*.whl --no-deps`. You must then ensure the runtime dependencies are complete yourself.
- **No ARM Linux version:** RKNN3-Toolkit is only provided for x86_64; there is no ARM Linux version.

Model Loading Failure

Error	Cause	Solution
Error parsing message	ONNX model corrupted	Re-download and verify MD5
Not support input data type 'float16'	PyTorch float16 weights not supported	Change to float32
Error with custom output nodes	Invalid outputs node names	Verify the node names
Error with operators of more than 4 dimensions	Not supported yet	Remove or simplify manually
Dynamic graph error (such as NonZero)	Causes a dynamic graph	Replace/remove the OP

It is recommended to preprocess the model with `onnx-simplifier` before conversion.

Model Conversion Issues

- **onnxsim simplification failure:** onnx-simplifier may fail to simplify models with failed shape inference or dynamic graphs. In this case, convert directly from the original ONNX; onnxsim is not mandatory.
- **mean_values channel count error** (such as `expect N`): When the model input is not a 3-channel image (for example 1x32 non-image data), the number of `mean_values` channels must match the actual input channels; if the model does not need mean/std, leave them unset.
- **RKNN model larger than the original model:** This is normal. In addition to weights and graph structure, an RKNN model contains NPU register configuration information and may split OPs to improve runtime efficiency.

Incorrect Inference Results

- **Differences between simulator and board:** Small differences caused by hardware driver variations are normal; report large differences to the NPU team.
- **Large differences between connected-board and on-device inference:** Check the consistency of pre-processing, data types, and layout (NCHW / NHWC).
- **Inference takes >20s with incorrect results:** Usually an NPU hang; update the Toolkit / Runtime.
- **Connected-board error with 3-dimensional input:** NPU support for 3 dimensions is incomplete; change to 4 dimensions or update to a newer version.
- **Results differ between runs:** An on-device NPU kernel driver bug; update the driver and the Toolkit.
- **All-nan results with `do_quantization=False`:** An intermediate layer exceeds the FP16 range; add a BN layer.
- **QAT model inconsistency:** Set `engine='qnnpack'` for PyTorch inference.

Simulator and Connected-Board Inference

- **Distinguishing the three inference methods:** Simulator inference (RKNN3-Toolkit simulator on Linux x86_64; output may not match the board), connected-board inference

(the PC calls the Toolkit Python API while the model runs on the board NPU), and on-device inference (the board calls the C API to run inference). Simulator results are for preliminary verification only; for accuracy/performance, refer to connected-board or on-device results.

- **Connected-board slower than on-device:** Connected-board inference incurs PC↔board data transfer overhead; the actual NPU inference performance is measured with the on-device C API.
- **Viewing connected-board logs:** Model initialization and inference run on the board, so logs are generated on the board. Enter the board system via the serial port, run `export RKNN_LOG_LEVEL=5`, then start connected-board debugging; on-device error messages appear in the serial console window.
- **data_format default value:** In connected-board mode, `rknn.inference()` defaults to `NHWC`, while ONNX 4-dimensional input defaults to `NCHW`; inference internally converts `NCHW→NHWC`. If `data_format='NCHW'` is set explicitly, no conversion is performed.

Model Evaluation Issues

- **Invalid RKNN format:** On the board, this error usually means `target_platform` in `rknn.config()` is unset or incorrect, or the Toolkit and Runtime versions are incompatible. Check `target_platform` and ensure version consistency.
- **Inference time inconsistent with eval_perf:** `rknn.inference()` connects to the board via PC + adb and incurs fixed data transfer overhead; for the real frame rate, test directly on the board with the C API.
- **Resize OP causes accuracy drop:** The NPU does not yet support hardware-level Resize; the conversion tool converts Resize to ConvTranspose, which loses accuracy. Avoid Resize where possible (replace it with ConvTranspose and retrain), or set `optimization_level=2` in `rknn.config()` to run Resize on the CPU (no accuracy loss but reduced performance).
- **perf_debug affects performance data:** Enabling `perf_debug` in `rknn.init_runtime()` adds debug code and may disable parallelism, so the measured time is higher than with it disabled. `perf_debug` is for locating layers that account for a large share of the time, not for absolute performance evaluation.

core_mask Mismatch

Error `core_mask ff is not match with npu core number 1!`: The `run_core_mask` of `rknn3_model_init()` does not match the `core_num` used at conversion. Configure according to the correct core number: `0x1~0xff` for RK1820 / RK1828, and only `0x1` for RK3572.

Insufficient Memory

- **Killed during quantization:** Toolkit quantization consumes a large amount of memory; increase the PC memory or virtual memory/swap partition.
- **Insufficient on-device memory:** Check Total with `rknn.eval_memory()` and optimize using the techniques in 5.3.

Return Value Checking

Check the return value of every API call; `RKNN3_SUCCESS` (`=0`) indicates success. Unified pattern:

```
int ret = rknn3_init(&ctx, NULL);
if (ret != RKNN3_SUCCESS) {
    printf("rknn3_init failed! ret=%d\n", ret);
    return -1;
}
```

`rknn3_session_init()` returns NULL on failure. Resource release order: destroy the session first (`rknn3_session_destroy()`), then destroy the context (`rknn3_destroy()`); for general model scenarios, also release self-allocated tensor memory (`rknn3_destroy_mem()`) before destroying the context.

C API Usage

- **`rknn3_create_mem()` allocation size:** Input memory is generally allocated using `aligned_size` and `core_id` from `rknn3_tensor_attr`, ensuring the size and alignment meet NPU requirements.
- **Input data filling order:** Fill 4-dimensional input in `NHWC` order (`[batch, height, width, channel]`); fill non-4-dimensional input according to the original model shape, and set `data_format` to `undefined`.

Logging and Debugging

- Connected-board inference logs are on the board; enter the board system via the serial port and run `export RKNN_LOG_LEVEL=5` to obtain detailed errors.
- Use `RKNN_LOG_LEVEL=4` for on-device performance profiling.
- Connected-board initialization stuck after Docker restart: `rknn3_transfer_proxy` exited abnormally, leaving the RKNN Server connection state unreset; restart the development board.

LLM and Session

- **Slow first token (high TTFT):** The prefill stage processes the entire input sequence and is compute-intensive. Reduce `max_context_len`, or enable KVCache Checkpoint (only for Linear/Sliding Attention models) to speed up long-prefix recovery.
- **Multi-turn conversation gets progressively slower:** Context growth enlarges the KVCache. Use the `RKNN3_KVCACHE_POLICY_RECURRENT` policy for automatic overwrite, and set `n_keep / n_keep_aligned` to keep the system prompt from being overwritten.
- **`rknn3_session_init()` returns NULL:** Check whether `vocab_info` is filled in correctly (`vocab_size / special_eos_id`, etc.), whether `max_context_len` is too large and causes insufficient memory, and whether the model is loaded correctly.
- **result_callback not triggered:** Confirm that the callback is registered via `rknn3_session_set_callback()` and that the `RKLLMCallback` structure is initialized correctly (fill it in after zeroing with `memset`).

Memory and Performance

- **Insufficient memory when loading an LLM model:** An overly large `kvcache_buffer_len` increases KVCache memory; set it according to actual needs (32-aligned). Choose `Int4_to_F16` or `Int8_to_F16` for `kvcache_dtype` for lower usage than

`Float16 ; position_embeddings_host_storage=True` stores position embeddings on the Host side to save device memory.

- **Insufficient memory with multiple models coexisting:** Multimodal serial inference can use Internal memory reuse (see 4.1.4); `llm_head.device` can offload `lm_head` computation to the main SoC to reduce coprocessor memory.

Deployment Modes

- **Cannot connect to the board in coprocessor mode:** Confirm that `rknn3_transfer_proxy` is started on the board, the PCIe / USB / Ethernet connection is working, and `device_id` is correct (query it via `rknn3_find_devices()` or `get_devices_id()`).
- **Whether RK3572 and RK182X models are interchangeable:** No. RKNN models for different platforms are mutually incompatible; re-convert for the target platform (`target_platform` parameter).

Model Encryption

- **Encrypted model fails to load:** Confirm that `rknn3_set_decrypt_key_from_path()` is called before `rknn3_load_model_from_path()` ; the key envelope file (`user_key_iv.enc`) must be paired with the encrypted model; the key envelope is generated by encryption with the corresponding `rk_public_key.pem` and must not be mixed with others.

Multimodal

- **Garbled output after image input in MLLM:** Check whether the placeholder tokens (`image_start` / `image_end` / `image_content`) match the special tokens of the model tokenizer, and whether the dimensions of `image_embed` (`n_image_tokens × embedding_dim`) and the count (`n_image`) match the number of `<image>` tags in the prompt.

Note: The error `Meet unsupported xxx operator` means the Runtime does not support that operator; update the toolchain to the latest version. The HTTP error codes of `rkllm3-server` follow the OpenAI specification (401 `authentication_error`, 400 `invalid_request_error`, 501 `not_supported_error`); for the complete error codes, see *RKLLM3_Server_User_Guide_and_API_Reference*.